ENERGY NOT ENEMY

Energy Not Enemy

A builder's guide to building-scale solar-thermal power,
from first principles to national scale

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Energy Not Enemy

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How to Read This Book

This book climbs the Technology Readiness Level scale from TRL 1 (basic principles) to TRL 9 (proven system). Each of the nine parts explains, demonstrates, guides, and evaluates: every part ends with a gate — a set of questions the reader must be able to answer before climbing to the next level.

The book's contract is stated here and kept throughout: **real data, real math, gaps named plainly.** Measurements, calculations, assumptions, and hypotheses are labeled as what they are and never mixed. Unresolved claims are marked **[VERIFY]**. Component maturity does not equal system maturity — the climb in these pages is the *integration's* climb, and it earns each level with evidence.

The system: a parabolic-profile Fresnel lens concentrating sunlight (with heliostat-style tracking mirrors), a closed-loop boiler flashing water to steam, a steam turbine driving a generator, and Tesla-valve sections directing water by phase — steam by day, liquid by night — in a sealed loop. No photovoltaics. Building-scale modules, repeated nationally: the building is the unit.

Three deployment tiers carry the whole book: the **home**, the **corporate building**, and the **apartment complex**. No component invention is claimed — the contribution is the integration of existing patented technologies, each traced to its patent and referenced.

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TRL 1

Basic Principles

Chapters 1–5

Chapter 1

The Building Is the Unit

TRL 1 · Basic principles. This chapter explains the demand side in numbers before any supply physics enters the book.

1.1 The problem, stated in numbers

Every energy discussion starts in the wrong place. It starts with power plants, grids, and national totals — abstractions no one can touch. This book starts where energy is actually consumed: the building.

A building is the natural unit of energy demand. People do not consume electricity as a nation. They consume it as a household running an air conditioner, an office running its lights and servers, a complex running a hundred of each. Every kilowatt-hour on every bill in the country resolves to a building. If you can serve the building, you can serve the nation — because the nation is buildings, repeated.

Here are the numbers, approximate and sourced at the end of this chapter:

- The average American home uses roughly **10,800 kilowatt-hours of electricity per year** — about 30 kWh per day.
- At the national average residential rate of about **16 cents per kWh**, that home pays roughly **$1,700 a year** for electricity.
- A typical home's electrical demand **peaks around 5 kilowatts** — air conditioner, dryer, and stove overlapping on a summer evening.

- Residential and commercial buildings together consume roughly **40% of all primary energy** in the United States and about **three-quarters of its electricity**.
- About **5% of generated electricity is lost** in transmission and distribution before it reaches anyone — burned as heat in wires between a distant plant and the building that paid for it.

Read those again. Three-quarters of the country's electricity goes to buildings. Five percent of it dies in transit. The building is not a small part of the energy problem. It is the energy problem, wearing a roof.

1.2 Why the building is the right unit

Three reasons, each checkable.

First, demand is metered at the building. The utility bill arrives per address, not per citizen. Every economic signal in the energy system — rates, demand charges, time-of-use pricing — resolves at the building meter. Any solution that wants to change the economics must change what happens at that meter.

Second, generation co-located with demand skips the losses. Electricity generated on the building's own footprint does not travel transmission lines. The ~5% T&D loss goes to zero for that power. It also skips the demand-charge structure commercial buildings pay for peak draw from the grid — often the largest line on a commercial bill.

Third, buildings are countable and repeatable. There are roughly 140 million housing units and several million commercial buildings in the United States. That sounds like an impossibly large deployment — until you recognize it as a manufacturing problem, not a construction problem. One design, built many times, installed per building. The nation is not powered by one heroic machine. It is powered by one good machine, repeated.

This is the scaling principle the entire book rests on: **the reason the method can power a nation is that it can power a**

building. An apartment complex. A corporate building. A trailer park. Solve one building completely — real power, real numbers, real operation — and national scale is replication, not reinvention.

1.3 Three tiers, in numbers

The book works three deployment tiers. Here is each one, quantified. Figures marked *(illustrative)* are worked examples with stated assumptions, not measurements.

Tier 1 — The single home

- Demand: ~30 kWh/day, ~5 kW peak.
- Footprint available: less space than a reasonable average US backyard.
- The design question this book answers: can a backyard-scale thermal unit generate 30 kWh on an average day? The method for answering it — sizing from local solar data — is developed in Chapters 8 and 12.

Tier 2 — The corporate building *(illustrative)*

- Take a mid-size office building: 50,000 square feet, 200 occupants.
- Assume 15 kWh per square foot per year for a typical office (a standard planning figure; the reader replaces it with the building's actual bills in Chapter 8's method).
- Demand: ~750,000 kWh/year, roughly 2,000 kWh/day, peak in the low hundreds of kilowatts.
- Note what changed from Tier 1: daily energy went up ~70×, but the *method* didn't change. Same equations, larger inputs. That is what a scaling law means, and Chapter 9 derives them.

Tier 3 — The apartment complex *(illustrative)*

- Take 100 units at the Tier 1 household figure: 100 × 30 kWh/day = 3,000 kWh/day, plus common areas (lighting, laundry, pool pumps) at ~10%.
- Demand: ~3,300 kWh/day, peak ~500 kW.
- The design question: one shared loop or per-building units? Chapter 9 works both and lets the numbers decide.

None of these are promises. They are the demand side of the ledger, stated so the supply side — everything from Chapter 2 onward — has something to answer to. Every sizing method in this book terminates in a check against numbers like these.

1.4 The method: read your own building

You do not need this book's permission to start. Take your last twelve utility bills and do this:

1. **Annual kWh** ÷ 365 = average daily kWh.
2. **Peak kW** — printed on most commercial bills; for homes, estimate from the largest appliance overlaps, or read it from a $30 clamp meter on the main feed.
3. **Cost per kWh** — total bill ÷ total kWh, which includes every fee and rider. This is the number any alternative must beat.
4. **Load shape** — when does the building use power? Day-heavy (office) or evening-heavy (home)? This determines how much of the demand a daytime-generating system serves directly.

Write those four numbers down. They are your building's energy identity. Every calculation in this book takes numbers like these as input and returns a system size as output. If you cannot state your building's four numbers, you are not ready for Chapter 8 — and that is fine; Chapters 2 through 7 build the physics you will feed them into.

1.5 What this chapter proved

- Energy demand resolves at the building. The bill, the meter, and the load all live there.
- Buildings are the dominant energy consumer in the country and the natural unit of any solution.
- Three tiers — home, corporate building, apartment complex — cover the deployment space, and the same methods size all three.
- National scale is replication of a solved building, not a separate invention.

The demand side is now on the table, in numbers. Chapter 2 begins the supply side: what happens when concentrated sunlight becomes heat, and how much heat a lens can actually deliver.

Self-evaluation — Gate 1, part 1

Answer before moving on. If any answer is a guess, re-read the section.

1. Why is the building — not the city, not the nation — the correct unit for this book's analysis? Give all three reasons.
2. A home uses 10,800 kWh/year at 16¢/kWh. What does it pay annually? What is its average daily use?
3. What are a building's four energy-identity numbers, and which one determines whether a daytime-generating system is a good fit?
4. A 100-unit complex is proposed with per-building units instead of one shared loop. Which chapter will decide that question, and what decides it — opinion or numbers?

Sources and honesty notes

- US residential consumption (~10,800 kWh/year), national generation totals, building share of energy use, and T&D losses (~5%): U.S. Energy Information

Administration (EIA). Figures are approximate national averages; your building's numbers come from its own bills (Section 1.4).

- Tier 2 and Tier 3 figures are illustrative worked examples with stated assumptions, not measurements. They demonstrate the method; they are not data.
- No claim in this chapter depends on the supply-side system working. The demand exists whether or not this book's machine ever gets built.

Chapter 2

Light into Heat

TRL 1 · Basic principles. This chapter explains concentration optics and states, once and unambiguously, what the lens in this book is.

2.1 What the lens is — stated without ambiguity

This book uses one lens. It is a **parabolic-profile Fresnel lens**: a flat, refractive lens — light passes *through* it — whose surface is cut into fine concentric prismatic grooves, with the groove angles following a parabolic focusing profile so that incoming parallel sunlight converges to a focus.

Read that again, because the terminology in this field is a swamp:

- It is **refractive** (a lens), not **reflective** (a mirror or dish). When solar literature says "parabolic concentrator," it almost always means a parabolic *reflector*. That is not this.
- It is **flat and thin** — millimeters of molded acrylic, not a thick glass lens. A true meter-scale solid-glass parabolic lens is essentially unprocurable: roughly 75–85 kg per square meter of aperture versus ~3 kg/m² for a molded Fresnel, plus severe light absorption in centimeters of glass. The book does not ask you to buy what cannot be bought.

- The word "parabolic" describes the **focusing intent** — the groove geometry converges light the way a parabola would. The word "Fresnel" describes the **manufacturable embodiment** — the thin, grooved, molded form you can actually order.

Wherever this book says "the lens," it means the parabolic-profile Fresnel lens. No other lens appears in this book.

2.2 What concentration means

Sunlight arrives dilute: about **1,000 watts per square meter** at peak on a clear day (the standard test-condition figure; the solar constant above the atmosphere is 1,361 W/m²). A square meter of sunlight will warm things, but it will not make steam at useful pressure. Concentration is the act of trading area for intensity.

The **geometric concentration ratio** is:

C = A_aperture / A_receiver

A 1 m² lens focusing onto a 1/60 m² receiver gives C = 60. The **flux concentration** — what the receiver actually feels — is C multiplied by the optical efficiency:

C_flux = C × η_optical

Optical efficiency is always below 1. Light is lost to surface reflection, groove imperfections, tracking error, and the receiver missing some rays. Measured Fresnel lenses in the literature land around **75–89% optical efficiency** depending on design and quality (sourced examples in Section 2.5). So a C = 60 lens with η = 0.75 delivers an effective flux concentration of 45 — the receiver sees 45,000 W/m² where the lens saw 1,000.

There is a hard ceiling. Conservation of étendue — the theorem that light cannot be squeezed into a smaller angular spread than it arrived with — caps concentration for the sun's angular size (about 0.265° half-angle):

- **Point focus (3D):** C_max ≈ 46,000
- **Line focus (2D):** C_max ≈ 212

Expressed as acceptance angle: θ_accept = arcsin(1/C) for point focus. Higher concentration means a narrower acceptance angle, which means the tracking must be more accurate. This is the fundamental trade of the whole optical design: **concentration buys temperature; tracking precision pays for it.** The home tier of this book works at C 40–80, which needs only ~1–2° of total tracking accuracy — well within ordinary tracker hardware, as Section 2.4 shows.

2.3 The heat ray, quantified

Follow the energy through a concrete case. Assumptions stated; you will replace them with your own in Chapter 12's method.

- Aperture: 1 m² parabolic-profile Fresnel lens
- Sunlight: DNI = 1,000 W/m² (clear midday)
- Geometric concentration: C = 60
- Optical efficiency: η_optical = 0.75 (mid-range of sourced measurements)

Power entering the lens: 1,000 W. Power arriving at the receiver: 1,000 × 0.75 = **750 W**, concentrated onto 1/60 m². Flux at the receiver: **45,000 W/m²** — forty-five times the intensity of direct sun.

What temperature does that make? The receiver radiates heat away as it gets hotter (proportional to T^4), convects heat to the air, and conducts it into its mount. The stagnation temperature — where losses equal input — is found by balancing the books: absorbed power in, all loss mechanisms out. The governing balance (Duffie & Beckman form) is:

Q_absorbed = Q_in Q_reflected Q_radiated Q_convected Q_conducted

with Q_in = DNI × A_aperture × η_optical. The book works this balance in full in Chapter 11; here is what the sourced operating points say a Fresnel concentrator actually achieves:

Concentration	Demonstrated temperatures (sourced)
C 60	200–260°C (NASA Marshall 1.8×3.7 m line-focus Fresnel, 34–43% collection efficiency)
C 30–100	~400°C (commercial trough operating band)
C of several hundred to 1,000+	700–1,000°C+

Focal spots of **930–1,064°C** have been demonstrated with Fresnel lenses at ordinary DNI (~790–860 W/m²). The home tier's target — steam hot enough for a small Rankine cycle, roughly 200–400°C — sits squarely inside demonstrated territory at C 40–80. Nothing in this chapter requires believing an unbuilt optic.

One more theorem, because it governs the small-scale economics of the whole book: **the loss fraction is proportional to 1/C.** A small receiver at low concentration loses a large share of its collected heat to radiation; raising C shrinks that fraction. This is why concentration is not optional for small systems — it is the mechanism that makes small receivers efficient. The building-scale unit survives thermodynamically *because* it concentrates.

2.4 The mirrors: the boost stage

Around the lens, the design places tracking mirrors — a heliostat boost stage. Their job is simple: intercept sunlight that would otherwise miss the lens and redirect it onto the aperture, raising the total power in.

Two honest statements about the boost stage:

1. **It adds power, not concentration beyond the étendue limit.** Mirrors enlarge the effective collection area; the focus still obeys the same optical limits. There is no closed-form concentration limit for a heliostat field feeding a lens — the étendue inequality bounds it, and real numbers require ray tracing the specific geometry (NREL's SolTrace does this; it is free).
2. **The tracking requirement is modest.** Buyable tracker hardware reaches 0.1°-class accuracy (sourced: Kipp & Zonen, Practical Solar, Stellio, CSIRO consensus ~1 mrad for tower heliostats). The home tier needs ~1–2°. The mirrors do not demand exotic hardware.

The mirrors move like solar-farm heliostats — dual-axis, sun-following. Their control, drives, and error budget are specified in Chapter 7's bill of materials and validated by ray trace in Chapter 12.

2.5 The method: from temperature target to lens specification

Given a target steam temperature, the reader sizes the optics in three steps:

1. **Set C from the temperature.** Write the receiver balance of Section 2.3 with your chosen receiver coating; solve for the C that reaches your target with 20% margin above stagnation. Cross-check against the sourced bands in the table above.
2. **Size the aperture.** A_aperture = P_thermal / (DNI_design × η_system), where η_system folds optical and receiver efficiency (0.45–0.55 for a non-evacuated receiver at 300°C; 0.55–0.65 evacuated), and DNI_design comes from local solar data (NREL's tools), not from optimism.
3. **Budget the tracking error.** Acceptance angle minus the sun's angular size leaves an error budget; allocate

it across sunshape, slope error, tracking, and alignment (root-sum-square); select a tracker using less than a third of what's left; confirm with a SolTrace ray trace before buying anything.

No step is a guess. Every step terminates in a number or a measurement.

2.6 What this chapter proved

- The lens is a parabolic-profile Fresnel lens: flat, refractive, grooved, molded — stated once, unambiguously, for the whole book.
- Concentration trades area for intensity; C = A_aperture / A_receiver, derated by measured optical efficiency.
- Étendue caps concentration and ties it to tracking accuracy; the home tier's C 40–80 needs only ~1–2° accuracy.
- Demonstrated Fresnel performance already reaches the temperatures this system needs. The heat ray is quantified, not imagined.
- Loss fraction 1/C: concentration is what makes small receivers efficient.

Chapter 3 takes the heat the lens delivers and turns it into work: the Rankine cycle, the steam tables, and the efficiency theorems.

Self-evaluation — Gate 1, part 2

1 State in one sentence what the lens in this book is, and in one more sentence what it is not.

2 Why is a meter-scale solid-glass parabolic lens not specified, even though "parabolic lens" sounds like one? Give the two quantitative reasons.

3 A 2 m^2 lens at C = 80 with η_optical = 0.75 receives DNI = 900 W/m^2. What power reaches the receiver, and what is the flux concentration?

4 Explain the trade between concentration ratio and tracking accuracy in your own words. Why does the home tier get away with ordinary trackers?

Sources and honesty notes

- Solar constant (1,361 W/m^2), STC irradiance (1,000 W/m^2): standard references; reader verifies against NREL data for their site.
- Étendue limits and acceptance-angle relations: standard nonimaging optics (Welford/Winston). The 46,000× / 212× figures are textbook theorems.
- NASA Marshall 1.8×3.7 m line-focus acrylic Fresnel (1977/79): peak C = 62–64, transmittance 78–81%, collection efficiency 43% at 200°C 34% at 260°C (OSTI biblio 5754330).
- Point-focus Fresnel optical efficiency 74.7%, mean flux C 90 (MDPI Energies 2018); up to 89.08% optical efficiency at C = 240–600 (2025 validated study); focal spots 930–1,064°C at DNI ~790–860 W/m^2 (MDPI Energies 2024).
- Solid-glass vs. Fresnel weight comparison (~75–85 kg/m^2 vs ~3 kg/m^2): calculated from geometry; the published data point is a 1,000 mm diameter, 2 mm PMMA Fresnel at 2 kg.
- Tracker accuracy figures: manufacturer/conference sources (Kipp & Zonen, Practical Solar, Stellio, CSIRO) — representative, not exhaustive; reader confirms current specs before procuring.
- The combined heliostat-field-plus-lens concentration limit has no closed form; the book states this openly and requires ray-trace validation per geometry.

Chapter 3

Heat into Work

TRL 1 · Basic principles. This chapter explains the Rankine cycle — the theorem that turns the lens's heat into shaft power — with the honest numbers for small scale.

3.1 The cycle in four states

Chapter 2 ended with heat arriving at the receiver. This chapter converts it. The machine that does it is the Rankine cycle: water pressurized, boiled, expanded through a turbine, and condensed back to water. Four states, four devices, one loop:

1. **State 1 → 2: the pump.** Liquid water at condenser pressure is pressurized to boiler pressure. Liquid barely compresses, so this costs little energy — but it is not free, and the book accounts for it.
2. **State 2 → 3: the boiler.** Pressurized water is heated to boiling, boiled to steam, and superheated — heated past the boiling point so it arrives at the turbine hot and dry.
3. **State 3 → 4: the turbine.** High-pressure superheated steam expands through the turbine, spinning the shaft. This is where heat becomes work.
4. **State 4 → 1: the condenser.** Exhaust steam is cooled back to liquid water at low pressure, closing the loop. The water is reused — the "no leaking" claim of Chapter 10's night loop begins here, in the thermodynamics.

The thermal efficiency — the fraction of heat in that becomes work out — is:

η_th = W_net / Q_in = 1 Q_out / Q_in

In the enthalpy form engineers actually compute with (h = specific enthalpy at each state):

$\eta_{th} = [(h_3 - h_4) - (h_2 - h_1)] / (h_3 - h_2)$

The numerator is turbine work minus pump work; the denominator is heat added in the boiler. Every efficiency number in this book comes from this equation or a stated refinement of it. There is no other way to compute it, which is exactly why it can be trusted.

3.2 The ceiling: Carnot

No heat engine beats Carnot. Stated in its cycle-meaningful form:

η 1 T_L / T_mean,H

where temperatures are absolute (kelvin), T_L is the condensing temperature, and T_mean,H is the entropy-averaged temperature at which heat is added. The design compass for the entire power block follows in one sentence: **raise the mean temperature of heat addition.** Superheat, reheat, regeneration — every advanced cycle is that sentence in different clothes. Hotter steam in, cooler condenser out, more work per unit of heat. Everything else is commentary.

This is why Chapter 2's temperature bands matter. A lens delivering 250°C steam and a lens delivering 400°C steam do not feed the same economics — the Carnot ceiling moves with the temperature, and the cycle efficiency follows it.

3.3 The honest numbers

Computed from the IAPWS-IF97 steam tables — the international standard formulation for water/steam properties, verifiable by any reader with the free `iapws` Python package or the NIST WebBook. Not hand-copied, not rounded into optimism:

Case	Conditions	Cycle efficiency	Carnot ceiling
Ideal	4 MPa, 400°C superheat, condenser 40°C	**36.2%**	53.5%
Realistic small-scale	Same, turbine 75%, pump 65%	**27.1%**	—
Pessimistic	Same, turbine 55%, pump 50%	**19.8%**	—

Read the pessimistic row carefully. Even with a mediocre small turbine and a mediocre pump, the cycle converts roughly one-fifth of its heat to shaft work. The physics does not require a good turbine to function — it requires one to be economical, which is a different question answered in Chapters 20 and 24.

Now the honesty this book owes you about small turbines, because the turbine is the weakest critical link in the power block:

- The EPA CHP catalog documents the scale penalty: ~50% isentropic efficiency at 500 kW, ~70% at 3 MW, ~80% at 15 MW. **Microturbines at or below 500 kW should be expected at 40–60%.**
- What punishes them: low blade Reynolds numbers (one study: efficiency falls 80% 75% shrinking a rotor from 127 mm to 12.7 mm), tip-clearance ratios, leakage, partial-admission losses, part-load operation.
- The honest engineering responses: a **de Laval impulse stage** (~80% potential in a single stage); a **Curtis velocity-compounded stage** (~70%, but swallows 8× the stage enthalpy drop — the standard small waste-heat choice: fewer stages, compact); **screw (volumetric) expanders** at 60–70% that tolerate wet steam, which answers the moisture problem saturated steam brings.

The book does not hide this. Chapter 20 tests it, Chapter 24 prices it. A theorem with a weak component is still a theorem — it just carries a named uncertainty, which is what TRL 3

through 6 are for.

3.4 Steam or something else: the decision rule

Below about 300°C, an organic Rankine cycle (ORC) — same cycle, different working fluid — is the thermodynamically honest choice; water's properties punish low-temperature operation. At 350–400°C superheat, steam Rankine is defensible. The founder's system targets steam temperatures where steam Rankine applies — but the book shows this crossover analysis instead of assuming it. Any reader whose lens delivers only 250°C should build the ORC variant; the methods transfer, the fluid changes. Truth first, preference never.

3.5 The condenser penalty

The cold end matters as much as the hot end. Every ~10 K rise in condensing temperature costs roughly **2 percentage points** of cycle efficiency. A Texas summer night condensing at ~55–60°C versus a mild night at ~40°C is ~25% versus ~28% — computed, not guessed. The rule the book enforces: **size the condenser for the design hot night, not the average night.** Chapter 15's night-loop experiments test exactly this, and Chapter 21 covers what hot-climate operation demands of the hardware.

3.6 The method: the Fixed State-Point Procedure

The named procedure for hand-calculating any Rankine cycle (Cengel & Boles, Chapter 10). Seven steps, no software required:

1 Fix state 1 at condenser pressure (saturated liquid): read h_1, v_1, s_1 from the tables.

2 Pump: $w_p \quad v_1 \cdot (P_2 \quad P_1)$ — the incompressible approximation; $h_2 = h_1 + w_p/\eta_p$.

3 Fix state 3 (boiler pressure, superheat temperature): read h_3, s_3.

4 Isentropic expansion: $s_4s = s_3$; at condenser pressure find h_4s; steam quality x = (s s_f)/(s_g s_f).

5 Actual turbine exit: $h_4 = h_3 \ \eta_t \cdot (h_3 \ h_4s)$.

6 Close the efficiency: $\eta = 1 \ (h_4 \ h_1)/(h_3 \ h_2)$.

7 Property sources: the **NIST WebBook** (free, authoritative) and the `iapws` **Python package** (IAPWS-IF97, free, scriptable).

Work this procedure once by hand for the ideal case in Section 3.3. You should land within rounding of 36.2%. If you don't, your tables or your arithmetic are wrong — not the theorem. That is the value of a worked standard: it calibrates the reader, not just the cycle.

3.7 What this chapter proved

- The Rankine cycle converts heat to work in four states; its efficiency has exactly one correct equation.
- Carnot caps every heat engine; the design compass is raising the mean temperature of heat addition.
- Computed efficiencies: 36.2% ideal, 27.1% realistic small-scale, 19.8% pessimistic — from standard steam tables, verifiable.
- Small turbines are the honest weak link (40–60% expected); the book names it, tests it, and prices it instead of hiding it.
- The steam/ORC crossover sits near 300–350°C; the reader applies the decision rule, not the author's preference.
- The condenser is sized for the design hot night.

Chapter 4 takes up the strangest component in the loop: the Tesla valve — flow control with no moving parts, and the phase-routing claim this book's night cycle depends on.

Self-evaluation — Gate 1, part 3

1. Write the Rankine efficiency equation in enthalpy form and name what each term represents physically.
2. State the design compass of the power block in one sentence, and explain why superheat serves it.
3. A colleague proposes a 150°C heat source with a steam Rankine cycle. What does the decision rule say, and why?
4. Run the Fixed State-Point Procedure for 1 MPa / 300°C, condenser at 10 kPa, ideal components. What efficiency do you get, and what does the Carnot ceiling say about it? (Check: ~27.8% thermal.)
5. Why does the book call the small turbine the weakest critical link, and in which chapters is that weakness tested and priced?

Sources and honesty notes

- Rankine relations, Carnot form, and the Fixed State-Point Procedure: Cengel & Boles, *Thermodynamics: An Engineering Approach*, Ch. 10 — the standard undergraduate reference. Reader verifies against any edition.
- Steam properties: IAPWS-IF97 formulation via the NIST WebBook and the `iapws` Python package. The efficiency figures (36.21%, 27.07%, 19.75%) are computed from these tables, not quoted from a secondary source.
- Small-turbine scale data: EPA Combined Heat and Power Catalog (500 kW 50%, 3 MW 70%, 15 MW 80%); rotor-size efficiency study via MDPI; Curtis-stage enthalpy-drop factor via MATEC 2021; screw-expander efficiencies via Kubawara — as compiled in the book's technical foundations. Reader confirms against originals before procurement decisions.

- The condenser-penalty figures (~2 points per 10 K; 25% vs 28% Texas-night example) are computed from IAPWS-IF97 for an ideal 1 MPa/300°C cycle — calculations with stated assumptions, not field measurements.
- The steam/ORC crossover (~300–350°C) is a design guideline from the literature, not a theorem; treat it as the book's stated rule, open to the reader's own analysis.

Chapter 4

Flow Without Moving Parts

TRL 1 · Basic principles. This chapter explains the Tesla valve — what it provably does, what it might do in this system, and the line between the two.

4.1 The mechanism

In 1920, Nikola Tesla patented a channel that behaves like a one-way valve with no moving parts — no flap, no spring, no gate. US Patent 1,329,559 (expired 1937; public domain for nearly a century). The geometry is fixed: a straight main corridor intersected by teardrop-shaped side loops.

Forward flow: the fluid stays mostly in the main corridor, deflects slightly around the loop entrances, and passes with low pressure loss.

Reverse flow: the fluid splits into the arc loops, which curve around and re-enter the main channel **head-on against the oncoming stream**. The colliding streams separate, recirculate, and shed intense vortices — compounding at every stage. The fluid itself becomes the obstruction.

That is the whole trick. There is nothing to wear out, nothing to power, nothing to control. The diode is carved into the wall shape.

4.2 Diodicity: the one number

The field standardizes on a single metric:

Di = ΔP_reverse / ΔP_forward — at the same flow rate.

Di > 1 means the device resists reverse flow more than forward flow: a forward diode. Di = 1 means no rectification. Everything in this chapter is a statement about Di under stated conditions.

The multi-study consensus on Di, honestly reported:

Flow regime	Diodicity
Re < ~100 (laminar, slow)	Di 1 — no rectification
Re 100–300 ("turn-on")	Di jumps ~1 ~2; the geometry trips turbulence at Re 200, two orders of magnitude earlier than a smooth pipe
Re 300–1,500	Di 2, plateaued (macro-scale)
Higher Re	Design-dependent; reported peaks Di 1.8–2.0 at Re 800–2,000 depending on geometry

Honest Di range: ~1.2 to ~4. Two is the robust macro-scale figure (Nguyen et al., *Nature Communications* 2021, ~11 loops, Tesla's original geometry). Four is the best credible experimental figure (Li et al. 2023, micro-scale, 18 stages, single-phase water). CFD-only claims above ~4.5 are unvalidated — the book flags them and moves on.

Note what Di 2 means practically: reverse flow suffers twice the pressure loss of forward flow. That is modest — industry skeptics have said so on the record (*Chemical Processing*: Di 2 is a lot of hardware length for the effect; "space is a valuable commodity in a piping system"). The valve is not a check valve; it will not hard-block reverse flow. It *biases* it. Whether a bias is enough depends on the application — which is why this chapter ends with the application stated as a hypothesis, not a fact.

4.3 What boiling does to the diode

The founder's system needs the valve to handle two-phase flow — steam and water, not just water. Here the literature gets thinner and more interesting.

The key paper is Li et al., *Nature Communications* 14:3996 (2023) — the only published demonstration of Tesla valves rectifying *boiling* two-phase flow, directly analogous to the flashing leg of a solar steam loop:

- Vapor backflow was **visualized being rectified**: at each Tesla island, roughly a third of the vapor was steered into the curved bend, turned with attenuated momentum; series compounding stagnated the backflow, then liquid reflushed — periodic rectification.
- **But stability required engineered capillary structures.** The valve alone produced unstable, discontinuous two-phase flow. Superhydrophilic micropillars added the capillary pressure that kept liquid renewing the surface. A bare Tesla valve in a boiler tube will not reproduce this result.
- The device acted as a **thermal switch**: below vapor quality $\chi = 0.25$, chaotic oscillation; above it, stabilized — with a heat-transfer switch ratio of ~6.

And the warnings, stated at full volume:

- **Two-phase Di is lower than single-phase Di** at higher Reynolds numbers (vapor–liquid interfaces carry lower friction).
- Gong et al. (*Aerospace*, 2024, CFD with volume-of-fluid): **Di = 0.78 at 0.2 m/s in slug flow — the diode inverts; reverse flows *easier* than forward.** At 0.8 m/s it recovers to 1.44. Low-velocity slug flow is the danger regime.
- The closest analog to a closed power loop — pulsating heat pipes (de Vries et al., 2017) — showed **25% velocity asymmetry and 14% thermal-resistance reduction** versus a valveless pipe. Real effect, modest magnitude.
- The only published Tesla-valve-plus-steam work (Gan, Pei & Li, *Physics of Fluids* 2025) is **CFD for**

nuclear-plant steam decompression — throttling and noise control, not a power loop. It shows the geometry survives steam conditions in simulation. Simulation is not operation.

4.4 The application: phase-directed cycling

Now the founder's proposition, stated with the care it deserves.

The system runs a daily cycle: **by day, water flashes to steam and steam flows one direction through the loop; by night, steam condenses to liquid and liquid returns the other direction.** The Tesla valve's job is to enforce that directionality — to make the loop prefer steam-forward by day and liquid-return by night, with no actuated valves, no control system, no parasitic power, and nothing to fail.

Why this is plausible:

- The valve provably rectifies single-phase flow at Di 2, and boiling two-phase flow under the right conditions (Li 2023).
- Phase change itself assists: steam and liquid have vastly different densities and velocities in the same pipe, so the valve's velocity-dependent rectification acts differently on each phase — the asymmetry the design exploits.
- Pulsating heat pipes prove the principle in a closed loop: passive geometry biasing two-phase circulation, measured.

Why it is a **hypothesis, not a theorem**:

- No published experiment tests Tesla-valve phase-routing at steam-pipe scale in a Rankine loop. The application is the book's novel integration — exactly what TRL 3 exists to validate.
- The inversion risk at low-velocity slug flow (Di < 1) must be designed around: flow velocities must stay in the rectifying regime, which constrains pipe sizing.

- Two-phase stability needed capillary structures at micro-scale; the macro-scale equivalent must be found experimentally.
- No two-phase loss correlations exist for Tesla valves — single-phase literature only. The pressure-drop math for the design loop must come from experiment.

This is the book's contract working as intended. The valve is real. Diodicity is measured. The *application* is a precise, falsifiable claim: **a Tesla-valve section in a solar steam loop will pass steam preferentially in the forward direction by day and liquid preferentially in the return direction by night, with zero water loss, at the flow velocities the design specifies.** Chapter 14 builds the rig that proves or kills it. If the data says no, the book says no — and the fallback is conventional check valves, priced in Chapter 24.

4.5 Design rules from the literature

For the reader sizing a valve section, the published geometric guidance:

- **Branch angle:** no universal optimum — reported optima range 30° to ~72° depending on the study. Angle trades diodicity against forward loss; the reader optimizes for their Reynolds regime, not someone else's.
- **Stage count N:** Di rises sub-linearly with diminishing returns (one study: 1.43 / 2.76 / 3.58 for N = 2 / 6 / 10 at Re = 450). A theoretical model predicts Di peaks at a critical N and degrades beyond — **unvalidated by experiment; treat as prediction.**
- **Scale rule of thumb:** keep the main channel narrow relative to the loops. Widening the main passage destroys rectification faster than any other change.
- **Operating rule:** stay out of low-velocity slug flow. Size the pipe so the working velocities sit in the rectifying regime (Re above turn-on, well into the Di 2 plateau) for both phases.

4.6 What this chapter proved

- The Tesla valve (Tesla, 1920, expired) rectifies flow with fixed geometry — no moving parts. Di = ΔP_reverse / ΔP_forward is the metric.
- Honest Di: ~1.2–4; ~2 is the robust macro figure; CFD-only claims above ~4.5 are unvalidated.
- Boiling two-phase rectification is demonstrated (Li 2023) but required capillary structures; the diode can invert at low-velocity slug flow (Di = 0.78).
- The phase-directed day/night application is the book's novel integration: plausible, precise, falsifiable — and validated in Chapter 14, not asserted here.
- The fallback is conventional check valves. The book never depends on a single unproven claim.

Chapter 5 closes TRL 1's physics: condensation thermodynamics and the sealed water cycle — the night half of the loop.

Self-evaluation — Gate 1, part 4

1. Explain in your own words how a Tesla valve rectifies flow without moving parts. What does the fluid collide with in reverse?
2. Define diodicity. What does Di = 2 mean for a pump trying to push fluid backward through the valve?
3. Name the two conditions under which published results say the diode weakens or inverts.
4. State the phase-routing hypothesis in one falsifiable sentence. What experiment in Chapter 14 tests it?
5. If the Chapter 14 experiments falsify the hypothesis, what does the book build instead, and where is that priced?

Sources and honesty notes

- Tesla valvular conduit: US Patent 1,329,559 (1920), expired 1937 — public domain. The book claims no invention here; the patent is cited as prior art.
- Diodicity regimes and the Re 200 early-transition: Nguyen et al., *Nature Communications* 2021 (macro, Tesla's original geometry).
- Di 4 micro-scale figure and all two-phase boiling results: Li et al., *Nature Communications* 14:399 "Tesla valves and capillary structures-activated thermal regulator" (2023). The capillary-structure requirement and the $\chi = 0.25$ switch criterion are from this paper — the book does not soften them.
- Diode inversion in slug flow (Di = 0.78 at 0.2 m/s): Gong et al., *Aerospace* 2024 — CFD (volume-of-fluid), flagged as simulation, not measurement.
- Pulsating-heat-pipe results (25% velocity asymmetry, 14% thermal-resistance reduction): de Vries et al., *Int. J. Heat and Mass Transfer* 2017.
- Steam-condition CFD: Gan, Pei & Li, *Physics of Fluids* 2025 — simulation only.
- Industry skepticism (*Chemical Processing* assessment) is cited deliberately: the book is stronger for carrying its critics.
- The phase-directed day/night application has no published precedent at steam-pipe scale. It is presented as a hypothesis with a defined validation experiment (Ch. 14), per the book's honesty contract.

Chapter 5

The Night Loop

TRL 1 · Basic principles. This chapter explains condensation — the physics of the night half — and turns "no leaking" from a slogan into an accounting procedure.

5.1 What condensation is

Boiling in reverse. In the boiler, heat entered water and left as latent heat carried by steam. In the condenser, that latent heat must go somewhere — into the night air — and the steam becomes water again. The energy is identical: h_fg 2,257 kJ/kg at atmospheric pressure. Every kilogram of steam the turbine exhausts must surrender that much heat to become liquid. There is no shortcut; the condenser is a heat-rejection device, and its size is set by how fast the night air can carry heat away.

The governing description is Nusselt's film-condensation theory (1916). Vapor condenses as a liquid film on a cool surface; the film's thickness — and therefore the heat-transfer coefficient — follows:

h = 0.943 · [g·ρ_l²·k_l³·h_fg / (μ_l·L·(T_sat T_s))]^(1/4) (vertical plate; 0.725 for horizontal tube with diameter D)

Typical magnitudes: film condensation runs **2,000–15,000 W/m²·K** — superb heat transfer. Which leads to the condenser theorem, the single most important sentence in this chapter:

The air side, not the steam side, sizes the passive condenser.

The steam condenses eagerly. The bottleneck is persuading ambient air to accept the heat. Forced-air steam condensation

manages overall U values of only **700–850 W/m²·K**; a fully passive radiator — natural convection plus thermal radiation, no fans — runs **U 5–20 W/m²·K**. Two orders of magnitude worse than the steam film itself. The condenser is not a steam device that happens to touch air. It is an air device that happens to touch steam.

5.2 The honest sizing

Follow the heat through a concrete case — the home tier's nightly duty. Assumptions stated:

- Day's generation to reject: 50 kWh of waste heat (the condenser rejects Q_out of the Rankine cycle — the majority of the heat in, per Chapter 3's efficiency)
- Night duration: 10 hours average rejection rate **5 kW**
- Log-mean temperature difference: ΔT_lm 20 K (condensing steam ~45°C against ~25°C night air)
- Passive overall coefficient: U 8 W/m²·K (natural convection + radiation)

The LMTD sizing relation — **Q = U · A · ΔT_lm · F**, with F = 1 whenever one side undergoes phase change at constant temperature — gives:

A = 5,000 / (8 × 20) 31 m²

Thirty-one square meters of radiator panel — roughly a small garage's roof area — to reject one home's nightly heat passively. State the implication plainly, because the book's contract requires it: **passive night rejection is area-hungry at building scale.**

The engineering responses, each honest:

1 **Forced convection.** A small fan raising the air-side coefficient to a few hundred W/m²·K collapses the area by an order of magnitude — at the cost of parasitic power and a moving part. The trade is priced in Chapter 24.

2 **The building as the radiator.** Walls, roof panels, and structural surfaces the building already has can carry the condenser duty — the complex tier's advantage, developed in Chapter 9's sizing.

3 **Cooler nights help.** ΔT_lm rises as ambient falls; the same panel rejects more on a cool night. The design point is the hot night, not the average one.

And the hot-night rule from Chapter 3 applies with full force: **size the condenser for the design hot night.** A Texas August night at 30°C ambient against 45°C condensing steam leaves ΔT_lm 15 K, not 20 — the required area grows by a third. The reader who sizes for the average night builds a machine that stalls on the nights it is needed most.

5.3 "No leaking," engineered

The founder's claim: the loop loses no water. Day after day, the same inventory flashes to steam and condenses back. Here is that claim converted from assertion to procedure.

The mass-conservation theorem: in a sealed loop, water inventory is constant except for measured, named streams. The nightly accounting is:

Inventory_evening = Inventory_morning + Makeup Blowdown Samples Fugitives

- **Makeup** is measured at the fill connection. In a healthy sealed loop it trends toward zero.
- **Blowdown** is deliberate: dissolved solids don't evaporate with the steam, so they concentrate in the boiler water. Holding total dissolved solids under the boiler's limit requires bleeding a small stream — the cycles-of-concentration arithmetic from water treatment, computed in Chapter 21. This is not a leak; it is chemistry, metered and logged.
- **Samples** are the chemistry lab's draws — milliliters, logged.

- **Fugitives** are the true losses: valve-stem weeps, flange seeps, pump seals. The design target is zero; the accounting detects nonzero.

The reader runs this balance nightly during testing (Chapters 17–19) and weekly in operation (Chapter 26). **A loop whose makeup trends to zero is a loop that does not leak — demonstrated by accounting, not claimed by adjective.** That is what "no leaking" means in this book: a measured inventory balance, not a promise.

Three physical realities the accounting must respect:

1. **Non-condensable gases.** Air leaking *in* (past seals at sub-atmospheric condenser pressure) blankets condenser surfaces and strangles heat transfer far out of proportion to its quantity. The loop needs a purge point and a purge procedure — specified in Chapter 16's build, operated in Chapter 21's chemistry doctrine. A condenser that mysteriously underperforms is usually an air problem, not a sizing problem.
2. **Freeze protection.** A water loop in a climate that freezes must drain, heat-trace, or use freeze-tolerant routing. The book does not pretend water is antifreeze; Chapter 19's relevant-environment testing includes the freeze case for sites that need it.
3. **The phase-routing assist.** By night, the loop's flow is liquid returning to the boiler — the Tesla valve's return direction from Chapter 4. The night loop and the valve are one design: the valve biases the return flow, condensation provides the liquid, gravity and the feed pump complete the circuit. Each component's chapter stated its piece; here they join.

5.4 The method: sizing the night loop

1. **Fix the nightly duty.** From the Rankine balance (Chapter 11's full version): Q_reject = Q_in W_net per day. For early sizing, Q_reject daily heat input × (1 η_th).
2. **Fix the design night.** The hottest night the site sees in a typical year (TMY data), not the average. Set condensing temperature ~10–15 K above it.
3. **Compute ΔT_lm** between condensing temperature and ambient across the night.
4. **Select U honestly:** 5–20 W/m²·K passive; forced-air per manufacturer data with measured verification in Chapter 17.
5. **A = Q_rate / (U · ΔT_lm).** If the area doesn't fit the site, iterate: forced convection, building-integrated surfaces, or a smaller daily duty.
6. **Write the inventory procedure** from Section 5.3 into the operating log before first fill. The loop's water gets a ledger on day one.

5.5 What this chapter proved — and what TRL 1 established

- Condensation is latent-heat rejection governed by Nusselt film theory; the air side sizes the condenser, not the steam side.
- Honest sizing: ~31 m² of passive radiator per 50 kWh nightly duty — area-hungry, with three legitimate engineering answers.
- The condenser is sized for the design hot night.
- "No leaking" is a nightly mass-balance accounting procedure, not a claim. Makeup trending to zero is the proof.
- Non-condensables, freeze, and chemistry are named realities with assigned chapters.

TRL 1 is complete. The physics the system stands on — concentration optics, the Rankine cycle, valvular fluid mechanics, condensation — has been stated as theorems with their equations, their sourced numbers, and their named limits. Nothing is integrated yet. That begins in TRL 2.

GATE 1 — Exit evaluation

The reader may not proceed to TRL 2 until every answer below is theirs, not the book's.

From Chapter 1:

1 Why is the building the correct unit of analysis? State the building's four energy-identity numbers.

From Chapter 2:

1 State what the lens is and what it is not, in two sentences. Compute the flux concentration for a 2 m^2 lens at C = 80, η_optical = 0.75, DNI = 900 W/m^2. *(Check: 54,000 W/m^2 at the receiver... work it: 900 × 0.75 × 80 = 54,000.)*

2 Explain the concentration–tracking-accuracy trade and why the home tier needs only ~1–2°.

From Chapter 3:

1 Write the Rankine efficiency equation in enthalpy form. State the design compass in one sentence.

2 A colleague proposes steam Rankine at 150°C. Apply the decision rule.

From Chapter 4:

1 Define diodicity. Name the two published conditions where the diode weakens or inverts.

2 State the phase-routing hypothesis in one falsifiable sentence, and name the chapter that tests it.

From Chapter 5:

1. State the condenser theorem. Size a passive night loop for 40 kWh over 8 hours at ΔT_{lm} = 18 K, U = 8 W/m²·K. *(Check: 5,000 W / (8 × 18) 34.7 m².)*
2. Write the nightly water-inventory equation and explain what "makeup trending to zero" proves.

Gate 1 exit criteria: all nine answered correctly; the reader can size a lens, a Rankine cycle, a valve operating point, and a night loop from the methods given, and can state what each component's chapter left unproven. Then — and only then — TRL 2.

Sources and honesty notes

- Nusselt film-condensation theory (1916): the classical reference; reader verifies against any heat-transfer text (Incropera, Cengel).
- Film-condensation h magnitudes (2,000–15,000 W/m²·K) and U sanity checks (steam-to-water 1,500–4,000; air-cooled steam 700–850; passive 5–20): handbook values via Engineers Edge overall-heat-transfer tables — representative, reader confirms against project-specific data.
- The 31 m² worked sizing is a calculation with stated assumptions, not a measurement. U = 8 W/m²·K is a mid-range passive assumption; the reader measures their own hardware in Chapter 17.
- h_{fg} = 2,257 kJ/kg at 1 atm: standard steam-table value (IAPWS-IF97).
- The cycles-of-concentration arithmetic is standard boiler water-treatment practice (ASME/ABMA guidelines); the book's full chemistry doctrine is Chapter 21.
- No standard transient model exists for a day-only full-condensation cycle — the book states this gap openly; Chapter 15's experiments generate the data.

TRL 2

Technology Concept

Chapters 6–10

Chapter 6

The Integration

TRL 2 · Technology concept formulated. This chapter assembles the TRL 1 physics into one machine, states precisely what is new, and opens the patent register.

6.1 The machine, end to end

TRL 1 gave you five proven pieces: the demand (Ch. 1), the lens (Ch. 2), the cycle (Ch. 3), the valve (Ch. 4), the night loop (Ch. 5). Now they join. Follow one kilogram of water through a full day.

Dawn. The loop sits charged with liquid water — last night's condensation, inventoried per Chapter 5's accounting. The tracking mirrors wake and find the sun. The lens begins concentrating.

Morning. Focused sunlight — the heat ray, quantified in Chapter 2 — strikes the receiver. The closed-loop boiler flashes water to steam, then superheats it. Pressure rises toward the design point (the 4 MPa / 400°C working example from Chapter 3, or whatever the reader's sizing in Chapter 8 specifies).

Midday. Superheated steam expands through the turbine. The shaft spins the generator; the building's meter slows, stops, or runs backward. Exhaust steam — still carrying most of the heat it entered with — flows to the condenser, where the day's first condensation begins. The Tesla valve sections bias every flow in its designed direction: steam forward through the power leg, against backflow.

Evening. The sun drops. The lens goes quiet. The boiler stops flashing. Residual steam continues through the turbine as

pressure decays, then the loop transitions: the remaining vapor condenses in the night condenser, and liquid water returns through the valve's return direction to the boiler — the phase-directed cycling of Chapter 4's hypothesis.

Night. Condensation completes. The inventory accounting from Chapter 5 runs: makeup measured, blowdown logged, fugitives investigated. By dawn the loop is liquid again, sealed, inventoried — ready.

No photovoltaics anywhere. No batteries. No lithium. Sunlight in, shaft work out by day; water recovered by night; the same water tomorrow.

6.2 What is new and what is not

The book's inventorship position, stated once and carried through every chapter:

Not claimed: the Fresnel lens, the Tesla valve, the Rankine cycle, the steam turbine, the condenser, or any other component. Each traces to real patents held by real holders — the register below. The components walk in at high TRL: proven, manufactured, bought off the shelf.

Claimed: the *integration* — this specific combination, arranged to do this specific thing:

> *A building-scale concentrated solar thermal power system comprising a parabolic-profile Fresnel lens with a heliostat mirror boost stage, a closed-loop boiler flashing water to superheated steam, a steam turbine driving a generator, and Tesla-valve sections directing flow by phase — steam forward by day, condensed liquid returning by night — in a sealed zero-makeup water loop, sized by the methods of Chapters 8–9 for one of three deployment tiers.*

The patent register's finding, which the book reports because it is true: **no verified patent covers a Tesla valve applied to phase-directed day/night cycling in a solar steam loop.** The components are crowded with prior art. The *application* is open

space. That is the contribution, and the book claims exactly that much — no more.

6.3 The patent register (abridged)

The full register — ~50 patents across 16 component sections, each with number, holder, dates, and legal status — is the book's Appendix A. The load-bearing entries:

Component	Key patent	Status
Tesla valvular conduit	US 1,329,559 (Tesla, 1920)	**Expired 1937** — public domain
Fresnel solar concentrator (foundations)	Sleeper 1962; O'Neill/E-Systems 1976; 3M 1988	**All expired**
Fresnel closed-loop steam turbine generator	US 5,632,147 (1997)	**Expired 2016**
Heliostat tracking (foundations)	US 4,317,031	**Expired 1999**
Closed-loop solar boiler	US 4,265,223 (Badger); US 6,957,536 (Boeing, expired 2023)	**Expired**
Small steam turbine	US 1,061,206 (Tesla, 1913); US 4,258,551 (Biphase)	**Expired**
Night condensation	US 4,905,474 (Larinoff)	**Expired 2007**
Fresnel array steam power, water recycling	EP 3,066,340 (Solar Steam Ltd, granted 2020)	**Possibly in force — verify before building**
Fresnel arrays + mirror beam-network + tracking + steam turbine	US 8,689,784 (Monsebroten)	**Base term ended Sept 2026 — verify maintenance/PTA status**
Vapor transfer via Tesla valve (closest active analog)	US 10,602,638 (Dell, to 2038)	Active — different application

Standing notice (from the front matter, repeated wherever patents are cited): referencing a patent is a formality of citation.

It points at the patent record as recorded. It does not certify that the accredited inventor is the true first inventor, and it does not certify claim validity.

What the register means for the builder: the foundational IP is overwhelmingly expired — public domain. A small number of active patents sit near the design space (the table's last three rows); their claim scope and in-force status must be confirmed by a patent attorney before building or publishing a prototype. The register is a research landscape, not a freedom-to-operate opinion. The book says this because the alternative — silence — would be the lie.

6.4 Where integrations fail: the interface list

Components don't fail. *Interfaces* fail. Every integration disaster in engineering history happened where two proven things met. So the concept document names every interface in the loop, the quantity that crosses it, and the chapter that owns it:

1. **Lens receiver:** concentrated flux distribution (W/m^2 map). Owned by Ch. 2/12. Failure mode: hot spots, spillage.
2. **Receiver boiler:** wall heat flux vs. critical heat flux. Owned by Ch. 3/13. Failure mode: dryout, tube failure.
3. **Boiler turbine:** steam pressure, temperature, dryness. Owned by Ch. 3/13. Failure mode: wet steam eroding blades.
4. **Turbine generator:** shaft speed, torque, coupling. Owned by Ch. 7. Failure mode: overspeed, misalignment.
5. **Turbine exhaust condenser:** mass flow, enthalpy, non-condensables. Owned by Ch. 5/15. Failure mode: air blanketing, back-pressure rise.
6. **Condenser valve pump boiler:** liquid return flow, valve diodicity, pump head. Owned by Ch. 4/5/14. Failure mode: the phase-routing hypothesis itself.

Six interfaces, six named owners, six failure modes stated before anything is built. This list is the integration's risk register, and TRL 3 through 6 retire it one experiment at a time.

6.5 What TRL 2 requires — and what would falsify the concept

TRL 2 is "technology concept formulated." Its exit criteria are documentation, not hardware:

- The complete system described, end to end (Section 6.1).
- The novel claim stated precisely and falsifiably (Section 6.2).
- Prior art cited; the contribution bounded (Section 6.3).
- Interfaces and failure modes named (Section 6.4).
- Sizing methods for all three tiers (Chapters 8–9). next

And the falsification conditions — what would kill the concept on paper, before a dollar is spent:

1. The end-to-end energy balance (Ch. 11) shows net output below usefulness at all three tiers.
2. The patent register finds the exact combination already claimed and in force, with no design-around.
3. A subsystem's required operating point falls outside every sourced demonstration band with no credible path (e.g., needing C = 500 at the home tier's tracking budget).

None of the three obtains — the balance is worked in Chapter 11, the register is open space, and every operating point sits inside demonstrated bands. The concept stands. TRL 2's remaining work is the sizing methods: Chapters 7 (the whole machine's bill of materials), 8 (home tier), 9 (corporate and complex tiers).

6.6 What this chapter proved

- The full day-cycle narrative: dawn charge midday generation evening transition night recovery, one kilogram of water followed throughout.
- The inventorship boundary: components cited, integration claimed, application space open per the register.
- The abridged patent register with statuses; the standing disclaimer; the attorney-verification requirement for the active entries.
- Six interfaces named with owners and failure modes — the integration's risk register.
- TRL 2's falsification conditions, checked and cleared on paper.

Chapter 7 names every part: the complete bill of materials, including the parts the vision didn't mention — generator, pump, receiver, condenser hardware, tracker, inverter, safety valves, piping.

Self-evaluation — Gate 2, part 1

1 Narrate the full day cycle in your own words: what happens to the water from dawn to dawn?

2 State the book's claimed contribution in one sentence. What is explicitly *not* claimed?

3 Which three register entries require attorney verification before building, and why?

4 Pick interface #6. Explain, using Chapters 4 and 5, why the phase-routing hypothesis lives or dies at that interface.

5 State one falsification condition for the concept. Why does none of the three currently obtain?

Sources and honesty notes

- Patent data: the book's Appendix A register, researched via Google Patents / USPTO / Espacenet; every number verified against its live record. Statuses are Google's displayed legal status with standard expiry rules applied. Fields that could not be confirmed are marked [VERIFY] in the register.
- "No verified patent covers Tesla-valve phase-directed day/night cycling in a solar steam loop" is a research finding from the September 2026 landscape scan — a finding, not a legal opinion. Absence of evidence in a search is not proof of absence; the attorney check stands.
- The day-cycle narrative is a design description, not an observed operation. It becomes observed operation in TRL 6 (Ch. 22).

Chapter 7

The Whole Machine

TRL 2 · Technology concept formulated. This chapter names every part — including the ones the vision didn't mention — and states the procurement doctrine: buy the proven, build only the integration.

7.1 Why a bill of materials belongs in TRL 2

A concept without a parts list is a wish. The bill of materials (BOM) is where the integration meets reality: every line is a thing someone must buy or make, with a specification someone must write. It is also where the book's TRL argument becomes concrete — nearly every line traces to a mature, high-TRL component with expired foundational patents. The reader builds the *arrangement*, not the parts.

Two doctrines govern the BOM:

1. **Buy the proven.** Any part with mature commercial supply and expired foundational IP is purchased, not invented. The book specifies it; the reader procures it.
2. **Build only the integration.** Custom fabrication is reserved for the parts that embody the novel combination: the valve sections, the receiver-boiler assembly, the control logic that runs the day cycle. Everything else is catalog hardware.

The BOM below is structured for the home tier (Chapter 8 sizes it fully); the corporate and complex tiers (Chapter 9) scale the same lines. Quantities marked *(1×)* are per home unit. No

prices are listed — prices are quoted at procurement time and decay in print; the economics live in Chapter 24 with their dates attached.

7.2 Optical subsystem

Part	Function	Specification driver	Register
Parabolic-profile Fresnel lens, ~1–2 m² aperture (1×)	Concentrates sunlight to the receiver	Aperture from Ch. 2 Method A; groove profile per focal length; PMMA, UV-stabilized	Fresnel foundations (expired)
Heliostat boost mirrors, N×	Redirects peripheral sunlight onto the lens aperture	Total boost area from Ch. 12 ray trace; flat second-surface mirrors	US 4,317,031 (expired)
Dual-axis tracker drives (1× lens + N× mirrors)	Sun-following	Accuracy ~⅓ of tracking error budget (Ch. 2); torque from wind-load calc	Tracker foundations (expired); slew-drive actives noted
Sun sensor + controller (1×)	Closed-loop tracking	0.1°-class; active correction	Buyable commodity
Support structure (1×)	Holds lens at focal distance, mirrors in field	Wind load per local code; focal-length rigidity (deflection focal tolerance)	Structural — no patent dependency

7.3 Receiver and boiler

Part	Function	Specification driver	Register
Receiver cavity / absorber (1×)	Converts concentrated flux to heat in the working fluid	Flux map from Ch. 12; wall heat flux 50% of CHF (Ch. 3 safety rule)	NREL receiver actives noted
Selective absorber coating (1× set)	Maximizes α, minimizes ε	α > 0.9, ε low at operating temp; proven at 400°C class	Black-chrome foundations (expired); ENEA actives noted
Boiler pressure vessel (1×)	Contains pressurized water/steam	Design pressure 1.5× operating (code); volume from Ch. 8 sizing	Badger/Boeing (expired)
High-temperatur e insulation (1× set)	Limits standby losses	Thickness from loss budget (Ch. 11)	Aerogel foundations (expired 2025)
Pressure relief valve(s) (2× minimum)	Overpressure protection	Set below vessel MAWP; sized for full boiler duty per code	Safety-valve commodity
Temperature + pressure instrumentation	Control and safety inputs	Range covers 0–150% of design; redundant on safety channels	Commodity

The receiver-boiler assembly is the highest-temperature, highest-pressure part of the machine. It is also the part most constrained by code (Chapter 27). The reader does not improvise here: the vessel is built or certified to the applicable pressure-vessel standard, full stop.

7.4 Power block

Part	Function	Specification driver	Register
Steam turbine / expander (1×)	Converts steam enthalpy to shaft work	Inlet conditions from Ch. 8; isentropic efficiency per honest Ch. 3 bands (40–60% micro-scale)	Tesla 1913 / Biphase (expired)
Generator / alternator (1×)	Converts shaft work to electricity	Matched to turbine speed and power; synchronous or induction per Ch. 26 grid interface	Generator commodity
Shaft coupling + overspeed protection (1×)	Transmits torque; guards the failure mode	Torque rating; overspeed trip independent of controls	Commodity

Note the turbine honesty carried forward: the BOM specifies the expander *class* (impulse, Curtis, screw) after Chapter 20's testing, not before. The TRL 2 BOM names the slot; TRL 5 fills it with a tested part number.

7.5 Condenser and night loop

Part	Function	Specification driver	Register
Condenser heat exchanger, finned panels (1× set)	Rejects latent heat to night air	Area from Ch. 5 Method (design hot night); passive or forced per Ch. 24 trade	Larinoff (expired); Holtec/Evapco actives noted
Condenser fan(s) (optional set)	Forced convection when area doesn't fit	Parasitic power budgeted in Ch. 11 balance	Commodity
Non-condensable purge valve + procedure (1×)	Removes air blanketing the condenser	Manual or automatic; operated per Ch. 21 doctrine	Procedure, not a part

Part	Function	Specification driver	Register
Condensate return piping	Carries liquid back to the boiler	Sized for return velocity in the valve's rectifying regime (Ch. 4 rule)	Piping commodity

7.6 Flow control and the valve sections

Part	Function	Specification driver	Register
Tesla valve sections (2×: power leg + return leg)	Phase-directed flow bias, no moving parts	Stage count N, branch angle, channel width per Ch. 4 rules; validated in Ch. 14	US 1,329,559 (expired 1937)
Boiler feed pump (1×)	Returns condensate to boiler pressure	Head = boiler pressure + loop losses; w_p $v_1(P_2P_1)$	Pump commodity; Sulzer actives noted (buy, don't clone, in EU)
Isolation + drain valves (set)	Maintenance, commissioning, freeze protection	Full-bore; rated to loop pressure/temperature	Commodity
Piping + insulation (set)	Connects everything	Diameter from velocity rules (stay out of slug flow); schedule per pressure code	Commodity

The valve sections are **build** items under the procurement doctrine — the integration's novel hardware, fabricated to the Chapter 4 design rules and proven in Chapter 14 before the loop depends on them. If Chapter 14 falsifies the phase-routing hypothesis, these two lines become conventional check valves (Chapter 4's named fallback), and the BOM is revised without touching any other line. That is what a risk register is for.

7.7 Electrical and controls

Part	Function	Specification driver	Register
Inverter / power electronics (1×)	Grid-compatible AC output	Power rating = generator peak; anti-islanding per local interconnection rule	SolarEdge/Enphase actives noted
Disconnects, metering, protection (1× set)	Code-compliant grid interface	Local electrical code; utility interconnection agreement	Code-driven commodity
Control system (1×)	Runs the day cycle: track, boil, generate, condense, account	State machine from Ch. 6's narrative; sensor inputs from §§7.3/7.5	Integration software — built
Pressure/temperature/level sensors (set)	Control + safety inputs	Redundant on safety channels	Commodity

7.8 Water and chemistry

Part	Function	Specification driver	Register
Fill/drain station (1×)	Initial charge with treated water; service drains	Distilled/DI fill per Ch. 21 spec	Procedure + commodity
Sampling points (set)	Chemistry monitoring for the Ch. 5 inventory	Accessible, valved, logged	Commodity
Blowdown valve + meter (1×)	Dissolved-solids control	Sized from cycles-of-concentration arithmetic (Ch. 21)	Commodity

7.9 The procurement doctrine, restated

Count the lines: roughly thirty parts. Of them, **two are built** (valve sections, control logic), **one is built-or-certified** (receiver-boiler assembly), and **the rest are bought**. That ratio is the TRL argument in BOM form — the machine is an arrangement of proven things, and the proven things stay proven because the book doesn't ask them to do anything their patents and catalogs don't already promise.

The BOM is a living document: Chapter 8 sizes the home tier's quantities, Chapter 14 revises or confirms the valve lines, Chapter 20 fills the expander slot, Chapter 24 attaches dated prices. What it never does is grow a line the concept didn't name — Chapter 6's interface list is the complete set, and the BOM covers exactly that set.

7.10 What this chapter proved

- Every part named, organized by subsystem, with specification drivers and register references.
- The missing parts from the original vision are now first-class lines: generator, feed pump, receiver, condenser hardware, tracker, inverter, safety valves, piping, water station.
- Procurement doctrine: buy the proven (~27 lines), build only the integration (~3 lines).
- The valve lines carry the named fallback; the expander slot awaits Chapter 20's test data.
- No prices in print — economics live in Chapter 24 with dates.

Chapter 8 sizes the home tier: the backyard unit, worked step by step from the methods of Chapters 1–5.

Self-evaluation — Gate 2, part 2

1 State the two procurement doctrines. Which BOM lines are *built* rather than bought, and why those?

2 Why does the BOM specify the expander's class but not its part number at TRL 2? Which chapter fills the slot?

3 If Chapter 14 falsifies phase-routing, which BOM lines change and which stay untouched? What does that say about the integration's risk isolation?

4 A supplier offers a "parabolic solar lens" that turns out to be a reflective dish. Using Chapter 2, explain why the BOM rejects it in one sentence.

5 Where do prices live, and why not in this chapter?

Sources and honesty notes

- Component patent statuses: Appendix A register (September 2026 landscape scan). "Buy, don't clone" flags (Sulzer pump EP, NREL receiver, ENEA coatings, slew-drive, inverter actives) mark where the builder procures rather than reproduces.
- No model numbers, prices, or supplier endorsements appear in the BOM. Specification *drivers* are given so the reader writes specs; the market fills them at procurement time.
- The 40–60% micro-turbine efficiency band is carried from Chapter 3's sourced data, not re-asserted here.
- Quantities are per home-tier unit (1×) unless marked; corporate/complex scaling is Chapter 9.

Chapter 8

Sizing the Home Tier

TRL 2 · Technology concept formulated. This chapter sizes the backyard unit end to end — a worked calculation with stated assumptions, a sensitivity table, and an honest footprint check.

8.1 The sizing chain

Sizing is working backward from demand to hardware through every chapter so far. Seven steps, each owned by its chapter:

1. **Demand** (Ch. 1): daily kWh, peak kW.
2. **Cycle** (Ch. 3): choose steam conditions → cycle efficiency → daily heat required.
3. **Sun** (Ch. 2): design-day beam resource from local data → aperture area.
4. **Concentration** (Ch. 2): choose C from the temperature band → receiver area.
5. **Tracking** (Ch. 2): error budget → tracker class.
6. **Night loop** (Ch. 5): nightly heat rejection → condenser area.
7. **Flow** (Ch. 3/4): pump head, pipe velocities, valve stage count.

No step is a guess. Each step's equation was derived in its chapter; this chapter only runs them in sequence.

8.2 The worked example — stated assumptions

A reference home, a reference site. Every assumption is labeled **(A)**; the reader replaces each with their own data in Section 8.4's worksheet.

- **(A1)** Demand: 30 kWh/day electrical, 5 kW peak (Ch. 1's national-average home).
- **(A2)** Steam conditions: 4 MPa / 400°C; small-turbine/pump efficiencies per Ch. 3's realistic case **η_{th} = 27%**.
- **(A3)** Site: design-day beam resource **5.5 kWh/m²/day** (a good-sun US site; the reader uses their TMY data — "the measurements scale to the environment").
- **(A4)** System optical-receiver efficiency: **η_{sys} = 0.50** (Ch. 2's mid-range: 0.75 optical × ~0.67 receiver).
- **(A5)** Concentration: **C = 60** (Ch. 2's demonstrated band for the target temperatures).
- **(A6)** Night: 10-hour rejection window; condensing ΔT_{lm} = 20 K; passive U = 8 W/m²·K (Ch. 5).

Step 1 — Demand: 30 kWh/day, 5 kW peak. Fixed by (A1).

Step 2 — Cycle: daily heat required = 30 / 0.27 **111 kWh thermal/day**.

Step 3 — Sun: aperture A = 111 / (5.5 × 0.50) **40 m²**. Forty square meters of lens aperture — the headline number of this chapter.

Step 4 — Concentration: receiver area = 40 / 60 **0.67 m²**. The heat ray from forty square meters of sky lands on two-thirds of a square meter of receiver.

Step 5 — Tracking: C = 60 acceptance budget needs ~1–2° total accuracy (Ch. 2). Ordinary dual-axis tracker class; no exotic hardware.

Step 6 — Night loop: nightly heat to reject = 111 × (1 0.27) **81 kWh** over 10 h 8.1 kW average. Condenser area = 8,100 / (8 × 20) **51 m²** passive. (Larger than Chapter 5's 31 m² example because this cycle rejects more heat — the accounting is

consistent, not optimistic.)

Step 7 — Flow: turbine/generator class **5–7 kW** (covers the 5 kW peak with margin); feed-pump head = 4 MPa plus loop losses; pipe diameters chosen so both steam and liquid velocities sit in the valve's rectifying regime (Ch. 4's operating rule); valve stage count per Ch. 4's diminishing-returns guidance, finalized after Ch. 14's tests.

8.3 The honest reading of these numbers

The aperture is large. Forty square meters of precision optics is the dominant footprint of the home unit — roughly a two-car garage of sky-facing collector. Against the founder's constraint — *less space than a reasonable average US backyard* — the verdict is: **feasible on a typical suburban lot, but it is the binding constraint of the tier.** The book does not shrink the number to flatter the vision.

Three legitimate levers, each quantified in the sensitivity table below:

1. **Better expander.** The aperture scales inversely with η_th — the small turbine is the economic lever of the whole tier (Ch. 3's weak link, Ch. 20's test program).
2. **Better sun.** The aperture scales inversely with the site's beam resource. At 6.5 kWh/m²/day the base case drops to ~34 m²; at 4.5 it rises to ~49 m². The method sizes to the environment; the environment answers.
3. **Daytime-load targeting.** Sizing to the full 30 kWh/day assumes the unit carries the whole home. Sizing to the daytime load fraction shrinks everything proportionally — a valid deployment choice the reader makes with their load-shape data (Ch. 1, §1.4).

The condenser is the second binding constraint. Fifty-one square meters passive is not a backyard afterthought — which is exactly why Chapter 5 gave three engineering answers (forced convection, building-integrated surfaces, cooler-night credit). The home tier almost certainly uses building-integrated condenser area (roof, walls) or a small fan; the fully-passive

ground array is the conservative bound, not the recommendation.

Sensitivity table — aperture (m^2) vs. cycle efficiency and site resource:

η_th \ sun	**4.5 kWh/m²/day**	**5.5 kWh/m²/day**	**6.5 kWh/m²/day**
20% (pessimistic)	66.7	54.5	46.2
27% (realistic small)	49.4	**40.4**	34.2
36% (ideal, reference)	37.0	30.3	25.6

Read the table as the tier's truth: the home unit's size is set by turbine efficiency and local sun, not by the lens. Anyone promising a dramatically smaller aperture is promising a better turbine or a sunnier site — both checkable claims.

8.4 The reader's worksheet

Replace every (A) assumption with your data and run the chain:

1. My home's daily kWh and peak kW (twelve bills + §1.4): _____ / _____
2. My steam conditions and η_th (Ch. 3 procedure): _____ heat/day: _____
3. My site's design-day beam from TMY (NREL): _____ aperture: _____
4. My C from the temperature band (Ch. 2): _____ receiver area: _____
5. My tracking budget (Ch. 2): _____ tracker class: _____
6. My design hot night (Ch. 5): _____ condenser area: _____
7. My expander class, pump head, pipe velocities (Ch. 3/4): _____

If the aperture doesn't fit your site, the sensitivity table tells you which lever to pull — and Chapter 20 tells you which lever is real.

8.5 What this chapter proved

- The seven-step sizing chain, each step owned by its chapter's method.
- The worked home tier: ~40 m^2 aperture, 0.67 m^2 receiver, 5–7 kW power block, ~51 m^2 passive condenser bound — all calculations with stated assumptions.
- The honest reading: the aperture is the binding constraint; the condenser is second; both have legitimate engineering answers.
- The sensitivity table: the tier's size is set by turbine efficiency and local sun.
- The reader's worksheet: the method transfers to any site.

Chapter 9 scales the method: the corporate and complex tiers, the scaling laws, and the shared-vs.-per-building decision.

Self-evaluation — Gate 2, part 3

1. Run the seven-step chain from memory for a home using 24 kWh/day at a 6.0 kWh/m^2/day site with η_{th} = 25% and η_{sys} = 0.5. What aperture? *(Check: heat = 96 kWh; A = 96/(6.0×0.5) = 32 m^2.)*
2. Explain why the aperture scales inversely with η_{th} and with site resource, using the equations.
3. The founder's constraint is "less space than a reasonable average US backyard." Does the 40 m^2 base case satisfy it? What is the chapter's exact verdict?
4. Why is the condenser area larger here (51 m^2) than in Chapter 5's example (31 m^2)? Show the accounting.

5 Name the three levers that shrink the aperture, and the chapter that determines which lever is real.

Sources and honesty notes

- All figures in the worked example are **calculations with stated assumptions (A1–A6)**, not measurements and not promises. The method is the deliverable; the numbers illustrate it.
- η_{th} = 27% is carried from Chapter 3's IAPWS-IF97-computed realistic small-scale case. The 20% and 36% sensitivity rows are the pessimistic and ideal computed cases from the same table.
- η_{sys} = 0.50 is an assumption compounded from Chapter 2's sourced optical efficiencies (~0.75) and a receiver efficiency estimate (~0.67); the receiver figure must be measured per design in Chapter 17.
- The 5.5 kWh/m²/day site is a reference assumption. The reader's TMY data governs their sizing — "the measurements scale to the environment."
- The backyard-constraint verdict ("feasible but binding") is the author's assessment of the calculated aperture against typical suburban lot sizes, not a surveyed fact.

Chapter 9

Scaling to Buildings

TRL 2 · Technology concept formulated. This chapter scales the sizing method to the corporate and complex tiers: the scaling laws, what changes, what doesn't, and the shared-vs.-distributed decision.

9.1 The scaling laws

Chapter 8's seven-step chain doesn't change with size — but the *answers* change in three distinct ways:

Linear scalers — grow in direct proportion to demand: aperture area, receiver area, daily heat, condenser area, water inventory, pipe diameters (by the square root of flow). Double the kWh, double the lens. No economies here; physics doesn't discount.

Sublinear improvers — get *better* per unit as size grows:

- **Turbine efficiency.** The home tier's weak link heals with scale. Micro-turbines (1–50 kW) manage 40–60% isentropic efficiency; mid-size machines (hundreds of kW) reach 70–85%. The cycle efficiency that was 27% at the home tier can exceed 32% at the corporate tier — and every point of η_th shrinks the aperture per the sensitivity table.
- **Concentration.** Bigger fields afford better tracking hardware per unit of aperture; higher C becomes economical, shrinking receivers.

- **Fixed costs.** Controls engineering, interconnection studies, permitting, commissioning — the soft costs that dominate small installations — amortize across more kilowatts. The dollars-per-kW fall even where the physics-per-kW doesn't.

Qualitative changers — new considerations appear:

- **Codes step up.** Pressure-vessel rules, electrical interconnection agreements, and fire codes have thresholds; crossing them changes the compliance path (Chapter 27 maps them).
- **The condenser becomes a product.** At home scale, rejected heat is waste to dispose of. At building scale, 81 kWh of nightly heat — and far more at the corporate tier — is domestic hot water and space heating. The waste stream becomes a revenue stream; Chapter 10 develops this as a competitive mechanism.
- **Maintenance professionalizes.** A homeowner maintains a unit; a corporate facility staffs it. The operating doctrine (Chapter 26) is written for both.

9.2 Worked sketch I — the corporate tier

Stated assumptions **(B)**: a mid-size office building, 2,000 kWh/day electrical, 400 kW peak; η_{th} = **32%** (the healed turbine, per §9.1); same site (5.5 kWh/m²/day) and η_{sys} = 0.5.

- Heat/day = 2,000 / 0.32 = **6,250 kWh thermal**.
- Aperture = 6,250 / (5.5 × 0.5) **2,270 m²** — about half an acre of collector.
- Nightly rejection = 6,250 × 0.68 **4,250 kWh** over 10 h 425 kW at forced-convection U = 300 W/m²·K and ΔT_{lm} = 20 K, condenser **71 m²** of finned coil — a rooftop mechanical penthouse, not a field.

Read it honestly: half an acre of optics is a campus-scale installation — parking canopies, rooftops, the grounds a corporate building already owns. The condenser, meanwhile,

shrinks relative to the home tier's passive bound because forced convection is economical at this scale. Scale giveth and scale taketh away: the aperture grows linearly, but the turbine heals and the condenser industrializes.

9.3 Worked sketch II — the apartment complex

Stated assumptions **(C)**: 200 units at 15 kWh/day each = **3,000 kWh/day**, 600 kW peak; η_th = 32%; same site.

- Heat/day = 3,000 / 0.32 **9,375 kWh thermal**.
- Aperture **3,400 m²**.
- The complex's advantage: **roof area**. Two hundred units under, say, ten buildings carry thousands of square meters of roof — the collector field the tier needs is the roof the tier already has. And the condenser heat (6,400 kWh nightly) is the complex's domestic hot water, nearly in full — the waste-to-product flip from §9.1, quantified.

9.4 The shared-vs.-distributed decision

A complex or campus faces a real architectural choice, and the book gives the decision framework, not the answer:

One central plant — single large lens field, single turbine, distribution of electricity (and possibly heat) to buildings.

- Wins when: land for one field exists; maintenance staff is centralized; the turbine-efficiency gain at larger unit size outweighs distribution losses.
- Loses when: distribution distances grow (thermal distribution bleeds heat per meter — priced in Chapter 11's balance); a single failure darkens every building (redundancy argument).

Per-building units — each building carries its own home-tier-derived unit, scaled to its load.

- Wins when: roofs are the available area (distributed mounting); redundancy matters; buildings come online in phases.
- Loses when: per-unit fixed costs (controls, interconnection, commissioning) dominate — the sublinear-improver argument in reverse.

The decision rule: compute both architectures through the seven-step chain, add distribution losses to the central case and duplicated fixed costs to the distributed case, and let the site's numbers decide. The method from Chapter 8 runs either way — that is what a sizing *method* is for.

9.5 What doesn't change

Across all three tiers, invariant:

- The lens definition (Ch. 2). The day cycle narrative (Ch. 6). The valve's operating rules (Ch. 4). The inventory accounting (Ch. 5). The interface list (Ch. 6). The procurement doctrine (Ch. 7).
- The honest bands: efficiencies stated as computed ranges, areas as functions of site data, economics dated (Ch. 24).
- The falsification structure: each tier's sizing is a checkable claim, and Chapter 11's end-to-end balance either closes or it doesn't.

And the national arithmetic from Chapter 1 now has its unit economics: one home unit 40 m² of aperture per 30 kWh/day. A million such units — the replication the founder described — is forty square kilometers of lens: a large number made of small, buildable, checkable pieces. The nation scales because the building scales; the building scales because the method does.

9.6 What this chapter proved

- The three scaling behaviors: linear (aperture, heat, condenser), sublinear-improving (turbine efficiency, concentration, fixed costs), qualitative (codes, heat-as-product, maintenance).
- Corporate sketch: ~2,270 m^2 aperture for 2,000 kWh/day at η_{th} = 32%; condenser industrializes to ~71 m^2 forced.
- Complex sketch: ~3,400 m^2 for 200 units; roofs carry the field; condenser heat becomes domestic hot water.
- The shared-vs.-distributed decision framework with its decision rule.
- The invariants across tiers, and the national replication arithmetic.

Chapter 10 closes TRL 2: why this category wins — the competitive mechanisms, with LCOE framed as the empirical question the prototypes will answer.

Self-evaluation — Gate 2, part 4

1. Name one linear scaler, one sublinear improver, and one qualitative changer — with the mechanism for each.
2. Why does the corporate tier's aperture per kWh shrink relative to the home tier's, even though aperture scales linearly with demand?
3. A 150-unit complex debates central plant vs. per-building units. State the decision rule and the two quantities that decide it.
4. Explain the waste-to-product flip: at what scale does condenser heat become hot water, and why not at the home tier? (Hint: it's about quantities and plumbing that already exists.)
5. Compute the national replication: 500,000 home-tier units at 40 m^2 aperture. How many km^2 of lens? *(Check: 20,000,000 m^2 = 20 km^2.)*

Sources and honesty notes

- Both sketches are **calculations with stated assumptions (B, C)**, not measurements. The corporate 2,000 kWh/day and complex 200-unit figures are reference cases; the reader substitutes real load data.
- η_{th} = 32% at the corporate tier reflects improved mid-size turbine isentropic efficiency (70–85% class, sourced in Ch. 3's bands) run through the same IAPWS-IF97 procedure — a computed projection, validated or corrected by Chapter 20's testing.
- The 70–85% mid-size turbine band is industry-typical for the size class; the reader confirms against specific machines at procurement.
- Forced-convection U = 300 $W/m^2 \cdot K$ is a mid-range design assumption for finned-coil condensers; measured per design in Chapter 17.
- Code thresholds and interconnection rules vary by jurisdiction — mapped, not assumed, in Chapter 27.

Chapter 10

Why This Category Wins

TRL 2 · Technology concept formulated. This chapter states the competitive mechanisms — the reasons the system can win — and frames LCOE as the empirical question the prototypes will answer. It closes TRL 2.

10.1 The category, not the competitor

This book is not an ad for solar panels, and it is not an attack on them either. The system plays a different game: **building-scale thermal power** — electricity *and* heat, generated where they are used, from hardware the building already has room for.

The distinction matters because categories set the terms of comparison. Against photovoltaics on pure daytime $/kWh-electric, the incumbent's measured cost curve is formidable — decades of data, bankable, silent, no moving parts. The book concedes nothing and pretends nothing: PV's LCOE is *measured*; this system's LCOE is *projected*. The race is between a measured incumbent and a mechanism-rich challenger. What follows are the mechanisms — the structural reasons the challenger can win — and the method by which the prototypes will settle it.

10.2 The five mechanisms

1. No batteries, no lithium. A PV array serving evening load pays the battery toll: electrochemical storage at on the order of $100+/kWh-electric (2026 figures — verify at procurement; they move). This system's thermal nature opens a cheaper door: storing *heat* is radically cheaper than storing *electricity*. An insulated hot-water tank stores thermal energy at roughly $1–5/kWh-thermal — one to two orders of magnitude cheaper per unit of energy. Heat is not electricity, and the book doesn't conflate them — but for the thermal half of a building's demand (hot water, space heating), the system stores energy in the cheapest medium physics allows, natively, with no battery chemistry involved.

2. Co-generated heat. Every kilowatt-hour of electricity the Rankine cycle produces comes with roughly two to three kilowatt-hours of heat at usable temperatures (the condenser stream from Chapters 5 and 9). At building scale that heat displaces fuel the building already buys — gas or electric water heating, space heating. The system's economics therefore credit *two* products against one capital cost. PV produces one.

3. Water reuse. The sealed loop's makeup trends to zero (Chapter 5's accounting). Conventional thermal generation — including concentrating solar plants with wet cooling — consumes on the order of liters per kWh to evaporation. This system consumes essentially none. In water-stressed regions, that is a siting advantage and a cost advantage; everywhere, it is one fewer consumable in the O&M ledger.

4. No wear in the flow path. The Tesla valve has no moving parts — nothing to fatigue, seize, or replace on a maintenance schedule. Pumps and the turbine have bearings and seals, honestly accounted in Chapter 26's O&M doctrine; but the phase-routing heart of the loop is solid geometry. Fewer moving parts is fewer failure modes is lower lifetime maintenance — a structural O&M advantage over systems whose core function depends on mechanisms.

5. Manufacturing scale on mature parts. The BOM (Chapter 7) is ~27 bought lines of catalog hardware. Cost reduction doesn't require inventing cheaper components — it requires manufacturing the *integration* at volume: standard balance-of-system learning curves (Wright's law: cost falls a fixed percentage with each doubling of cumulative production) apply to assembly, optics, and the three built lines. The learning-curve *rate* is measured, not assumed — Chapter 24's economics carry the first data, Chapter 29 the mature figures.

10.3 LCOE: the empirical question

Levelized cost of energy is the comparison's common currency, defined:

LCOE = (CAPEX × CRF + annual O&M) / annual kWh

where **CRF = $r(1+r)^n$ / $((1+r)^n - 1)$** annualizes capital at discount rate r over lifetime n years.

The book's position, stated plainly: **no LCOE is computed in this chapter**, because the inputs don't exist yet. An LCOE computed from assumed costs would be precisely the lie the book's contract forbids. Instead, the measurement plan — which chapter produces each input:

LCOE input	Produced by
Installed CAPEX (dated $)	Ch. 24 (pilot build accounting)
Annual O&M (cleaning, water treatment, bearing/seal service)	Ch. 26 (operating doctrine + measured)
Annual kWh (capacity factor, availability)	Ch. 22–23 (measured generation)
Lifetime n, degradation rate	Ch. 28–29 (qualification + proven operation)
Discount rate r	Reader's cost of capital

Chapter 24 runs the first full LCOE with pilot data. Chapter 29 closes it with proven-operation figures. Between here and there, the five mechanisms are the *reasons to run the experiment* — structured, quantitative reasons to believe the

answer can favor the system. Belief is not a number. The prototypes supply the number.

10.4 What the book never concedes

The founder's directive stands as the chapter's spine: *this isn't an ad for solar panels* — and the book never writes the system's obituary in advance. Every mechanism above is a real, physical, checkable path to winning. But the book also refuses the mirror-image lie: it does not claim victory before measurement. The honest sentence, carried from TRL 2 through TRL 9, is:

The mechanisms favor the system; the prototypes decide.

A reader who finishes this chapter knows exactly what would prove the system the better option (the five mechanisms, quantified in operation), exactly what would disprove it (an LCOE that won't clear the bar at measured costs), and exactly where each number comes from. That is what "it's not to lie, it's to prove what is possible with actual data" means in practice.

10.5 What this chapter proved — and what TRL 2 established

- The category positioning: building-scale thermal power, not panel competition; the measured incumbent vs. the mechanism-rich challenger.
- Five competitive mechanisms, each with quantitative logic: thermal-vs.-electrochemical storage costs, co-generated heat credited against one CAPEX, near-zero water consumption, no moving parts in the flow path, learning curves on mature parts.
- The LCOE definition, the refusal to compute it from assumptions, and the chapter-by-chapter measurement plan that produces every input.
- The spine: the mechanisms favor the system; the prototypes decide.

TRL 2 is complete. The concept is formulated: the machine described end to end (Ch. 6), every part named (Ch. 7), the home tier sized (Ch. 8), the larger tiers scaled (Ch. 9), and the competitive case stated with its empirical test defined (Ch. 10). TRL 3 begins the proving: critical functions demonstrated, starting with the lens.

GATE 2 — Exit evaluation

The reader may not proceed to TRL 3 until every answer below is theirs.

From Chapter 6:

1 Narrate the day cycle from dawn to dawn. State the claimed contribution in one sentence.

From Chapter 7:

1 State the two procurement doctrines. Which lines are built, and what is the fallback if Chapter 14 falsifies phase-routing?

From Chapter 8:

1 Run the seven-step chain for the reference home: what are the aperture, receiver area, and condenser bound, and which assumption drives each most?

From Chapter 9:

1 Explain the three scaling behaviors. Apply the shared-vs.-distributed decision rule to a 12-building campus.

From Chapter 10:

1 Name all five competitive mechanisms and the quantitative logic of each.

2 Write the LCOE equation. Explain why the book refuses to compute it here, and name the chapter that produces each input.

3 State the chapter's spine sentence. Explain what would prove the system the better option and what would disprove it.

Gate 2 exit criteria: all seven answered; the reader can describe the complete concept, size any tier, argue the competitive case without conceding or overclaiming, and state exactly which future chapters supply which evidence. Then TRL 3 — where the proving starts.

Sources and honesty notes

- Storage-cost orders of magnitude (hot-water thermal ~$1–5/kWh-thermal; Li-ion ~$100+/kWh-electric) are 2026 approximate figures for illustration — they move with markets and must be verified at procurement. The *mechanism* (thermal storage electrochemical storage per unit energy) is structural and stable.
- Wet-cooled thermal water consumption (~liters/kWh evaporated) is standard industry figures for CSP/fossil steam plants; the system's near-zero makeup is by design (Ch. 5), to be demonstrated in operation (Ch. 26).
- Wright's-law learning curves are the standard empirical regularity of manufacturing economics; the *rate* for this system is measured in Chapters 24 and 29, not assumed here.
- No LCOE figure appears in this chapter by design. Any LCOE the reader encounters for this system before Chapter 24 is someone else's assumption, not this book's.

TRL 3

Critical Functions Proven

Chapters 11–15

Chapter 11

The End-to-End Energy Balance

TRL 3 · Critical functions proven. This chapter accounts for every joule from sunlight to socket — the analytical proof that the chain closes, with parasitics subtracted and the gap to demand stated honestly.

11.1 The balance principle

Chapter 6's falsification condition #1: *the end-to-end energy balance shows net output below usefulness at all three tiers.* This chapter runs that check. The method is a stage-by-stage account — beam sunlight in, losses itemized at every conversion, net electricity and useful heat out. Every stage's efficiency was sourced or derived in TRL 1; this chapter only chains them, because a chain is only as strong as the honesty of its multiplication.

The reference case: Chapter 8's home tier on its design day (assumptions A1–A6 carried forward).

11.2 The worked balance — one design day

Stage	In (kWh/day)	Efficiency	Out (kWh/day)	Lost (kWh/day)	Loss mechanism
Beam on aperture	220	—	220	—	—

Stage	In (kWh/day)	Efficiency	Out (kWh/day)	Lost (kWh/day)	Loss mechanism
Lens + boost optics	220	0.75	165	55	Reflection, absorption, spillage, cosine
Receiver	165	0.67	111	54	Reradiation, convection, conduction
Rankine cycle	111	0.27	30 (shaft)	81 (heat)	Condenser rejection — Ch. 5
Generator	30	0.95	28.5	1.5	Windage, copper/iron losses
Parasitics	—	—	—	—	—
Feed pump	—	—	0.25	0.25	w_p $v_1(P_2 P_1)$; small by physics, not by wish
Tracking drives	—	—	0.15	0.15	Intermittent DC motors (estimate)
Controls + instrumentation	—	—	0.5	0.5	Tens of watts continuous (estimate)
Condenser fan (optional)	—	—	2.0	2.0	200 W × 10 h if forced convection chosen
Net electric (passive condenser)			**27.6**		

Stage	In (kWh/day)	Efficiency	Out (kWh/day)	Lost (kWh/day)	Loss mechanism
Net electric (forced condenser)			**25.6**		

Against the 30 kWh/day reference demand: **the design day delivers ~92% passively, ~85% with a fan.** The balance closes — the machine is useful on paper — but it does not flatter: there is a measured gap, and the book names the levers that close it (Chapter 8's sensitivity table): a better expander, a sunnier site, or a few more square meters of aperture. A balance that lands at 92% with named levers is worth more than a balance fudged to 100%.

Note what the parasitics teach: the feed pump — the part novices worry about — costs a quarter of a kilowatt-hour because pump work scales with liquid volume, and liquids barely compress. The fan — the part novices forget — costs eight times more. **Size your attention by the balance, not by intuition.** That sentence is the whole chapter in miniature.

11.3 Useful heat: the second product

The 81 kWh/day of rejected heat is not waste in the building's ledger — it is low-grade heat at condenser temperatures, available for domestic hot water preheating (Chapter 9's waste-to-product flip at scale; at the home tier a modest but real credit). The energy balance therefore reports two products:

- **27.6 kWh/day electricity** (the metered product)
- **~81 kWh/day low-grade heat** (the thermal product, partially capturable)

Any economic comparison that counts only the electricity is undercounting the machine by design. Chapter 24's economics credit both.

11.4 Uncertainty: which stage to measure first

Multiply the chain's uncertainties and the balance wobbles. Ranked by leverage on the net result:

1. **Turbine/cycle efficiency** (±7 points of η_th swings net output by ±26%) — the weak link, measured in Ch. 13/20.
2. **Optical efficiency** (±5 points swings net ±7%) — measured in Ch. 12.
3. **Receiver efficiency** (±5 points swings net ±7%) — measured in Ch. 12/13.
4. **Parasitics** — small individually, but the fan decision (Ch. 5/24) is a 7% swing by itself.

The measurement priority for TRL 3 is therefore: **cycle first, optics second, receiver third, parasitics metered always.** Chapters 12–15 follow exactly this order.

11.5 The reader's balance worksheet

1. My design-day beam on aperture (A_aperture × local kWh/m²/day): _____
2. My optical efficiency (measured Ch. 12, else 0.70–0.80 assumed): _____ to receiver: _____
3. My receiver efficiency (measured Ch. 13, else 0.60–0.70 assumed): _____ to cycle: _____
4. My cycle efficiency (Ch. 3 procedure at my steam conditions): _____ shaft: _____
5. My generator efficiency: _____ gross electric: _____
6. My parasitics, each metered or estimated with its basis: _____ **net:** _____
7. My demand: _____ coverage: _____% levers (Ch. 8 table): _____

If coverage is under 100%, the worksheet doesn't ask you to believe harder — it points at the sensitivity table. The balance is an instrument, not a verdict.

11.6 What this chapter proved

- The full stage-by-stage energy balance for the reference home tier: 220 kWh beam 27.6 kWh net electric (passive) with every loss itemized and mechanized.
- Falsification condition #1 checked: the balance closes at ~92% of reference demand with named levers — useful, not fudged.
- The parasitics lesson: pump negligible by physics, fan significant by choice — attention sized by the balance.
- The two-product ledger: electricity plus ~81 kWh/day low-grade heat.
- Measurement priority for TRL 3: cycle, optics, receiver, metered parasitics.

Chapter 12 proves the flux: the lens and heliostat-boost experiments — what the focus actually delivers.

Self-evaluation — Gate 3, part 1

1 Reproduce the balance table's chain from memory: 220 165 111 30 28.5 27.6. State each stage's efficiency and loss mechanism.

2 Compute the feed-pump parasitic from w_p $v_1(P_2 P_1)$ for the reference case. Why is it small? (Answer in terms of liquid incompressibility, not hand-waving.)

3 The fan costs 2 kWh/day and the pump 0.25. Explain what "size your attention by the balance" means for a builder's test plan.

4 Coverage is 92%. Name the three levers and the chapter containing their quantitative table.

5 Rank the four measurement priorities and justify the order from the uncertainty leverage.

Sources and honesty notes

- All stage efficiencies are carried from TRL 1 chapters (optical 0.75, receiver 0.67, cycle 27%, generator 0.95); none are newly asserted here.
- Parasitic figures: the feed pump is computed from the stated relation; tracking, controls, and fan figures are **engineering estimates**, labeled as such — the reader meters their own hardware from Chapter 17 onward.
- The 92%/85% coverage figures are calculations on the reference case, not performance claims. Real coverage is measured in Chapters 22–23.
- The uncertainty ranking is a sensitivity analysis of the worked balance; actual measurement uncertainties are established by the experiments of Chapters 12–15.

Chapter 12

Proving the Flux

TRL 3 · Critical functions proven. This chapter specifies the optical experiments: how the reader measures what the focus actually delivers, and replaces the assumed 0.75 with a measured number.

12.1 From assumed to measured

Every efficiency in Chapter 11's balance was carried forward from TRL 1's sourced bands — except the optical 0.75, which was an *assumption* wearing a sourced number's clothes. TRL 3 is where assumptions get replaced by measurements, in the priority order Chapter 11 set: optics second only to the cycle. This chapter specifies four experiments. Each states its objective, apparatus, procedure, measured quantities, pass criteria — and what happens if it fails, because a test plan without failure branches is a ritual, not an experiment.

Standing safety notice: concentrated sunlight blinds and burns. Every experiment in this chapter runs under the optical-safety protocol: no personnel in the beam path, beam dumps rated for full flux, interlocked shutters, and eye protection specified for the actual irradiances involved — not sunglasses. The builder who skips this doesn't get data; they get injuries. Chapter 27 covers safety qualification formally.

12.2 Experiment 12A — Flux mapping: what the focus actually delivers

Objective: measure the true irradiance distribution at the receiver plane and the true optical efficiency.

Apparatus: a water-cooled Lambertian (diffuse white) target placed at the receiver plane; a calibrated camera imaging the target; a pyrheliometer measuring DNI beside the lens; thermocouples on the target cooling loop (calorimetric cross-check).

Procedure:

1. Track the sun; record DNI continuously.
2. Image the target; convert pixel brightness to irradiance via the camera's radiometric calibration (calibrated against a reference lamp — the calibration, not the camera's spec sheet, is the instrument).
3. Integrate irradiance over the target: $P_{focus} = \int E \, dA$.
4. Optical efficiency: $\eta_{optical} = P_{focus} / (DNI \times A_{aperture})$.
5. Cross-check: the cooling loop's calorimetry ($\dot{m} \cdot c_p \cdot \Delta T$) must agree with the optical integration within the uncertainty budget — two independent methods, one number.

Measured quantities: flux map E(x,y), peak flux, spillage fraction (power missing the receiver aperture), $\eta_{optical}$.

Pass criteria: $\eta_{optical}$ measured with stated uncertainty; flux map shows peak flux within the receiver's material limits and spillage quantified.

If it fails: if $\eta_{optical}$ measures 0.62 instead of 0.75, Chapter 11's balance is *revised*, not defended. The book's numbers update; its wishes don't. A measured 0.62 with a diagnosed cause (groove scatter, soiling, misalignment) is worth more than an assumed 0.75 — it tells the builder exactly what to fix.

12.3 Experiment 12B — Heliostat boost: each mirror earns its place

Objective: verify that every boost mirror delivers its designed contribution to the aperture.

Apparatus: the 12A setup plus individual mirror shutters (or sequential shading).

Procedure:

1. With all mirrors tracking, record P_focus (baseline).
2. Shade mirrors one at a time; record the drop ΔP per mirror.
3. Each mirror's delivered power must match its predicted contribution (mirror area × DNI × reflectivity × cosine × spillage) within uncertainty.

Measured quantities: per-mirror ΔP, total boost factor, individual mirror pointing error (from the flux shift when unshaded).

Pass criteria: total boost within 10% of design; no mirror contributing less than 70% of its prediction without a diagnosed cause.

If it fails: underperforming mirrors get root-caused — reflectivity loss (clean it), misalignment (recalibrate it), shading by structure (redesign the field layout). The experiment doesn't just verify the boost; it *commissions* the field.

12.4 Experiment 12C — Tracking accuracy: staying inside the budget

Objective: verify pointing accuracy against Chapter 2's error budget (~1–2° total for C = 60).

Apparatus: sun-position algorithm (NREL SPA — the standard) as truth reference; encoder readouts on both axes; the flux centroid from 12A's camera as the optical truth.

Procedure:

1. Log commanded vs. actual axis positions and the flux centroid position across a full solar day.

2 Decompose the error: sensor bias (constant offset), drive backlash (hysteresis on reversal), structural deflection (wind- and elevation-dependent), algorithm error (vs. SPA).

3 Total RMS pointing error vs. the budget.

Measured quantities: per-source error contributions, total RMS error, worst-case excursion.

Pass criteria: total error within the Chapter 2 budget with margin; every error source attributed.

If it fails: the budget tells you which fix applies — backlash means drive preload or software compensation; deflection means structure; bias means sensor recalibration. An over-budget tracker with a diagnosed dominant source is one fix away from passing; an over-budget tracker with unknown sources is a redesign.

12.5 Experiment 12D — Soiling: the efficiency you actually keep

Objective: measure how fast dust and grime degrade optical efficiency, establishing the cleaning doctrine's data basis.

Apparatus: the 12A setup; a reference clean coupon beside the lens.

Procedure:

1 Clean the lens to reference state; measure η_optical.

2 Operate normally without cleaning; re-measure η_optical weekly; log weather (rain, dust events).

3 Plot efficiency vs. time; extract the soiling rate for the site.

Measured quantities: soiling rate (%/week), rain-recovery fraction, cleaning-restored efficiency.

Pass criteria: a site-specific soiling curve and a cleaning interval derived from it (clean when the value of recovered energy exceeds the cleaning cost — the doctrine of Chapter 26, fed by this data).

If it fails: "fails" here means the soiling rate is worse than the economics tolerate — which is data for Chapter 24, not a defect in the machine. Some sites need weekly washing; some need coatings research. The experiment's job is to say which.

12.6 The measured-vs.-assumed ledger

TRL 3 maintains a running ledger — every assumption from TRL 2, struck through and replaced by a measurement or carried forward explicitly:

Assumption (Ch. 11)	Experiment	Status after TRL 3
η_optical = 0.75	12A	Measured: ___ ± ___
Boost factor = design	12B	Measured per mirror: ___
Tracking within budget	12C	RMS error: ___ vs. budget
Soiling negligible	12D	Soiling rate: ___; cleaning interval: ___

The reader fills the right column with their numbers. A book full of filled ledgers is a technology at TRL 4. A book full of assumptions is a proposal.

12.7 What this chapter proved

- Four specified optical experiments: flux mapping (12A), boost verification (12B), tracking accuracy (12C), soiling (12D).
- Each with objective, apparatus, procedure, measured quantities, pass criteria, and failure branches.
- The calorimetric cross-check: two independent methods, one number — the standard against which all TRL 3 measurements are judged.
- The optical-safety protocol as a standing requirement.
- The measured-vs.-assumed ledger: TRL 3's instrument for replacing belief with data.

Chapter 13 proves the steam: boiler experiments, heat flux, and the critical-heat-flux safety margins.

Self-evaluation — Gate 3, part 2

1. Why must the camera's radiometric calibration — not its spec sheet — be the instrument? What does the calorimetric cross-check guard against?
2. In 12B, a mirror delivers 55% of its predicted power. Walk the root-cause branches in order.
3. Decompose a 2.5° total tracking error into its four sources. Which fix applies to each?
4. Derive a cleaning interval from a soiling curve: state the economic condition in one sentence.
5. Your 12A measures η_optical = 0.66 ± 0.03. Recompute Chapter 11's net electric (passive) with this number. Does the concept survive? (Work it; don't assert.)

Sources and honesty notes

- The NREL Solar Position Algorithm (SPA) is the standard sun-position reference; the reader implements it from NREL's published code, not from this book's description.
- Pyrheliometer calibration and the uncertainty budget follow standard radiometry practice (ISO 9060 instrument classes); the book specifies the *structure* of the experiments, and the reader's metrology lab supplies the traceability.
- All pass criteria are the author's specified thresholds for the reference design — the reader justifies their own thresholds against their design's error budgets.
- No measured optical data for this system exists yet; the right column of the ledger is empty until the reader's experiments fill it.

Chapter 13

Proving the Steam

TRL 3 · Critical functions proven. This chapter specifies the boiler and steam experiments: receiver efficiency, the boiling regime, the critical-heat-flux safety margin, steam quality, and the daily start-up transient.

13.1 What "proving the steam" means

Chapter 11's balance assumed a receiver efficiency of 0.67 and a cycle fed with dry superheated steam at its design point. Chapter 12 measured what the focus delivers. This chapter measures what the boiler does with it — and, more importantly, proves the boiler does it *safely*, inside margins that survive the worst day, not just the design day.

Standing safety notice: this chapter works with pressurized steam at hundreds of degrees. Every experiment runs under the pressure-safety protocol: vessels hydrotested to code before first firing, relief valves sized and bench-tested, personnel outside the blast radius during pressurized operation, and a written operating envelope posted at the rig. Steam doesn't negotiate. Chapter 27 covers qualification formally; nothing here runs without it.

13.2 Experiment 13A — Receiver efficiency: heat in vs. heat captured

Objective: measure the fraction of focused power that reaches the working fluid.

Apparatus: the 12A flux station feeding the receiver-boiler test article; calorimetric measurement on the working-fluid loop (ṁ·Δh across the receiver, with enthalpies from IAPWS-IF97 at measured pressures and temperatures).

Procedure:

1. Establish steady focused power P_focus (measured per 12A, not assumed).
2. Flow water/steam at the design mass flux; measure inlet/outlet states.
3. η_receiver = ṁ(h_out h_in) / P_focus, at several flux levels spanning the operating range.

Measured quantities: η_receiver vs. incident flux, wall temperatures, loss breakdown (reradiation vs. convection inferred from the flux dependence).

Pass criteria: η_receiver measured with uncertainty across the operating range; the design-point value recorded in the ledger.

If it fails: low receiver efficiency gets root-caused by its flux dependence — reradiation-dominated losses grow with temperature (coat it better, per Ch. 7's coating line), convection-dominated losses grow with wind (shield it). The measurement doesn't just grade the receiver; it prescribes the fix.

13.3 Experiment 13B — The boiling regime: staying in nucleate boiling

Objective: verify the boiler operates in the nucleate-boiling regime with margin — the efficient, stable mode — and characterize its boiling curve.

Apparatus: instrumented boiler test section with wall thermocouples; heat flux varied stepwise; pressure controlled.

Procedure:

1. Raise heat flux q'' in steps; at each step record wall superheat ΔT_e = T_wall T_sat.

2 Plot the boiling curve; verify the nucleate-boiling slope against the Rohsenow correlation (Chapter 3, §3.2) — agreement within the correlation's known scatter confirms the regime.

3 Identify the onset of any departure: rising ΔT_e at fixed q'' is the warning sign.

Measured quantities: boiling curve q'' vs. ΔT_e, Rohsenow C_sf for the actual surface-fluid pair (a measured constant, not a handbook guess), margin to the transition.

Pass criteria: operation confirmed in nucleate boiling across the full design flux range; C_sf measured for the reader's actual materials.

If it fails: unexpected boiling behavior (fouling-altered C_sf, flow instabilities) triggers the surface-preparation and water-chemistry review of Chapter 21 — the boiler's behavior is a chemistry diagnostic, not just a thermal one.

13.4 Experiment 13C — The CHF margin: proving the safety rule

Objective: demonstrate the Chapter 3 safety rule — design heat flux 50% of critical heat flux — without destroying the boiler proving it.

Apparatus: a separate sacrificial test section, geometrically representative of the boiler tubes, instrumented to detect the CHF excursion.

Procedure:

1 On the *sacrificial* section only, raise q'' until departure from nucleate boiling is detected (the wall-temperature excursion that marks CHF). Record q''_CHF,measured at the operating pressure.

2 Compare against Zuber's prediction (Chapter 3, §3.3). Agreement validates the correlation for the reader's geometry; disagreement investigates surface and flow effects.

3 Verify: q''_design / q''_CHF,measured 0.5 at every operating point, including the peak-flux transient (cloud-edge brightening can spike flux above the steady value — the margin must cover the spike, not the average).

Measured quantities: q''_CHF,measured vs. pressure, Zuber agreement factor, design margin ratio including transients.

Pass criteria: margin ratio 0.5 everywhere in the envelope, demonstrated by measurement on representative geometry.

If it fails: the margin is not negotiated — the design changes. Lower the flux (larger receiver area, lower C), improve the boiling surface, or reduce the operating pressure. Chapter 3's safety rule is a hard gate: TRL 3 does not pass without it.

13.5 Experiment 13D — Steam quality: protecting the turbine

Objective: verify the turbine inlet receives dry superheated steam — wet steam erodes blades, and the turbine is the tier's economic lever.

Apparatus: throttling calorimeter or conductivity-based quality measurement at the boiler outlet; temperature/pressure confirming superheat.

Procedure:

1 At each operating point, measure dryness fraction x (must be 1.0 — superheated, not merely saturated) and degrees of superheat.

2 During transients (start-up, cloud passages), log the minimum quality reached.

Measured quantities: superheat vs. operating point, worst-case transient quality.

Pass criteria: superheated at all steady operating points; transients never drop below the turbine manufacturer's minimum quality — with a moisture-separator or superheater trim specified if they do.

If it fails: persistent wet steam means the boiler/superheater design is wrong for the flux profile — resize the superheater section or revise the operating procedure. The turbine's blades are not the place to discover this.

13.6 Experiment 13E — The daily start-up: proving the transient

Objective: verify the boiler survives its actual duty cycle — cold every morning, hot every midday, cooling every night — without accumulating damage.

Apparatus: the full boiler test article under automated day-cycle control.

Procedure:

1. Run repeated thermal cycles: ambient → operating → ambient, at the design ramp rates.
2. Monitor: time to operating conditions, peak thermal stresses (strain gauges or computed from temperature gradients), and any drift in performance across cycles.
3. Target: hundreds of cycles minimum — a year of mornings — with inspection intervals per the pressure-vessel code.

Measured quantities: start-up time, thermal-stress peaks, performance drift per 100 cycles, inspection findings.

Pass criteria: no measurable degradation across the test campaign; start-up time compatible with the solar day (a boiler that needs four hours to warm up misses the morning); stresses within the fatigue analysis.

If it fails: slow start-up revises the operating strategy (overnight warm-keeping vs. cold start — priced in Chapter 26); stress or drift findings revise the design before TRL 4 builds on it. Daily cycling is the application's defining load case — it is tested as such, not as an afterthought.

13.7 The ledger grows

Assumption (Ch. 11)	Experiment	Status after TRL 3
η_receiver = 0.67	13A	Measured: ___ ± ___
Nucleate boiling, CHF margin 2×	13B/13C	Regime: ___; margin ratio: ___
Dry superheated steam to turbine	13D	Min transient quality: ___
Survives daily cycling	13E	Cycles tested: ___; drift: ___

13.8 What this chapter proved

- Five specified steam experiments: receiver efficiency (13A), boiling regime (13B), CHF margin on sacrificial geometry (13C), steam quality (13D), daily start-up cycling (13E).
- The safety rule made testable: design flux 50% of *measured* CHF, transients included — a hard TRL 3 gate.
- The pressure-safety protocol as a standing requirement.
- The turbine protected by measurement: quality verified before blades ever see steam.

Chapter 14 proves the valve: diodicity tests and the phase-routing experiments — the book's central hypothesis under test.

Self-evaluation — Gate 3, part 3

1 Why is CHF demonstrated on a sacrificial section rather than the boiler itself? What does agreement with Zuber's prediction license, and what would disagreement trigger?

2 A cloud-edge transient spikes flux 40% above steady design. Your measured margin ratio at steady design is 0.45. Do you pass? Show the reasoning.

3 The boiling curve disagrees with the Rohsenow correlation's slope. Name two physical causes and the chapter that owns each.

4 During start-up, quality dips to 0.92 for 90 seconds. State the decision procedure — what gets specified, and what would force a redesign instead?

5 Your 13E campaign shows 2% performance drift per 100 cycles. Project to one year (365 cycles) and state whether this passes, with the reasoning.

Sources and honesty notes

- Zuber's CHF equation and the Rohsenow correlation are the established references (Chapter 3); the experiments *validate them for the reader's geometry* rather than re-deriving them.
- The 50% CHF design rule is the author's specified safety margin for this application — conservative by intent, and a hard gate by design.
- No boiler test data for this system exists yet; the ledger's right column is empty until the reader's experiments fill it.
- Pressure-vessel and relief-valve requirements are code-driven (ASME BPVC or local equivalent); Chapter 27 maps the compliance path — this chapter specifies the *functional* proofs, not the legal ones.

Chapter 14

Proving the Valve

TRL 3 · Critical functions proven. This chapter puts the book's central hypothesis under test: the phase-directed day/night cycling claim, with the falsification branch specified in advance.

14.1 The hypothesis, in falsifiable form

From Chapter 4, restated exactly — because a hypothesis that drifts between chapters can't be tested:

> *In the proposed loop geometry, steam preferentially flows through the power leg toward the turbine while condensed liquid preferentially returns through the return leg, with reverse leakage in each leg small enough that the day/night cycle operates without mechanical check valves.*

Two quantities make this testable: **diodicity** (Di = ΔP_reverse / ΔP_forward at matched flow rate) and **phase selectivity** (the fraction of each phase taking its intended leg). The chapter specifies five experiments. And because honesty requires the failure branch *before* the data, it is stated here, in advance:

If the experiments falsify the hypothesis, the valve sections become conventional check valves — Chapter 4's named fallback, Chapter 7's BOM revision, no other line touched. The book survives its hypothesis failing. That is what the risk register was for.

14.2 Deriving the pass criteria from the loop

The literature reports diodicities; the loop requires a *leakage*. The pass criteria are derived from what the system needs, not from what papers celebrate:

- **Day operation:** steam entering the power leg must reach the turbine, not bypass backward through the return leg. Requirement: reverse mass flow through the return leg 5% of design forward steam flow, at the loop's actual operating ΔP.
- **Night operation:** returning liquid must reach the boiler, not bypass through the power leg. Requirement: reverse liquid flow 5% of design return flow.
- **Translated to diodicity:** at the operating Reynolds numbers, the valve geometry must deliver Di sufficient to hold those leakages — the reader computes the required Di from their loop's ΔP-flow characteristic (Chapter 4's operating rules), then measures whether their geometry delivers it.

A valve with Di = 10 that holds 2% leakage passes. A valve with Di = 50 that holds 8% leakage fails. **The loop grades the valve, not the literature.**

14.3 Experiment 14A — Single-phase diodicity: the baseline

Objective: measure the valve section's diodicity with liquid water, both directions, across the operating Reynolds-number range.

Apparatus: a flow loop with reversible pumping, calibrated flowmeter, differential-pressure transducers across the test section, temperature control.

Procedure:

1. Pump water forward through the valve section at stepped flow rates spanning the design range; record ΔP_fwd(Q).

2. Reverse the flow; record ΔP_rev(Q).
3. Compute Di(Q) = ΔP_rev / ΔP_fwd at matched Q.
4. Repeat for each candidate geometry (stage count N, branch angle, channel width per Chapter 4's design rules).

Measured quantities: Di vs. Re for each geometry, the design-rule validation (does measured Di follow the predicted stage-count trend?).

Pass criteria: measured Di meets the §14.2-derived requirement at design Re, with the geometry frozen for the system build.

If it fails: geometry iterates per Chapter 4's rules — more stages, revised branch angle — until the requirement is met or the rules are exhausted. Exhaustion without meeting the requirement is a *finding*, and it triggers the fallback branch (§14.7).

14.4 Experiment 14B — Steam forward: the power leg under fire

Objective: verify the valve section passes the design steam flow in the forward direction without choking, excessive pressure drop, or condensation-induced hammer.

Apparatus: steam supply at design pressure/temperature; the valve test section; downstream calorimetry/condenser.

Procedure:

1. Flow superheated steam forward at stepped rates to 120% of design.
2. Record ΔP, outlet superheat (any desuperheating indicates unwanted condensation in the valve), and acoustic monitoring for two-phase hammer signatures.

Measured quantities: forward steam ΔP vs. flow, outlet state, acoustic baseline.

Pass criteria: ΔP within the Chapter 11 balance's allowance; outlet remains superheated; no hammer signatures.

If it fails: excessive forward resistance revises the geometry (the diode's forward direction must be nearly free — a Tesla valve that throttles its own forward flow is a bad valve, however good its diodicity).

14.5 Experiment 14C — Phase routing: the hypothesis test

Objective: the core experiment — demonstrate that steam and liquid self-separate into their intended legs.

Apparatus: a two-leg test loop replicating the system topology: boiler simulator feeding a junction, power leg (valve section + turbine simulator orifice) one way, return leg (valve section + condenser simulator) the other; phase-discriminating instrumentation (void-fraction probes, or thermal tracing) in each leg.

Procedure:

1. **Day simulation:** inject steam at the junction at design rate. Measure the split: fraction to the power leg vs. the return leg.
2. **Night simulation:** inject liquid at the junction. Measure the split: fraction to the return leg vs. the power leg.
3. **Transition:** ramp steam down while ramping liquid up, simulating the evening transition; verify no regime where both phases significantly take the wrong leg.
4. Repeat across the operating envelope (pressures, flow rates, subcooling).

Measured quantities: phase split fractions per leg per phase, transition behavior, envelope map.

Pass criteria: ≥95% of steam to the power leg in day simulation; ≥95% of liquid to the return leg in night simulation; no transition instability — the §14.2 requirements, measured

directly.

If it fails: this is the falsification event the book prepared for. The failure gets fully characterized (which phase, which leg, at what conditions) — because even a failed hypothesis teaches the fallback design its requirements — and §14.7 executes.

14.6 Experiments 14D & 14E — Stability and endurance

14D — Two-phase stability. Flow boiling/condensing mixtures through the valve section; monitor for destructive oscillations (density-wave, pressure-drop oscillations). The valve's two-phase stabilization evidence (Chapter 4) is confirmed or bounded for the reader's geometry. Pass: no sustained oscillations in the operating envelope; any observed modes mapped and avoided by the operating procedure.

14E — Endurance. Thermally and flow-cycle the valve sections through the equivalent of years of day/night operation. Inspect for erosion (steam-side) and scaling/fouling (liquid-side); re-measure Di afterward. Pass: Di within 10% of the 14A baseline; no geometry change beyond the fouling allowance that Chapter 21's chemistry doctrine manages.

14.7 The fallback branch, specified

If 14C falsifies phase-routing, or 14A exhausts the design rules without meeting §14.2:

1. The two valve-section BOM lines (Chapter 7, §7.6) are revised to conventional check valves — spring or swing type, rated for the loop's pressure, temperature, and both phases.
2. The control system (Chapter 7, §7.7) gains the valve-actuation logic the passive design didn't need — an added line, honestly costed in Chapter 24.

3 Chapters 15–31 proceed unchanged: the day/night cycle, the energy balance, and every downstream experiment are identical, because the *function* (one-way phase-appropriate flow) is preserved while the *mechanism* changes.

4 The book records the falsification openly — a failed hypothesis, tested fairly, is a contribution, not an embarrassment. The patent-office reader in particular needs to see exactly this: what was claimed, how it was tested, what the data said.

14.8 The ledger

Assumption / claim	Experiment	Status after TRL 3
Di meets loop-derived requirement	14A	Measured Di: ___; required: ___
Steam forward without choking/hammer	14B	ΔP: ___; outlet state: ___
95% phase selectivity both legs	14C	Steam split: ___; liquid split: ___
Two-phase stability in envelope	14D	Modes observed: ___
Endurance across years of cycling	14E	Di drift: ___; inspection: ___
Fallback if falsified	14.7	Triggered? ___

14.9 What this chapter proved

- The hypothesis restated falsifiably, with the failure branch specified *before* data.
- Pass criteria derived from the loop's leakage requirements (§14.2), not from literature diodicity records.

- Five specified experiments: single-phase diodicity (14A), steam forward (14B), phase routing — the hypothesis test (14C), two-phase stability (14D), endurance (14E).
- The fallback branch: check-valve revision, control-logic addition, unchanged downstream chapters, open recording of the result.

Chapter 15 proves the night: condensation and water-recovery experiments — the "no leaking" claim, measured.

Self-evaluation — Gate 3, part 4

1 Restate the hypothesis in one falsifiable sentence. What two quantities make it testable?

2 Derive a diodicity requirement from a loop requirement: if reverse leakage must be 5% and the loop's ΔP-flow characteristic is given, outline the computation (no numbers needed — the method).

3 In 14C's transition test, 12% of liquid takes the power leg during the ramp. State the verdict and the next three steps per §14.7.

4 A valve measures Di = 40 but forward steam ΔP exceeds the Chapter 11 allowance. Does it pass? Explain which experiment fails it and the design implication.

5 Argue for or against: "The fallback branch means the hypothesis test doesn't matter." (The book's position is in §14.1 and §14.7 — engage it, don't quote it.)

Sources and honesty notes

- Diodicity data and two-phase stabilization evidence are the published results reviewed in Chapter 4; the experiments here *test the application*, not the literature.

- The 95% selectivity and 5% leakage thresholds are the author's specified requirements for the reference design — derived from the loop's tolerance for misplaced phase, and the reader justifies their own against their design.
- No phase-routing test data for this application exists yet; the ledger's right column is empty until the reader's experiments fill it.
- The fallback to conventional check valves is a standard engineering redundancy, not a concession — its specification in advance is what makes the hypothesis test scientifically honest.

Chapter 15

Proving the Night

TRL 3 · Critical functions proven. This chapter specifies the condensation and water-recovery experiments — the "no leaking" claim as a measured mass balance. It closes TRL 3.

15.1 What "proving the night" means

Chapter 5 turned "no leaking" from a slogan into an accounting procedure. This chapter turns the procedure into data. Three things must be demonstrated: the condenser rejects heat at its designed rate, the water inventory survives the night intact, and the loop's chemistry stays within its operating envelope across many cycles. Each gets its experiment; each experiment states what would falsify it.

15.2 Experiment 15A — Condenser performance: measuring U

Objective: measure the actual overall heat-transfer coefficient of the condenser hardware against the Chapter 5 design assumption (U = 8 W/m²·K passive).

Apparatus: the condenser test article (finned panels per Chapter 7, §7.5); steam supply at the design condensing temperature; ambient weather station; air-side velocity and temperature traverses.

Procedure:

1 Flow steam at measured rate ṁ; measure inlet/outlet states to get Q_rejected = ṁ(h_in h_out).

2 Measure ambient conditions and air-side temperatures; compute ΔT_lm.

3 U_measured = Q / (A · ΔT_lm) across ambient temperatures spanning the design range — especially the design hot night.

4 If forced convection: repeat at each fan setting, producing the U-vs.-fan-power curve that prices Chapter 5's trade.

Measured quantities: U vs. ambient temperature, U vs. fan power, the design-hot-night verification point.

Pass criteria: U_measured at the design hot night meets or exceeds the Chapter 5 sizing assumption with margin; the full curve recorded for the operating procedure.

If it fails: the condenser is resized per Chapter 5's method with the *measured* U — larger area, forced convection, or building-integrated surfaces. The method doesn't change; its input gets honest.

15.3 Experiment 15B — The full night recovery: "no leaking," measured

Objective: run the complete day/night cycle and close the water inventory — the book's most quoted claim, as a measurement.

Apparatus: the integrated test loop (boiler simulator or the Chapter 13 article, condenser per 15A, valve sections per Chapter 14, feed pump); precision inventory measurement (load cells on the loop, or calibrated sight-glass volumetrics with temperature correction).

Procedure:

1 Charge the loop with a measured inventory; log every gram.

2 Run the day simulation: heat input, steam generation, power-leg flow.

3 Run the night simulation: heat rejection, condensation, liquid return.

4 At dawn, measure the inventory. Run Chapter 5's accounting: Inventory_evening = Inventory_morning + Makeup Blowdown Samples Fugitives.

5 Repeat for a minimum campaign (tens of cycles); trend the makeup.

Measured quantities: per-cycle inventory closure, makeup trend, fugitive identification (tracer or pressure-decay localization if nonzero).

Pass criteria: inventory closes within measurement uncertainty every cycle; makeup trends toward zero across the campaign; any nonzero fugitive is localized and eliminated, not averaged away.

If it fails: a persistent nonzero makeup is not a tuning problem — it is a leak, and it gets found (pressure-decay testing, tracer gas, joint-by-joint inspection) and fixed. The claim "no leaking" is earned one eliminated leak at a time. A loop whose makeup won't trend to zero doesn't proceed to TRL 4.

15.4 Experiment 15C — Non-condensables: the invisible blanket

Objective: quantify how air ingress degrades condensation, and verify the purge procedure restores it.

Apparatus: the 15B loop with a controlled air-injection point and the purge valve from Chapter 7, §7.5.

Procedure:

1 Establish baseline condenser performance (15A data).

2 Inject measured quantities of air; record the degradation in U and in condensation completeness.

3 Execute the purge procedure; verify restoration to baseline.

4 Determine the maximum tolerable air fraction — the operating limit posted at the rig.

Measured quantities: U vs. air fraction, purge effectiveness, the tolerable limit.

Pass criteria: purge restores baseline within the procedure's specified time; the tolerable air fraction is quantified and monitored in operation (Chapter 26).

If it fails: an ineffective purge redesigns the purge point location or procedure — air traps at high points the designer didn't anticipate are the classic finding. The experiment's value is finding them on the test loop, not on the customer's roof.

15.5 Experiment 15D — Freeze: the climate the book doesn't skip

Objective: verify freeze protection for sites that need it (not all do — the procedure is climate-gated).

Apparatus: the loop or a representative section in a controlled cold chamber.

Procedure:

1. Subject the charged loop to the design freeze event (temperature and duration from the site's climate data).
2. Verify the protection strategy: drain-down completeness (no trapped water), heat-trace coverage (no cold spots by thermal imaging), or freeze-tolerant routing — whichever the design specifies.
3. Return to ambient; pressure-test; verify no damage.

Measured quantities: coldest point in the loop during the event, drain-down residual, post-event integrity.

Pass criteria: no damage, no trapped water where the strategy requires none, protection verified by measurement rather than by brochure.

If it fails: the freeze strategy is redesigned before any cold-climate deployment. A water loop that can't survive its site's winter is a seasonal machine — the book says which sites those are instead of pretending otherwise.

15.6 Experiment 15E — Water chemistry: the slow experiment

Objective: verify the blowdown and cycles-of-concentration arithmetic across extended operation.

Apparatus: the 15B loop run for an extended campaign; water sampling per Chapter 7, §7.8; dissolved-solids analysis.

Procedure:

1. Operate with the Chapter 21 chemistry doctrine: fill spec, blowdown rate from the cycles-of-concentration calculation.
2. Trend dissolved solids, pH, and conductivity in the boiler water across the campaign.
3. Verify: solids stay under the boiler's limit; blowdown volume matches the accounting; no scaling or corrosion signals on inspection.

Measured quantities: solids trend vs. prediction, actual blowdown volume, inspection findings.

Pass criteria: chemistry stays in envelope for the full campaign; the blowdown arithmetic confirmed by measurement.

If it fails: chemistry failures are slow and expensive — scaling that takes months to matter. A failed 15E revises the water-treatment spec (Chapter 21) before TRL 4 multiplies the operating hours. This is the cheapest chapter in the book to get wrong early and the most expensive to get wrong late.

15.7 The TRL 3 ledger — complete

Assumption (Ch. 11)	Experiment	Status after TRL 3
η_optical = 0.75	12A	Measured: ___ ± ___
Boost factor = design	12B	Measured per mirror: ___
Tracking within budget	12C	RMS error: ___ vs. budget

Assumption (Ch. 11)	Experiment	Status after TRL 3
η_receiver = 0.67	13A	Measured: ___ ± ___
Nucleate boiling, CHF margin 2×	13B/13C	Regime: ___; margin ratio: ___
Dry superheated steam	13D	Min transient quality: ___
Survives daily cycling	13E	Cycles: ___; drift: ___
Di meets requirement	14A	Measured: ___; required: ___
Phase selectivity 95%	14C	Steam split: ___; liquid split: ___
U_condenser = 8 (passive)	15A	Measured: ___ ± ___
Makeup trends to zero	15B	Trend: ___; fugitives: ___
Purge restores baseline	15C	Tolerable air fraction: ___
Freeze protection (cold sites)	15D	Verified / N/A: ___
Chemistry in envelope	15E	Solids trend: ___

Fourteen assumptions entered TRL 3. Fourteen measurements leave it — or the gate doesn't open.

15.8 What this chapter proved — and what TRL 3 established

- Five specified night-loop experiments: condenser U (15A), full-cycle inventory closure (15B), non-condensable purge (15C), freeze protection (15D), water chemistry (15E).
- "No leaking" as a measured mass balance: inventory closes within uncertainty, makeup trends to zero, leaks found and eliminated — not averaged.

- The complete TRL 3 ledger: every assumption from the energy balance now has a measurement procedure and a recorded result.
- TRL 3's exit standard: critical functions demonstrated by measurement, safety margins demonstrated (not assumed), failure branches executed where triggered.

TRL 3 is complete. The critical functions are proven: the flux measured (Ch. 12), the steam proven safe (Ch. 13), the valve tested against its hypothesis (Ch. 14), the night closed by accounting (Ch. 15) — all chained by the energy balance (Ch. 11). TRL 4 builds the breadboard: the first integrated laboratory assembly, where the proven functions meet for the first time.

GATE 3 — Exit evaluation

The reader may not proceed to TRL 4 until every answer below is theirs.

From Chapter 11:

1 Reproduce the balance chain and state the coverage result with its honest reading. Why is the fan 8× the pump in the parasitics?

From Chapter 12:

1 Your 12A measures η_optical = 0.66 ± 0.03. Recompute the net electric and state whether the concept survives, with the levers.

From Chapter 13:

1 State the CHF safety rule as a measured criterion. Your margin ratio at the cloud-edge transient is 0.62. Verdict and next step?

From Chapter 14:

1 Restate the valve hypothesis falsifiably. Your 14C shows 12% liquid misrouting in transition. Walk the fallback branch's four steps.

From Chapter 15:

1. Your 15B campaign shows makeup stabilizing at 40 mL/day, not trending to zero. Is this "no leaking"? State the exact procedure the chapter requires.
2. Explain why 15E is "the cheapest chapter to get wrong early and the most expensive to get wrong late."

Gate 3 exit criteria: all six answered; the reader's ledger (§15.7) filled with their measurements; every safety margin demonstrated rather than assumed; every falsified hypothesis processed through its specified branch. Then TRL 4 — the breadboard, where the proven functions meet.

Sources and honesty notes

- All experimental structures in this chapter are specified by the author; metrology traceability (ISO 9060, IAPWS-IF97, NREL SPA) comes from the cited standards, not from this book.
- The "tens of cycles" campaign minimums and the 10% endurance-drift threshold are the author's specified test depths for the reference design — the reader justifies their campaign depth against their reliability targets (Chapter 28).
- No night-loop test data for this system exists yet; the ledger's right column is empty until the reader's experiments fill it.
- The freeze experiment is climate-gated by design: sites without freeze risk record N/A with the climate data attached, rather than performing a ritual.

TRL 4

Laboratory Validation

Chapters 16–18

Chapter 16

Building the Lab Loop

TRL 4 · Laboratory breadboard. This chapter specifies the first integrated assembly: what gets built, in what order, and how every interface gets its sensor.

16.1 What TRL 4 is — and isn't

TRL 3 proved the functions separately. TRL 4 introduces them to each other: the first integrated loop, assembled in a laboratory, where the lens feeds the receiver, the receiver feeds the valve, the valve feeds the condenser, and the instrumentation watches every interface from Chapter 6's list.

What the breadboard is: a reduced-scale, heavily instrumented integration test article that proves the *system behavior* — control logic, interface dynamics, start-up/shutdown sequencing, the day/night transition.

What it isn't: a efficiency demonstration at scale. The breadboard's job is to find the integration failures — the ones that live at interfaces, per Chapter 6 — while they are cheap. Full-scale performance is TRL 5 and 6.

16.2 The scale decision

The reference breadboard: **2 m² lens aperture** (1/20th of the home tier's 40 m²), with all other subsystems scaled to match the resulting ~5.5 kW thermal design-day input. The scale is chosen by three rules:

1 **The valve geometry stays representative.** Diodicity is geometry- and Reynolds-dependent — a valve

scaled arbitrarily teaches nothing. The breadboard's valve sections use the *actual* channel geometry from Chapter 14's frozen design, at matched Reynolds numbers (flow rates scaled to hold Re). This is non-negotiable: scale the loop around the valve, not the valve around the loop.

2 **The optics stay real.** A 2 m² Fresnel lens with a proportional heliostat set — real sun, real tracking, real flux mapping per Chapter 12's methods. No solar simulator: the optical integration (tracker + lens + receiver alignment) is exactly what TRL 4 must test. (An electric receiver-heater *bypass* is included for thermal-only testing — decoupling optics from thermals during commissioning, per §16.5.)

3 **Everything else scales by the energy.** Receiver area, boiler volume, condenser area, pump head — all recomputed through Chapter 8's seven-step chain at the 2 m² aperture.

16.3 The lab bill of materials

Adapted from Chapter 7's BOM, annotated for the breadboard:

- **Optical:** 2 m² parabolic-profile Fresnel lens; 4–6 heliostat mirrors (proportional boost field); dual-axis tracker (same drive class as the design, so 12C's error data transfers); sun sensor + controller.
- **Receiver/boiler:** instrumented receiver-boiler test article from Chapter 13 (reused — TRL 3 hardware graduates to TRL 4); relief valves (2×); full thermocouple rake.

- **Power block:** *turbine simulator* — a calibrated orifice/throttle valve standing in for the turbine. Honest reason: the breadboard proves the *loop integration*, not the expander; the expander gets its own test program (Chapter 20) and joins at TRL 5. The simulator presents the design back-pressure and mass flow so the loop dynamics are real.
- **Condenser:** scaled finned-panel array (passive + instrumented fan for the forced-convection curve); purge valve; air-injection point (from 15C).
- **Flow control:** the frozen valve-section geometry from 14A (two legs); feed pump (positive displacement, instrumented); isolation/drain valves; full-bore piping sized by Chapter 4's velocity rules.
- **Electrical/controls:** the control system running the actual day-cycle state machine from Chapter 6's narrative — dawn charge, track, boil, generate (simulated), transition, condense, account. This is the breadboard's real product: *the control logic, proven*.
- **Water/chemistry:** fill station with treated water; sampling points; blowdown valve + meter; load-cell inventory (from 15B).
- **Instrumentation (the breadboard's superpower):** every Chapter 6 interface gets sensors — flux (pyranometer + target camera), receiver in/out states, valve-leg pressures/flows/void fraction, condenser in/out, pump head, ambient weather. The lab loop is *more* instrumented than the final product will ever be; the sensors that prove the integration are removed (or kept as diagnostics) later.

16.4 The build sequence

Order matters — pressure boundary before optics, always:

1. **Pressure boundary:** assemble boiler, piping, condenser, valves. Hydrotest to code. Install relief valves; bench-test them.
2. **Flow verification (cold):** pump water through the loop; verify valve-section orientation, check-valve fallback compatibility, drain-down completeness, instrumentation readings against known flows.
3. **Thermal (electric):** power the receiver-heater bypass; bring the loop to temperature and pressure *without* the sun. Verify the control state machine through a full simulated day/night cycle. Find the leaks, the air traps, the sensor errors — with a heater switch, not a star.
4. **Optical integration:** install lens, mirrors, tracker. Run Chapter 12's flux mapping on the actual receiver. Align, calibrate, verify the error budget end-to-end.
5. **First solar day:** the loop runs the full Chapter 6 narrative on real sun — the integration's first breath. Heavily attended, heavily logged.
6. **Commissioning campaign:** repeated day/night cycles per the test plan of Chapter 17.

Skipping step 3 is how builders learn about leaks at 4 MPa instead of at atmospheric pressure. The sequence is the safety protocol in build form.

16.5 What the breadboard proves — and what it explicitly doesn't

Proves: the control state machine across the full day cycle; interface dynamics (does the flux transient upset the boiler? does the evening transition oscillate?); the valve sections in a real loop (not a test rig); the inventory accounting in operation; the commissioning procedure itself.

Doesn't prove: full-scale efficiency (that's TRL 5/6 with the real expander); economics (Chapter 24); long-term reliability (Chapter 28); any site-specific performance (Chapter 19). The

breadboard chapter of the book is honest about being a laboratory chapter.

16.6 What this chapter proved

- TRL 4 defined: first integrated laboratory assembly; integration behavior proven, scale performance deferred.
- The scale decision: 2 m^2 aperture, valve geometry held representative at matched Re, real sun with an electric thermal bypass for commissioning.
- The lab BOM: Chapter 7 adapted, turbine honestly replaced by a calibrated simulator, instrumentation exceeding the final product.
- The six-step build sequence: pressure first, optics last, electric-thermal commissioning between.
- The breadboard's real product named: the proven control logic and the commissioning procedure.

Chapter 17 runs the loop: test procedures, measurement methods, and the efficiency testing the breadboard can honestly perform.

Self-evaluation — Gate 4, part 1

1 State TRL 4's definition in one sentence. What does the breadboard prove, and what does it explicitly defer?

2 Why must the valve geometry stay full-representative while everything else scales? What goes wrong if the valve is scaled geometrically with the loop?

3 Justify the turbine simulator: what does it preserve about the loop dynamics, and what would be dishonest about claiming expander performance from it?

4 Walk the build sequence: what failure does step 3 catch that step 5 would catch expensively?

5 A builder proposes skipping the electric bypass to "save time." State the objection in terms of leak-discovery pressure.

Sources and honesty notes

- The 2 m² breadboard aperture is the author's reference scale for the laboratory article — the reader justifies their scale against their lab's constraints and the valve-representativeness rule.
- Reynolds-number matching for the valve sections follows standard fluid-dynamic similitude; the reader verifies their scaled flow rates hold the operating Re range.
- The turbine simulator is a calibrated orifice — its calibration, not its presence, makes the loop dynamics representative; the calibration procedure is part of the commissioning campaign.
- No breadboard has been built for this system yet; this chapter specifies the article to build, not one that exists.

Chapter 17

Running the Loop

TRL 4 · Laboratory breadboard. This chapter specifies the test campaign: the procedures, the measurement methods, and the efficiency testing the breadboard can honestly perform.

17.1 The campaign structure

The breadboard's testing runs in four phases, each with entry criteria — no phase starts until the previous one's exit criteria are met:

1. **Functional checkout** (days): every sensor reads, every actuator moves, the state machine steps through its states on command.
2. **Control verification** (1–2 weeks): the day-cycle logic proven, including fault injection.
3. **Performance mapping** (3–4 weeks): steady-state efficiency at mapped operating points.
4. **Endurance cycling** (4+ weeks): repeated day/night cycles; inventory accounting; drift watch.

The discipline: *no data is taken on a loop whose control isn't verified, and no endurance claim is made on a loop whose performance isn't mapped.* Each phase's exit criteria are stated below.

17.2 Test 17A — Control verification: the state machine under fault

Objective: prove the Chapter 6 day-cycle logic handles reality, not just sunshine.

Procedure:

- Step the controller through every state — dawn charge, track, boil, generate (simulated), transition, condense, account, night idle — verifying each transition's guard conditions.
- **Fault injection**, each with the required safe response:
- Cloud passage (flux collapse): boiler holds pressure, loop doesn't trip on a transient it was designed for.
- Sensor failure (each critical sensor, one at a time): controller detects, falls back to redundant channel or safe state — never to an unmonitored operating point.
- Pump trip: boiler safes before dryout; the sequence is timed and verified.
- Tracker fault: lens parks to the safe (defocused) position; verified by flux measurement, not by command echo.
- Power loss: fail-safe positions assumed (valves, tracker brake); restart procedure verified.

Pass criteria: every fault produces its specified safe response; no fault produces an unmonitored operating point; the fault log from the campaign is complete.

If it fails: control logic is revised and the fault re-injected. A controller that hasn't been fault-injected hasn't been tested — it has been *demonstrated*, which is a different and lesser thing.

17.3 Test 17B — Performance mapping: the honest efficiency

Objective: measure the breadboard's thermal efficiency across its operating envelope.

Measured quantity: η_thermal = ṁ(h_out − h_in)_receiver / (DNI × A_aperture) — beam in, heat captured out. *Not* cycle efficiency: the breadboard has a turbine simulator, not a turbine, and the book doesn't pretend otherwise. The cycle efficiency comes from Chapter 20's expander tests; the *product* of the two — validated by the system model — is the predicted system efficiency.

Procedure:

1. At each operating point (flux level × ambient condition bins), hold steady state; record all interface measurements.
2. Map η_thermal vs. flux and vs. ambient; compare against the Chapter 11 balance's stage efficiencies (0.75 × 0.67 = 0.50 predicted).
3. Every point carries its uncertainty budget — instrument calibrations, not spec sheets (Chapter 12's discipline).

Pass criteria: mapped η_thermal with uncertainties across the envelope; agreement-or-diagnosed-disagreement with the Chapter 11 prediction at every point.

If it fails: disagreement is diagnosed by stage — the interface instrumentation from Chapter 16 exists precisely so a shortfall can be *localized* (optics? receiver? piping losses?) rather than merely observed. A mapped shortfall with a localized cause is the most valuable data TRL 4 produces.

17.4 Test 17C — Transition dynamics: the evening, recorded

Objective: verify the day/night transition in the integrated loop — Chapter 14's rig result, now with real boiler, real condenser, real control.

Procedure:

1. Run the evening transition under instrumented control; record valve-leg flows, void fractions, pressures, and the phase split continuously through the ramp.
2. Repeat across the envelope: fast ramp (clear-evening), slow ramp (overcast evening), and interrupted ramp (evening cloud followed by clearing — the awkward case).
3. Verify against the 14C pass criteria (95% selectivity) in the integrated context.

Pass criteria: selectivity criteria met in the loop, not just the rig; no transition instability in any ramp case; the control guards (from 17A) engage correctly if a transition goes outside envelope.

If it fails: the integrated loop is where rig simplifications die — piping volumes, thermal masses, and real condenser dynamics all participate. Failure here revises the operating procedure (ramp-rate limits are a legitimate control output) or, if the physics disagrees, triggers Chapter 14's fallback branch with integrated-loop data.

17.5 Test 17D — Inventory in operation

Objective: Chapter 15's mass balance, running on the integrated breadboard across the endurance campaign.

Procedure: the 15B accounting, executed every cycle of Phase 4 — inventory logged at dawn, makeup/blowdown/samples metered, fugitives investigated the day they appear.

Pass criteria: makeup trends to zero across the campaign; the accounting closes within uncertainty every cycle.

If it fails: same rule as 15B — a leak gets found and fixed, not averaged. The breadboard's plumbing is the design's plumbing at small scale; a leak here is a leak in the drawings.

17.6 Test 17E — Parasitics, metered

Objective: replace every Chapter 11 parasitic estimate with a measurement.

Procedure: dedicated power metering on each parasitic — feed pump, tracker drives, controls, instrumentation, fan (at each setting). Logged across the campaign; reported as daily kWh with the operating conditions attached.

Pass criteria: a metered parasitic budget that either confirms or revises Chapter 11's balance. (The fan measurement, in particular, prices Chapter 5's central trade with data.)

If it fails: "fails" here means the parasitics exceed the balance's allowance — which revises the balance and possibly the design (a tracker drive that eats 3% of output gets re-specified). Small loads, honestly metered, beat large loads approximately estimated.

17.7 Measurement discipline

Every test in this chapter runs under three rules:

1. **Calibrate, don't assume.** Every instrument's calibration is current and traceable; the uncertainty budget is computed, not inherited from a datasheet.
2. **Two methods where it matters.** The Chapter 12 calorimetric cross-check is the pattern: critical quantities (flux, heat, flow) get independent verification.
3. **Log everything, publish the ugly.** The campaign log records the failed runs, the aborted transitions, the sensor that lied for a week. Chapter 18 exists because of this rule.

17.8 What this chapter proved

- The four-phase campaign with entry/exit criteria: checkout, control verification, performance mapping, endurance cycling.

- Five specified tests: fault-injected control verification (17A), honest thermal-efficiency mapping (17B), integrated transition dynamics (17C), operational inventory accounting (17D), metered parasitics (17E).
- The efficiency honesty: thermal efficiency measured; cycle efficiency deferred to Chapter 20; the product validated by the system model.
- The measurement discipline: calibration, cross-checks, and the ugly log.

Chapter 18 logs the failures: what breaks, what it teaches, and how the models get revised with data.

Self-evaluation — Gate 4, part 2

1. State the four campaign phases with their entry/exit logic. Why can't performance mapping precede control verification?
2. A cloud passage collapses flux mid-boil. Walk the required controller response and how 17A verifies it.
3. Your 17B maps $\eta_{thermal} = 0.44 \pm 0.03$ against a predicted 0.50. Using the interface instrumentation, outline the localization procedure — what do you check first, second, third?
4. Explain the efficiency honesty: what does the breadboard's $\eta_{thermal}$ license the reader to claim, and what would be dishonest?
5. The metered tracker parasitic is 4× the Chapter 11 estimate. State the exact consequences per §17.6.

Sources and honesty notes

- The four-phase campaign structure and fault-injection list are the author's specified test program for the reference breadboard; the reader scales the campaign depth to their reliability targets.

- η_thermal as defined here is a thermal collection efficiency, not a system efficiency — the book maintains this distinction rigorously because conflating them is the most common dishonesty in solar-thermal reporting.
- No breadboard test data for this system exists yet; all pass criteria await the reader's campaign.
- The "log the ugly" rule is a procedural requirement, not a suggestion — Chapter 18's failure log is only as good as Phase 1–4's honesty.

Chapter 18

Failure Logging

TRL 4 · Laboratory breadboard. This chapter makes failure a instrument: what breaks, what each break teaches, and how the models get revised with data. It closes TRL 4.

18.1 The log as an instrument

Every first integration fails. The difference between a failed project and a successful one isn't the failure count — it's whether each failure was *converted* into a design change, a procedure, or a revised number. The failure log is that conversion mechanism, and it has a required structure:

Field	Content
Date / cycle #	When in the campaign
Subsystem / interface	Which Chapter 6 interface
Symptom	What was observed, with data attached
Root cause	Found by investigation, not by first guess
Class	Build defect / design error / physics disagreement (§18.3)
Fix	Design change, procedure change, or guard added
Model revision	Which number in which chapter changed, old new

A log entry without a model revision is an anecdote. A log entry *with* one is engineering. TRL 4's deliverable isn't a working breadboard — it's a working breadboard *plus the log that made it work*.

18.2 The expected failure taxonomy

First integrations fail in predictable places. The taxonomy below isn't pessimism — it's the checklist the builder works through, so each failure arrives as a recognized event with a known diagnostic path rather than a crisis:

Seals and joints. The most common first-integration failure in any pressurized loop. Symptom: the 17D inventory won't close. Diagnostic: pressure-decay per isolated section, then tracer. Teaches: which joint designs survive thermal cycling (13E's data, now at loop scale). Model revision: the fugitive allowance in the inventory procedure, and possibly the joint spec in the build drawings.

Sensor lies. A thermocouple drifts; a flowmeter reads low because of a partially blocked tap. Symptom: the energy balance won't close, or two methods disagree (Chapter 12's cross-check discipline exists for exactly this). Diagnostic: redundant channels, calibration checks against references. Teaches: which measurements the control *actually* depends on — those get the redundant channels permanently. Model revision: the instrumentation plan, and the uncertainty budgets.

Air traps. The loop has a high point the drawings didn't flag; non-condensables accumulate where the purge can't reach. Symptom: condenser performance degrades over days (15C's signature, now unplanned). Diagnostic: thermal imaging, sequential venting. Teaches: the as-built geometry differs from the as-drawn — the purge procedure gets the new vent point. Model revision: the piping layout and the purge procedure.

Thermal expansion binding. Hot piping grows; a support that was fine cold binds hot, stressing a joint. Symptom: a leak that appears only at temperature. Diagnostic: inspection hot vs. cold, expansion calculations checked against measured

movement. Teaches: the support design. Model revision: the mechanical drawings.

Control timing. The evening transition oscillates because the controller's ramp rate fights the condenser's thermal lag. Symptom: 17C instability. Diagnostic: the logged time series — the data shows which lag the controller didn't model. Teaches: the control model. Model revision: ramp-rate limits, feedforward terms, the state machine itself.

Fouling and chemistry. Weeks in, heat-transfer coefficients drift — the slow failure from 15E arriving on schedule. Symptom: performance decay with clean sensors and tight joints. Diagnostic: water analysis, inspection. Teaches: the chemistry doctrine's real-world adequacy. Model revision: Chapter 21's treatment spec.

Tracker and optical drift. Structure settles, mirrors need re-aiming, soiling accumulates per 12D's curve. Symptom: flux map drift. These aren't failures — they're the *operating reality* the maintenance doctrine (Chapter 26) is built from. Model revision: the commissioning and maintenance procedures.

18.3 Three classes, three responses

Build defects (a bad weld, a miswired sensor): fix the article, note the inspection gap that let it through, revise the build procedure so the *next* article doesn't repeat it. No model changes — the design was right; the hands weren't.

Design errors (the support that binds, the unvented high point): fix the design, revise the drawings and the affected chapter's numbers. The log's model-revision field earns its keep here.

Physics disagreements (the boiling curve that doesn't match Rohsenow, the diodicity below prediction): the most valuable class. The procedure: re-verify the measurement first (sensor lies are commoner than physics surprises), then characterize the disagreement fully, then revise the model — and flag every downstream chapter that used the old number. A physics disagreement that survives re-measurement is a

finding, publishable in its own right (Chapter 31's dissemination includes negative and anomalous results).

18.4 The revision procedure

When a model revision is warranted, it propagates by rule:

1. **Record:** old value new value, with the measurement and its uncertainty that forced the change.
2. **Trace:** every chapter that used the old value gets flagged — the energy balance (Ch. 11), the sizing chain (Ch. 8/9), the economics inputs (Ch. 24's future data).
3. **Recompute:** the dependent quantities are re-run with the new value. If the recomputation breaks a downstream requirement (coverage drops below usefulness, a margin goes negative), that's a design iteration trigger, not a footnote.
4. **Version:** the book's numbers carry their revision history. A reader who builds from revision C of the balance isn't blindsided by revision B's optimism.

This is the unglamorous machinery that makes "prove what is possible with actual data" operational. Data without propagation is trivia; propagation without versioning is confusion.

18.5 The "no surprises twice" rule

Every logged failure gets a guard — one of three kinds:

- **A sensor** that watches for the symptom (the leak detector on the joint class that leaked).
- **A procedure** that prevents the cause (the venting step added to commissioning).
- **A design change** that removes the possibility (the expansion loop that replaces the binding support).

The rule is literal: the same failure mode may not recur unguarded. The second occurrence of a logged failure is a process failure, and it gets its own log entry — against the

process, not the hardware.

18.6 What this chapter proved — and what TRL 4 established

- The failure log's required structure: seven fields, with the model-revision field mandatory.
- The expected failure taxonomy: seals, sensor lies, air traps, expansion binding, control timing, fouling, optical drift — each with its diagnostic path and teaching.
- Three failure classes with three distinct responses: build defects, design errors, physics disagreements.
- The revision procedure: record, trace, recompute, version — data propagated to every dependent number.
- The "no surprises twice" rule: every failure gets a sensor, a procedure, or a design change.

TRL 4 is complete. The breadboard was specified (Ch. 16), run through its campaign (Ch. 17), and its failures converted to design knowledge (Ch. 18). The integration works in the laboratory — with the log to prove what "works" cost. TRL 5 takes it into the relevant environment: the full-scale home-tier article, outdoors, on a real site.

GATE 4 — Exit evaluation

The reader may not proceed to TRL 5 until every answer below is theirs.

From Chapter 16:

1 State the breadboard's scale decision and the valve-representativeness rule. Why does the turbine simulator preserve loop-dynamics honesty?

From Chapter 17:

1. Your 17B maps η_thermal = 0.44 ± 0.03 vs. 0.50 predicted. Walk the localization procedure using the interface instrumentation.
2. Explain the efficiency honesty: what the breadboard licenses the reader to claim.

From Chapter 18:

1. A leak appears only at operating temperature. Classify it, diagnose it per the taxonomy, and state the model revision and the guard.
2. Your measured diodicity is 30% below the Chapter 4 prediction, and re-measurement confirms it. Classify, then walk the four-step revision procedure — including which chapters get flagged.
3. State the "no surprises twice" rule. A logged air-trap failure recurs six months later. What gets the log entry?

Gate 4 exit criteria: all six answered; the reader holds a filled failure log with model revisions propagated and versioned; every interface from Chapter 6 has been exercised in the integrated loop; the control logic is fault-injection proven. Then TRL 5 — the relevant environment.

Sources and honesty notes

- The failure taxonomy is the author's synthesis of first-integration experience patterns in thermal-fluid test programs; the reader's log will contain their own entries, and the taxonomy is a checklist, not a prediction.
- The three failure classes and the revision procedure are the author's specified engineering process for this program — presented as process requirements, not as empirical claims.

- No breadboard failure data for this system exists yet; the log is empty until the reader's campaign fills it. An empty log at the end of TRL 4 means the campaign wasn't run, not that nothing broke.

TRL 5

Relevant Environment

Chapters 19–21

Chapter 19

Taking It Outside

TRL 5 · Relevant-environment validation. This chapter specifies the outdoor validation campaign: real sun, real weather, full scale — and the honest list of what the lab didn't catch.

19.1 What TRL 5 is

TRL 4 proved the integration in the laboratory. TRL 5 proves it in the **relevant environment**: outdoors, at full home-tier scale, under the sky it was designed for. The relevant environment isn't "outside" as a location — it's a set of conditions the lab couldn't provide:

- **Real solar variability:** cloud transients, hazy days, the DNI distribution of an actual site — not the lab's clear-sky bias.
- **Real thermal cycling at scale:** the full-size boiler and full-size structure breathing through real days and nights.
- **Real weather:** wind on the tracker, dust on the lens, rain, the design hot night, the freeze event (for cold sites).
- **Real time:** weeks to months, so soiling, fouling, and drift accumulate the way they will in service.

The validation article: the full-scale home-tier system from Chapter 8's sizing, built from TRL 4's revised drawings, running TRL 4's proven control logic. Nothing about the article is

exploratory anymore — the exploration happened in the lab. TRL 5 is where the design meets the world it claimed to serve.

19.2 Site selection: choosing the relevant environment

The site is part of the experiment. Requirements:

1. **Representative solar resource** for the target market — the validation means little at a site sunnier than any customer will see. The site's TMY data is published with the results.
2. **The design hot night, reachable.** The site must experience condenser design conditions during the campaign — or the campaign waits for the season that provides them.
3. **Instrumented and accessible.** Full weather station (DNI, GHI, ambient, wind, humidity, soiling station per 12D); daily access for the test team; secure enough for months of hardware.
4. **At least one harshness the lab lacked.** Every site has one — dust, wind, humidity, freeze. The site is chosen partly *for* its harshness, because TRL 5's job is to find what the lab didn't catch.

19.3 Test 19A — Real-sun performance vs. the lab

Objective: quantify the environment's tax — the difference between laboratory-mapped performance and outdoor reality.

Procedure:

1. Run the Chapter 17 performance-mapping procedure on the full-scale article across the site's actual DNI distribution.
2. Compare measured η_thermal against the lab's 17B map, binned by the same operating points.

3 Attribute every systematic difference: wind-driven convection on the receiver, real soiling state, tracker performance under wind load, ambient effects on the condenser.

Pass criteria: the outdoor performance map with uncertainties; every lab-vs.-field difference attributed to a physical cause; the system model updated with the field corrections.

If it fails: "fails" here means the field shortfall has no attributable cause — an unattributed gap is a model error, and it blocks TRL 5 until it's found. Attributed gaps just revise the numbers (Chapter 18's revision procedure, now at field scale).

19.4 Test 19B — Weather: the transients the lab simulated

Objective: verify the control and the hardware against real weather events.

Procedure — the event catalog, each ridden out under full instrumentation:

- **Cloud transients:** the fast passages 17A fault-injected, now real. Verify the boiler rides through without tripping and without quality violations (13D's limits, at scale).
- **Wind:** tracker holds pointing under gusts (12C's budget, now with wind load); structure verified against the design loads; the wind-speed operating envelope confirmed — including the stow condition and its trigger.
- **Rain:** soiling recovery measured (12D's rain-recovery fraction, at the site); drainage verified — no water pooling on optics or in enclosures.
- **Dust events:** the soiling spike ridden out; the cleaning trigger from 12D exercised for real.

- **Hail (where applicable):** the lens and mirrors survive the design hail event, or the stow/cover procedure does. Verified by the event or by representative testing — not by assumption.

Pass criteria: every cataloged event ridden out within the operating envelope, or the envelope honestly revised (a lower wind-stow threshold is a revision, not a failure).

If it fails: weather failures revise the operating envelope or the hardware — storm-rated mounts, faster stow, hail covers. The relevant environment doesn't negotiate; the design adapts, and the adaptation is logged.

19.5 Test 19C — The design hot night, for real

Objective: verify condenser performance at the actual design condition — the sizing case from Chapter 5, now measured rather than assumed.

Procedure: during the campaign's hottest nights, run the full night-rejection duty and measure U against the 15A curve. If the site's campaign doesn't include the design hot night, the test waits — or a second site-season covers it.

Pass criteria: the condenser meets its duty at the design hot night with the Chapter 5 margin intact.

If it fails: the condenser gets the Chapter 5 treatment with measured inputs — more area, forced convection, building integration. Hot nights are when the machine is needed most (Chapter 3's rule); the test that verifies them doesn't get skipped for schedule.

19.6 Test 19D — Seasons: the annual energy, measured

Objective: measure performance across seasons and validate the annual energy prediction.

Procedure:

1. Operate through a full seasonal cycle (minimum: the campaign spans the site's high-sun and low-sun seasons).
2. Log daily energy in, daily electricity out (thermal, until the expander joins in Chapter 20 — honest as ever), daily heat rejected, availability (uptime fraction with causes coded).
3. Compare the measured annual projection against the Chapter 11 balance run on the site's TMY data.

Pass criteria: the annual energy model validated within its uncertainty; availability measured with every downtime cause coded and addressed.

If it fails: seasonal shortfalls get the attribution treatment (19A's method, extended in time). A machine that works in spring and stalls in summer has a condenser problem; one that works in summer and stalls in winter has a freeze or low-sun problem. The seasons diagnose by themselves — if the logging is honest.

19.7 What the lab didn't catch — the honest list

TRL 5 exists because laboratories are kind. The field is not. The standing list of lab-blindness, to be checked off or added to:

- Wind-induced tracking error and structural dynamics.
- Real soiling rates, compositions, and cleaning realities.
- Biological ingress: insects, nesting, vegetation — the maintenance doctrine's unglamorous chapter.
- UV and weather degradation of the lens and coatings over months.
- The design hot night the lab couldn't stage.

- Human factors: the procedures that work with engineers present and fail with operators alone (Chapter 26's training doctrine is built from these observations).

Every item either gets a test, a design change, or a written reason it doesn't apply. "The lab didn't test it and neither did we" is not a status the gate accepts.

19.8 What this chapter proved

- TRL 5 defined: full-scale home-tier article, validated in the relevant environment — real sun, real weather, real time.
- Site selection as experimental design: representative, instrumented, harsh on purpose.
- Five specified tests: real-sun vs. lab (19A), the weather event catalog (19B), the design hot night (19C), seasonal annual-energy validation (19D), and the lab-blindness checklist (19.7).
- The attribution discipline at field scale: every lab-vs.-field gap gets a physical cause or blocks the gate.

Chapter 20 answers the small-turbine question: honest numbers at 1–50 kW, expander options compared — the tier's economic lever, tested.

Self-evaluation — Gate 5, part 1

1 Define the relevant environment as a set of conditions, not a location. Why is "outside" insufficient as a definition?

2 Your 19A comparison shows field η_thermal 8% below lab at matched operating points. Walk the attribution procedure — what do you check, in what order?

3 A dust storm hits mid-campaign. Which test covers it, what gets measured, and what decision does the data feed?

4 The campaign's hottest night is 4 K cooler than the design hot night. Can 19C pass? State the options.

5 Name three items from the lab-blindness list and, for each, whether it's a test, a design change, or a procedure — with the reasoning.

Sources and honesty notes

- The site-selection requirements and event catalog are the author's specified validation program; the reader's site and its harshness profile determine the specific catalog.
- TMY data (NREL and equivalents) is the standard solar-resource reference; the campaign's measured resource is reported alongside the TMY for comparability.
- The "minimum: high-sun and low-sun seasons" campaign depth is the author's specified minimum for annual-energy validation — full-year operation strengthens the claim and Chapter 29 requires it.
- No field validation data for this system exists yet; every test in this chapter awaits the reader's campaign.

Chapter 20

The Small-Turbine Question

TRL 5 · Relevant-environment validation. This chapter asks the hardest honest question in the book: what does steam expansion actually achieve at 1–50 kW — and what does the answer do to the energy balance?

20.1 Why this chapter exists

Every number in Chapter 11's energy balance downstream of the boiler depends on one assumption: the Rankine cycle converts heat to shaft work at 27%. Chapter 3 derived that figure from established cycle thermodynamics — at utility scale. But small expanders are not small versions of big turbines. They are a different engineering regime, with different losses, and the book's honesty contract requires the assumption to be tested at the scale the tiers actually use: 1–50 kW shaft.

This is the chapter where the book risks its own economics. If the measured expander efficiency is 15% instead of 27%, the balance changes, the sizing changes, and the reader learns it here — from the book's own test program — not from a skeptic's blog post later. Theorems, not theories: the number gets measured.

20.2 Why small expansion is hard

Three scaling penalties, all physics, all unavoidable:

1. **Reynolds number.** Small blades run at low Reynolds numbers; boundary layers thicken; profile losses rise. A blade row that achieves 90% stage efficiency at utility scale achieves materially less at 50 kW and less again at 5 kW.
2. **Tip clearance.** Clearance gaps don't scale down with the machine — manufacturing tolerances have a floor. Leakage flow becomes a larger fraction of total flow as the machine shrinks.
3. **Heat loss.** Surface-to-volume ratio rises as machines shrink; more of the steam's energy leaks through the casing as heat instead of leaving as shaft work. Small machines also spend a larger fraction of their output on their own friction and auxiliaries.

None of this is news to turbomachinery engineers. It is news to readers who assumed the 27% traveled with the cycle. It doesn't — it travels with the *scale*.

20.3 The candidate expanders

Five options, evaluated on the same terms — measured isentropic efficiency at the design steam conditions, maturity, cost, maintenance:

Single-stage impulse steam turbine. The classical choice. Mature, simple, tolerant of wet steam relative to multi-stage machines. At 5–50 kW, realistic isentropic efficiencies run well below utility practice — the reader measures their unit; the book doesn't quote a brochure. Strengths: simplicity, high-speed direct drive possible, no oil in the steam path with proper design.

Screw expander. Positive displacement; handles wet steam and two-phase flow gracefully — a genuine advantage in a cycle that may flirt with the saturation dome. Lower speed, noisier, oil management needed (oil-free variants exist at a price). Mature in geothermal and waste-heat service at hundreds of kW; the 1–50 kW range is thinner but real.

Scroll expander. Quiet, few moving parts, tolerant of liquid slugs — attractive for the home tier where noise matters and the steam may be wet. Efficiency is modest; the technology is mature from the compressor world run in reverse, but steam-scroll operating history is shorter than air-scroll.

Piston (reciprocating) steam engine. The oldest option and not an embarrassment: excellent part-load behavior, tolerant of wet steam, maintainable by any mechanic. Low speed, vibration, and the maintenance burden of rings and valves. For a building that already maintains machinery, an honest candidate.

Tesla turbine (boundary-layer turbine). It belongs in this book — Tesla's name is on the valve, and readers will ask. The honest evaluation: elegant, simple, nearly unkillable — and historically poor measured efficiency in most implementations, sensitive to disc spacing and operating point. The book tests it like everything else. If it wins on the test stand, it wins. If it doesn't, the data says so, and the book's Tesla content remains the valve — which was always the actual contribution.

20.4 Test 20A — The expander test program

Objective: measure isentropic efficiency for each candidate at the design steam conditions, on the same test stand, by the same procedure.

Procedure:

1. Each candidate runs against the steam conditions from Chapter 13's boiler spec (pressure, temperature, mass flow range) — the same inlet, the fair fight.
2. Map isentropic efficiency vs. pressure ratio and vs. mass flow (part-load matters — the sun doesn't do full load all day).
3. Measure durability-relevant quantities: vibration, bearing temperatures, seal leakage, oil carryover (where applicable), noise at the property line (home tier).

4 Run the endurance leg: hundreds of start-stop cycles — the expander sees a daily cycle, not baseload; start-stop wear is its life.

Pass criteria: measured efficiency maps with uncertainties for every candidate; a ranked selection with the ranking justified by the system-level metrics (§20.5); the winner's endurance leg complete without disqualifying wear.

If it fails: "fails" here means *no* candidate meets the cycle-efficiency allowance the balance needs. The responses, in order: re-examine the cycle (higher pressure? superheat? — Chapter 3's tools), re-examine the tier sizing (bigger machine per unit — the efficiency scale curve is the reader's friend), and only then re-examine the concept. The book doesn't hide this branch — it's drawn on the map before anyone needs it.

20.5 The system-level selection metric

Expander efficiency alone doesn't pick the winner. The selection metric is system-level:

> ***Net daily electricity per unit of aperture, minus the maintenance burden in the operator's currency*** *— dollars for the commercial tiers, hours and attention for the home tier.*

A scroll expander at modest efficiency that runs ten years unattended beats a turbine at higher efficiency that needs quarterly bearing service — *for the home tier*. The corporate tier, with maintenance staff, may rank them oppositely. The book gives the method and the measured inputs; the tier's reality picks the winner.

20.6 Re-running the balance with measured numbers

This is the chapter's load-bearing calculation. Take the measured expander efficiency, combine it with the generator efficiency (measured, not 95%-assumed — Chapter 13's instrumentation discipline applies to the dynamometer), and re-run Chapter 11's balance:

- If the measured cycle efficiency lands near 27%: the balance stands, and the book's economics rest on measured ground.
- If it lands at 15–20%: the aperture grows, the tiers re-size (Chapters 8–9 re-run with the measured number), and the economics chapter (24) works with the truth. A system that needs 60 m^2 instead of 40 m^2 is still a system — it's just an honestly sized one.

The rule: **the balance is re-run with measured numbers the day they exist, and every chapter that consumed the assumption gets its revision entry** (Chapter 18's procedure). The 27% was an honest assumption clearly labeled; the measured number is better; the bookkeeping is the proof of the honesty.

20.7 The expander joins the loop

With the winner selected and mapped, the turbine simulator (Chapter 16) retires. The real expander integrates into the full-scale article, and the Chapter 17 campaign re-runs in abbreviated form — the control verification especially, because the expander brings its own dynamics (spin-up, trip, part-load) that the simulator never had. This is the loop becoming the machine: the last placeholder leaving the test program.

20.8 What this chapter proved

- The small-expander scaling penalties: Reynolds, tip clearance, heat loss — why 27% doesn't travel with the cycle.

- Five candidates evaluated on identical terms, including the Tesla turbine's honest assessment.
- Test 20A: the same-steam, same-stand, same-procedure efficiency program with part-load mapping and start-stop endurance.
- The system-level selection metric: net energy per aperture minus maintenance in the operator's currency.
- The balance re-run rule: measured numbers replace the assumption the day they exist, with revision entries in every consuming chapter.
- The simulator's retirement: the real expander integrates and the campaign re-verifies.

Chapter 21 covers the operating reality no brochure mentions: water chemistry and the sealed loop, week after week.

Self-evaluation — Gate 5, part 2

1 State the three small-scale expansion penalties. Which one is a manufacturing-tolerance floor, and why can't better engineering remove it?

2 Your 20A maps the impulse turbine at 18% isentropic where the balance assumed 27% cycle. Walk the exact consequences: which chapters get revision entries, and what recomputation runs first?

3 Justify the Tesla turbine's inclusion and its honest evaluation in one paragraph each.

4 Two candidates: turbine at higher efficiency needing quarterly service, scroll at lower efficiency running unattended for years. Pick the home-tier winner and the corporate-tier winner, using §20.5's metric.

5 The expander's start-stop endurance leg shows seal wear at 400 cycles. Classify per Chapter 18's three classes and state the response.

Sources and honesty notes

- The small-expander scaling penalties are established turbomachinery fundamentals; the reader verifies candidate performance on their own test stand — the book specifies the test, not the results.
- No expander efficiency figures are quoted as fact in this chapter; every number the reader uses must come from their 20A campaign. Brochure efficiencies are not measurements.
- The Tesla turbine assessment reflects the documented historical record of measured implementations; the book's test program gives any candidate — including Tesla's — the chance to win on data.
- The 27% cycle assumption from Chapter 3 was labeled as a utility-scale-derived assumption; this chapter is its scheduled reckoning.

Chapter 21

Water Chemistry and the Sealed Loop

TRL 5 · Relevant-environment validation. This chapter covers the operating reality no brochure mentions: what happens to water inside a sealed high-temperature loop, week after week — and the chemistry doctrine that keeps it honest.

21.1 The finite problem

Chapter 15 proved the loop doesn't leak. This chapter asks the next question: the water that stays inside — what does it *do* in there?

Here the sealed architecture pays its quiet dividend. An open cooling tower fights an infinite problem: fresh water in, concentrated chemistry out, endless treatment against an endless supply of contaminants. A sealed loop fights a *finite* problem: a fixed inventory, treated once, maintained against slow drift. The chemistry program for a sealed loop is smaller, cheaper, and more forgiving — one of the genuine operating advantages of this architecture, and the book states it plainly because it was earned by the design, not claimed for it.

Finite doesn't mean trivial. It means winnable.

21.2 The enemies

Five, in order of how fast they hurt:

1. **Dissolved oxygen.** Corrodes steel and copper, fastest at temperature. The sealed loop's first defense is simply being sealed — no continuous air ingress like an open system — but every maintenance opening, every makeup addition, is an oxygen event.
2. **Dissolved solids.** Calcium, magnesium, silica — scale on heat-transfer surfaces, where even a thin layer measurably degrades the coefficients Chapter 13 mapped. Scale is the slow tax on every assumption the balance makes about heat transfer.
3. **pH excursions.** Too low corrodes; too high with the wrong chemistry scales. The loop wants its pH in the band its materials were chosen for — and the band is a design decision, not a universal constant.
4. **Corrosion products.** Iron and copper oxides don't just damage surfaces — they migrate and foul the tight passages, including the valve sections whose geometry Chapter 14 froze. A fouled Tesla valve is a valve with different diodicity.
5. **Microbiological growth.** In the cool sections — the condenser, the makeup tank — where temperature permits it. Biofilms foul heat transfer and shelter corrosion. The hot sections self-sterilize; the cool ones need the doctrine.

21.3 The chemistry doctrine

Fill water: demineralized or deionized, to a specified conductivity — the loop starts clean because cleaning it later is harder. The fill spec is part of the commissioning procedure (Chapter 16's sequence gains a water-quality gate).

Oxygen control: mechanical deaeration at commissioning, chemical scavenger maintained at its residual, and — the procedural part — every maintenance opening followed by the re-deaeration procedure. Oxygen events get logged like failures (Chapter 18's log has a chemistry section).

pH and solids: maintained in the design band by the treatment program; verified by the sampling schedule, not by assumption.

The sampling schedule: weekly in the first months (the loop is young and the doctrine is unproven), relaxing to monthly once the trends are flat and understood. The analysis panel: conductivity, pH, dissolved oxygen, iron, copper, silica — each with its action limit. An action limit is a tripwire with a procedure attached, not a number in a table.

Blowdown: the controlled release that bounds dissolved solids — metered, logged, part of the 15B/17D inventory accounting. Blowdown is not a leak; it's a *measured* discharge, and the distinction matters to the "no leaking" claim.

21.4 Materials: the loop is a galvanic system

The loop contains multiple metals — at minimum, the boiler's pressure boundary, the condenser's extended surfaces, the piping, the valve sections. Mixed metals in hot water are a galvanic system, and the doctrine accounts for it:

- **Material selection** follows the chemistry, not the catalog: the pH band, the oxygen residual, and the temperature range pick the alloys. Carbon steel wants oxygen scavenging done right; copper alloys want the pH kept out of their corrosion range; stainless buys forgiveness at a price.
- **Dielectric isolation** where dissimilar metals meet, so the loop doesn't build batteries at its joints.
- **Corrosion coupons** in the loop from day one — weighed on schedule, the ground truth of the chemistry program. Coupons don't lie about corrosion rates; water analyses infer them.

The home tier gets the forgiving material set (fewer metals, wider chemistry tolerance, cheaper treatment). The corporate tier can afford the higher-performance set with its tighter doctrine — Chapter 9's scaling includes the chemistry program's scaling.

21.5 Test 21A — Chemistry endurance

Objective: prove the chemistry doctrine over operating time, not just on paper.

Procedure:

1. Run the full-scale article through months of operation on the sampling schedule; trend every analysis parameter.
2. Weigh the corrosion coupons on schedule; compare rates against the design allowances.
3. Trend the heat-transfer coefficients (receiver, condenser) against their Chapter 13/15 baselines — the earliest warning of fouling or scaling.
4. Execute at least one deliberate chemistry upset (a missed treatment interval, within safe bounds) and verify the recovery procedure works — the doctrine gets fault-injected like the controller did in 17A.

Pass criteria: parameters inside their bands for the campaign; coupon rates within allowances; no unexplained heat-transfer decay; the upset recovered by procedure.

If it fails: the doctrine tightens (more frequent sampling, different treatment, material upgrade) — or the design changes (a section re-specified in a more forgiving alloy). Chemistry failures are slow, which makes them cheap to catch and expensive to ignore. The trending exists so they're always caught.

21.6 The garden-hose failure mode

The honest failure mode for the home tier: the operator tops up the loop with untreated water — because the makeup procedure was unclear, or the treated supply ran out on a Sunday. The doctrine's answer is layered:

1. **Make the right thing easy:** the makeup station is plumbed to the treated supply; the procedure is one page.

2. **Make the wrong thing visible:** every makeup addition is metered and logged — the inventory accounting (17D) sees the garden hose.
3. **Make the system forgiving:** the chemistry has headroom for occasional small additions; the action limits catch a pattern before it becomes damage.

A chemistry doctrine the operator can't follow is a chemistry fiction. The home tier's doctrine is designed for the operator it will actually have.

21.7 What this chapter proved — and what TRL 5 established

- The sealed loop's finite chemistry problem — the architecture's quiet operating advantage, stated because it was earned.
- The five enemies: oxygen, solids, pH, corrosion products, biology — in order of how fast they hurt.
- The chemistry doctrine: fill spec, oxygen control, pH/solids bands, sampling schedule with action limits, metered blowdown.
- Materials as a galvanic system: selection follows chemistry, dielectric isolation, corrosion coupons as ground truth.
- Test 21A: chemistry endurance with trending, coupon weighing, and a deliberate upset with recovery.
- The garden-hose failure mode: the doctrine designed for the operator it will actually have.

TRL 5 is complete. The full-scale home-tier article was validated outdoors (Ch. 19), the expander question was answered with measured numbers (Ch. 20), and the sealed loop's chemistry was proven over operating time (Ch. 21). The technology works in the relevant environment. TRL 6 demonstrates it in the *operational* environment: the pilot installation, serving a real building.

GATE 5 — Exit evaluation

The reader may not proceed to TRL 6 until every answer below is theirs.

From Chapter 19:

1. Define the relevant environment as conditions. Your field η_thermal trails the lab by 8% at matched points — walk the attribution.
2. The campaign's hottest night misses the design hot night by 4 K. State the options for 19C.

From Chapter 20:

1. The 20A campaign measures cycle efficiency at 18% against the 27% assumption. Walk the revision propagation — which chapters, which recomputation first?
2. Justify the system-level selection metric with a home-tier vs. corporate-tier example.

From Chapter 21:

1. State the five chemistry enemies in order and the doctrine's answer to each.
2. The operator tops up with untreated water twice in a month. Which layers of §21.6 respond, and what does the trending show?

Gate 5 exit criteria: all six answered; field validation complete with attributed lab-vs.-field differences; expander selected on measured data with the balance re-run; chemistry doctrine proven over operating time; the failure log current through the field campaign. Then TRL 6 — the pilot.

Sources and honesty notes

- The chemistry enemies and treatment principles are established boiler/closed-loop water-treatment practice; the doctrine's specific bands, limits, and schedules are set by the reader's materials and operating conditions — the book specifies the doctrine's structure, not its numbers.
- Corrosion coupons and the trending discipline are standard practice in industrial water treatment; their application to this sealed loop follows directly.
- The "finite problem" framing is the author's characterization of the sealed architecture's genuine operating advantage — an honest consequence of "no leaking," not a marketing claim.
- No chemistry endurance data for this system exists yet; Test 21A awaits the reader's campaign.

TRL 6

Operational Prototype

Chapters 22–24

Chapter 22

The Home Prototype

TRL 6 · Operational-environment demonstration. This chapter specifies the pilot: the full home-tier build, installed in a real backyard, serving a real home, operated by its owner.

22.1 What TRL 6 is

TRL 5 validated the technology in the relevant environment — the test team's site, the test team's hands. TRL 6 demonstrates it in the **operational** environment: a real home, real loads, and — the decisive change — the homeowner as the operator. The test team monitors remotely. The homeowner lives with the machine.

This is the TRL where technologies meet the truth about themselves. The lab forgives; the field tests; the home *judges* — on noise, on appearance, on whether the operator's life got better or worse. A machine the homeowner resents is a failed product no matter what its efficiency map says.

22.2 The pilot build

The pilot article: the full home-tier system from Chapter 8's sizing, incorporating every revision from TRL 4 and 5 — the revised drawings, the proven control logic, the selected expander (Ch. 20), the chemistry doctrine (Ch. 21), the field corrections (Ch. 19).

The site: a real backyard in the target market — real shading, real setbacks, real neighbors, real HOA or permitting

authority. The site selection from 19.2 narrows to one property, and the property's constraints (shading hours, available area, the electrical interconnection point) size the final installation. If the reference 40 m² doesn't fit this real yard, the installation fits the yard — and the sizing chain (Ch. 8) re-runs honestly for the actual aperture. The pilot proves the *method*, not the reference numbers.

Permitting and interconnection: the pilot goes through the real process — building permit, electrical interconnection agreement with the utility, inspections. The permitting experience is itself a deliverable: every requirement, every delay, every inspector's question gets logged, because Chapter 27's deployment doctrine is built from this log. A technology that can't be permitted can't be deployed, and the pilot finds out which one this is.

Installation: performed by the crew that the deployment model will actually use — not the design engineers. The installation manual is drafted *before* the pilot and revised *by* the pilot: every step the crew stumbled on, every tool they lacked, every drawing that lied. The crew's feedback is a first-class test result.

22.3 The handover: from test team to homeowner

The most important day of TRL 6 isn't first steam — it's the day the test team leaves. The handover has three parts:

1 **Training.** The homeowner learns the operator's doctrine: the daily non-routine (there is almost none — the machine runs itself), the weekly glance (the dashboard's three numbers), the monthly five minutes (chemistry check per Ch. 21's home-tier doctrine), and the emergency procedure (one page, large type, on the wall). Training is verified by demonstration — the homeowner performs each procedure, not just watches.

2 **The operator's manual.** Written for the homeowner, not the engineer. Pictures, not schematics. "If this light is red, do this" — the full decision tree fits on laminated cards. The manual is Chapter 26's first draft, tested on its actual reader.

3 **The monitoring that stays.** Remote telemetry keeps flowing to the program team — every sensor from the validation campaign stays live. The homeowner gets the simple dashboard; the program team gets the full data. This dual view is permanent: it's how the fleet learns (Chapter 29).

22.4 The operating doctrine

The homeowner's role: near-zero. The machine runs its Chapter 6 day cycle autonomously; the homeowner's job is the monthly chemistry five-minutes, the occasional cleaning per the soiling trigger (12D's doctrine, now lived), and calling the service line when the dashboard says so. The pilot measures operator burden in minutes per month — honestly, with the homeowner's own log, not the program team's estimate.

The program team's role: watch everything, touch nothing unless the procedures say so. Every remote intervention is logged with its cause — each one is either a procedure working as designed or a design gap. The intervention log is the most honest document TRL 6 produces.

The loads are real: the house draws what it draws — the 30 kWh/day reference is replaced by the actual meter. The pilot measures served load, unmet load (if any), and grid interaction (import/export per the interconnection agreement). Chapter 23 turns these into the performance verdict.

22.5 What the pilot proves that TRL 5 didn't

- **Real loads**, not test profiles — including the 5 kW peak from Chapter 8's sizing, arriving when the household (not the test plan) demands it.

- **The homeowner as operator** — Chapter 21's garden-hose doctrine, Chapter 26's training, all of it, in the field, with a real person's attention budget.
- **Installation reality** — what the backyard, the permit office, and the crew actually required. The gap between the installation manual's draft and its revised version is measured in crew-hours and change orders.
- **Neighborhood reality** — noise at the property line (the expander selection from Ch. 20, judged by neighbors), appearance (the HOA's opinion, fairly or not), the curious-visitor factor. Technologies live in communities, not test sites.
- **A full year** — the pilot runs twelve months minimum, through every season, because Chapter 19's seasonal validation was on the test site and this is the operational confirmation.

22.6 Success criteria

The pilot succeeds when, over its year:

1. **Availability** meets the design target with every downtime cause coded (the intervention log is the evidence).
2. **Energy delivered** matches the Chapter 11 balance (as revised with measured numbers) within its uncertainty — Chapter 23's verdict.
3. **Operator burden** stays within the doctrine's budget — the homeowner's log, not the team's.
4. **The installation and operations doctrines** are revised from pilot experience and ready for the next article.

A pilot that hits its energy numbers but burns out its homeowner is a failure with good data. All four criteria, or TRL 6 isn't done.

22.7 What this chapter proved

- TRL 6 defined: the operational environment — real home, real loads, homeowner as operator, test team remote.
- The pilot build: full home-tier article, real backyard, real permitting (logged as a deliverable), installation by the deployment crew with the manual revised by their experience.
- The handover: training by demonstration, the homeowner's manual, the dual-view monitoring that stays permanently.
- The operating doctrine: near-zero homeowner role, program team watches without touching, real loads on the meter.
- What the pilot proves beyond TRL 5: real loads, the homeowner-operator, installation reality, neighborhood reality, a full year.
- Four success criteria — including the homeowner's burden, measured by the homeowner.

Chapter 23 measures the pilot: performance vs. prediction, and the scaling factors the builder actually gets.

Self-evaluation — Gate 6, part 1

1 State the decisive difference between TRL 5's environment and TRL 6's. Why does the homeowner-as-operator change what gets proven?

2 The reference 40 m^2 doesn't fit the pilot property's real yard. What happens — and what doesn't happen — to the pilot's validity?

3 The installation crew takes twice the manual's estimated hours. Classify the findings per Chapter 18's system and state what gets revised.

4 A pilot hits its energy numbers but the homeowner's log shows 3 hours/month of unplanned attention. Does TRL 6 pass? Cite the criteria.

5 Why is the permitting experience a deliverable? What future chapter consumes it?

Sources and honesty notes

- The TRL 6 definition follows the standard TRL scale's "system/subsystem model or prototype demonstration in a relevant [here: operational] environment" — the homeowner-operator is the author's specified operational condition for this technology.
- The twelve-month minimum pilot duration is the author's specified requirement for seasonal confirmation in the operational environment.
- The four success criteria are the author's specified gate conditions; the reader's program may add criteria but may not drop these.
- No pilot has been built for this system yet; this chapter specifies the pilot to build and the terms of its judgment.

Chapter 23

Measuring the Pilot

TRL 6 · Operational-environment demonstration. This chapter renders the verdict: pilot performance vs. every prediction the book made — and the scaling factors the builder actually gets.

23.1 The verdict structure

The pilot's year of data gets compared against every prediction the book made, in order:

1. **The energy balance** (Ch. 11, as revised with measured expander efficiency and field corrections): predicted annual electricity vs. metered.
2. **The sizing** (Ch. 8, as built for the actual yard): predicted coverage of the home's load vs. served/unmet load on the meter.
3. **The availability assumption**: the design target vs. the coded downtime record.
4. **The operator burden**: the doctrine's budget vs. the homeowner's log.

Each comparison carries uncertainties on *both* sides — the prediction had error bars (Chapter 12's discipline propagated forward) and the measurement has them too. A prediction of 10.5 ± 0.8 MWh against a measurement of 9.9 ± 0.3 MWh is agreement. Against 8.1 ± 0.3 MWh it's a gap — and gaps get attributed, not averaged.

23.2 The metrics, defined

- **Annual energy delivered:** metered kWh at the interconnection point, by month. The number the economics will use.
- **Capacity factor:** annual energy / (rated power × 8760). Reported honestly — solar-thermal capacity factors are modest, and the book doesn't dress them up.
- **Performance ratio:** measured output / (incident beam × aperture × predicted stage efficiencies). The diagnostic metric — a PR below prediction localizes the shortfall by stage.
- **Availability:** uptime fraction with every downtime cause coded — weather-stow, maintenance, faults, grid outages (counted separately; the machine isn't blamed for the grid).
- **Served load fraction:** the home's metered consumption vs. the pilot's delivered energy — the number the homeowner actually feels.

23.3 The attribution: every gap gets a cause

The 19A discipline, now for the full system: for every predicted-vs.-measured gap beyond the combined uncertainty, the investigation assigns a physical cause using the pilot's instrumentation:

- Optical shortfall? The flux maps and soiling log answer.
- Thermal shortfall? The stage efficiencies from the interface measurements answer.
- Expander shortfall? The 20A maps vs. the field operating points answer.
- Availability shortfall? The coded downtime answers — and each code has an owner (design, procedure, environment, grid).

An unattributed gap blocks the gate. The pilot report doesn't get to say "8% below prediction (cause unknown)" — that's a TRL 5 result wearing TRL 6 clothes.

23.4 The scaling factors you actually get

This is the chapter's unique contribution — the reason the pilot was full-scale at the home tier rather than another breadboard. Three scale steps now have measured data at both ends:

- **Breadboard → field article** (Ch. 16–19): how the lab integration's numbers translated outdoors.
- **Field article → pilot** (Ch. 19–22): how the test-site numbers translated to the operational home.
- **The composite**: the measured scaling law from laboratory to rooftop.

Each step's ratio is reported per stage — optics, receiver, expander, condenser — because stages scale differently (the expander's scale curve from Ch. 20 is the steepest). These measured factors are what license Chapter 9's building tiers: the corporate and apartment-complex sizings were projections; the pilot's scaling factors turn them into *extrapolations with measured bases*. The difference between a projection and a measured extrapolation is the difference between TRL 5 and TRL 6.

23.5 The availability accounting

Downtime, coded and costed:

- **By cause:** design faults (the log's entries), maintenance (planned and unplanned), weather-stow (the envelope working as designed — counted, not blamed), grid outages (excluded from machine availability, reported alongside).

- **MTBF and MTTR**, honest: computed from the coded record, reported per subsystem. The inverter-grade electronics will have different numbers than the boiler — the book doesn't average them into meaninglessness.
- **The availability the economics can assume:** the measured figure, with its trend (improving as the infant-mortality failures from Ch. 18's taxonomy burn off, or not — the trend is data either way).

23.6 The homeowner's verdict

Numbers the program team can't generate: the homeowner's log of burden-minutes, the satisfaction interview (structured, recorded, published in summary), the neighbors' experience (noise, appearance, curiosity), the answer to the only question that matters commercially — *would you recommend this to your neighbor?*

A "yes" with caveats is data — the caveats become requirements. A "no" is data too — the reasons become the redesign brief. The pilot doesn't get to dismiss its operator.

23.7 The pilot report: published, including the ugly

The pilot report is a public document — the open-IP strategy (Aries's directive) demands it, and the patent-office audience requires the enabling detail. Its structure:

1. The article as built (drawings, BOM, the yard it fit).
2. The year of data: every metric from §23.2, by month, with uncertainties.
3. Predicted vs. measured: every comparison from §23.1, every gap attributed (§23.3).
4. The scaling factors (§23.4) and the availability accounting (§23.5).
5. The homeowner's verdict (§23.6), in their words as well as the team's.

6 The failure log's pilot entries and every resulting revision.

7 **The claims the program can now make — and the ones it still can't** (§23.8).

Nothing redacted except genuine personal information. A pilot report that hides its gaps is an advertisement; this one's a document.

23.8 What the pilot licenses

After the pilot year, the program can claim — with the report as evidence:

- The measured annual energy and capacity factor at the home tier.
- The measured availability with coded causes.
- The measured scaling factors from lab to rooftop.
- The measured operator burden.
- The validated annual-energy model (Ch. 11's balance, now with field lineage).

It still can't claim: building-tier performance (that's TRL 7 — the next section), long-term reliability (Chapter 28's program), or economics (Chapter 24 computes them from *this chapter's* numbers — the sequence is deliberate).

23.9 What this chapter proved

- The verdict structure: four predictions, each compared with uncertainties on both sides.
- The metrics defined: annual energy, capacity factor, performance ratio, availability, served load fraction.
- The attribution discipline at full-system scale: every gap gets a physical cause or blocks the gate.
- The measured scaling factors: breadboard field pilot, per stage — the license for the building-tier extrapolations.

- The availability accounting: coded downtime, honest MTBF/MTTR, the figure economics may assume.
- The homeowner's verdict and the public pilot report — ugly included.
- The claims ledger: what the pilot licenses, and what it explicitly doesn't.

Chapter 24 is the LCOE reckoning: measured cost vs. rooftop PV and grid power — reported honestly, whichever way it lands.

Self-evaluation — Gate 6, part 2

1. Your pilot meters 9.2 ± 0.4 MWh against a predicted 10.5 ± 0.8 MWh. Is this agreement or a gap? State the rule and apply it.
2. The performance ratio trails prediction. Walk the stage-by-stage localization using the pilot's instrumentation.
3. Explain the difference between Chapter 9's projections and the pilot's measured extrapolations — and why the gate cares.
4. Downtime coding shows 40% of unavailability from weather-stow. How is this counted, and what decision does it feed?
5. The homeowner's verdict is "yes, but it's ugly and the monthly check annoys me." State the program's exact obligations in response.

Sources and honesty notes

- The predicted-vs.-measured comparison method with two-sided uncertainties is standard experimental practice; the "unattributed gap blocks the gate" rule is the author's specified gate condition.

- Capacity factor, performance ratio, MTBF/MTTR, and availability coding follow standard energy-industry definitions; the reader applies them to their metered data.
- The "measured extrapolation vs. projection" distinction is the author's framing of TRL 6's epistemic contribution — the scaling factors are only as good as the pilot's instrumentation, which is why the instrumentation stayed live (§22.3).
- No pilot data for this system exists yet; every metric in this chapter awaits the reader's pilot year.

Chapter 24

The LCOE Reckoning

TRL 6 · Operational-environment demonstration. This chapter computes the levelized cost of energy from the pilot's measured numbers — against rooftop PV and grid power — and reports it honestly, whichever way it lands.

24.1 The promise, kept

Chapter 10 laid out the mechanisms by which this system could win — co-generated heat, no battery or lithium costs, manufacturing scale — and refused to compute the number, because the number didn't exist yet. It exists now: the pilot's metered energy (Ch. 23), the pilot's accounted costs, the measured availability. This chapter does the arithmetic the book has been walking toward for twenty-three chapters. Whatever the number is, it's the number.

24.2 LCOE, defined without mercy

Levelized cost of energy: the lifetime cost of the system divided by its lifetime energy output, discounted to present value. The inputs, each with its provenance:

- **CAPEX:** the pilot's accounted build cost — every invoice, every crew-hour, the permitting fees from Chapter 22's log. Not an estimate: the ledger.

- **OPEX:** measured maintenance (the intervention log costed), the chemistry program (Ch. 21's actual consumables), the operator burden (Ch. 23's minutes/month, costed at a fair rate — the homeowner's time isn't free just because it's theirs).
- **Annual energy:** the pilot's metered figure (Ch. 23), not the balance's prediction.
- **Lifetime:** the design life, stated as an assumption with Chapter 28's reliability program as its verification path — the book doesn't get to assume 25 years because PV does.
- **Degradation:** the measured year-one decay rates (soiling-adjusted performance, heat-transfer trending from 21A) extended by stated assumption, not by hope.
- **Discount rate:** stated explicitly — the number moves with it, so the reader sees the sensitivity.

24.3 The pilot-cost honesty

Pilot articles cost more than production articles — always, and substantially. The book handles this in two steps, both explicit:

1. **The pilot LCOE**, computed straight from the ledger: this is the cost of energy from *the pilot*, and it's almost certainly uncompetitive. It's reported anyway — it's the anchor of honesty the rest of the analysis hangs from.
2. **The production LCOE**, projected via the learning-curve treatment from Chapter 10 — Wright's law applied to the pilot's real cost as the starting point, with the learning rate *stated as an assumption* and the cumulative-volume scenario named. The reader sees exactly how much learning the competitiveness claim requires: "at X% learning rate and N units, the production LCOE is $Y."

No step hides. A reader who distrusts the learning rate can recompute with their own — the pilot anchor is public.

24.4 The comparison

Three numbers, same methodology, same discount rate:

- **The system's production LCOE** (from §24.3).
- **Rooftop PV LCOE** at current installed-cost benchmarks for the same market — with PV's own honesty applied (its degradation, its inverter replacement, its capacity factor at the site).
- **Grid power** at the retail rate — the number the homeowner actually pays, by tariff, because the system's first competitor is the utility bill, not another generator.

Then the value streams the LCOE doesn't capture, stated alongside — not smuggled in:

- **Co-generated heat** (Ch. 3, Ch. 10): the system's waste heat is a product — domestic hot water, space heating — with a displaced-fuel value. Credited separately, at the displaced fuel's price, so the reader can accept or reject the credit.
- **The storage the system doesn't buy:** the day/night cycle runs on water and steel, not lithium. PV-plus-battery pays for its night; this system condenses. The comparison runs both with and without storage on PV's side — because "PV is cheaper" means something different at noon than at midnight.
- **Resilience:** the outage-ride-through value — real, hard to price, stated qualitatively with the pilot's islanding data if the interconnection permitted the test.

24.5 The sensitivity: what moves the needle

The tornado analysis — each lever varied alone, LCOE re-computed:

- **Expander efficiency** (Ch. 20's measured number ± its uncertainty): the steepest lever — every point of cycle efficiency is worth more than every point of optical efficiency, because it multiplies the whole output.
- **CAPEX learning rate:** the claim's fragility made visible — at 10% learning the story differs from 20%.
- **Lifetime and degradation:** what the economics assume that Chapter 28 must verify.
- **Capacity factor:** the site's sun, honestly — the system doesn't get to borrow Arizona's DNI for a Ohio customer's economics.

The sensitivity analysis is the chapter's real product for the skeptical reader: even if they reject the headline number, the levers tell them *what would have to be true* for the system to win. That's a falsifiable economic claim — the book's honesty contract, extended to money.

24.6 The branches

If the system wins: the chapter states where (which markets, which tariffs, which tier), by how much, and on which value streams the win depends — so the win can be checked, and so the reader knows which assumptions to attack first.

If the system loses: the chapter states where, by how much, and exactly what has to change — the sensitivity analysis becomes the development priority list. A loss with a diagnosed cause is a roadmap; the book has never hidden from a roadmap. (And the positioning from the start holds regardless: this isn't an ad for solar panels — it's the measured economics of a different category, winning or losing on its own numbers.)

If it's close: the most likely outcome, and the most useful — the chapter states the conditions under which each option wins, which is precisely the decision support the builder, the college,

and the patent office each need.

24.7 What this chapter proved — and what TRL 6 established

- LCOE computed from the pilot's ledger and meter — the number the book walked toward for twenty-three chapters.
- The pilot-cost honesty: the uncompetitive pilot anchor reported anyway, production projected via stated learning assumptions.
- The three-way comparison on identical methodology, with non-LCOE value streams (heat, no-battery nights, resilience) credited separately and rejectably.
- The sensitivity analysis: the levers that move the needle, making the economic claim falsifiable.
- The three branches — win, lose, close — each with its obligations. No branch gets to hide.

TRL 6 is complete. The pilot was built in a real backyard (Ch. 22), measured against every prediction (Ch. 23), and its economics computed from the ledger (Ch. 24). The home tier works, in operation, at a known cost. TRL 7 takes the building tiers: the corporate and apartment-complex demonstrations.

GATE 6 — Exit evaluation

The reader may not proceed to TRL 7 until every answer below is theirs.

From Chapter 22:

1 State TRL 6's decisive condition. The pilot hits energy and availability but the homeowner logs 3 unplanned hours/month — pass or fail, and why?

From Chapter 23:

1 Define performance ratio and explain why it's the diagnostic metric. Your PR trails prediction — walk the localization.

2 Explain measured extrapolation vs. projection, and which building-tier claims it licenses.

From Chapter 24:

1 Your pilot LCOE is uncompetitive but the production projection wins at 20% learning and 10,000 units. State exactly what the book claims — and what a skeptic is entitled to recompute.

2 The system loses to PV-plus-battery on LCOE but wins on the no-battery night comparison. Walk how §24.4 presents both without contradiction.

3 Name the steepest economic lever from the sensitivity analysis and explain why it multiplies the whole output.

Gate 6 exit criteria: all six answered; pilot year complete with the public report published; economics computed from the ledger with stated assumptions; the scaling factors measured; the homeowner's verdict recorded. Then TRL 7 — the buildings.

Sources and honesty notes

- LCOE methodology follows standard energy-economics practice (lifetime discounted cost over lifetime discounted energy); every input's provenance is specified — ledger, meter, or stated assumption.
- Wright's-law learning-curve treatment follows Chapter 10's specification; the learning rate and volume scenario are assumptions, labeled as such, with the pilot cost as the measured anchor.
- Rooftop PV benchmarks and retail tariffs are market data the reader sources current at their time of analysis — the book specifies the comparison method, not the market numbers, which move.

- The win/lose/close branches are the author's specified reporting obligation; the chapter's number will be whatever the reader's pilot measured — the book's integrity doesn't depend on the outcome, only on the reporting.

TRL 7

Operational Demonstration

Chapters 25–27

Chapter 25

The Corporate Building Installation

TRL 7 · Prototype demonstration in the operational environment. This chapter specifies the building-tier demonstration: the full-scale corporate system, in real operation, serving a real building.

25.1 What TRL 7 is

TRL 6 proved the home tier in operation. TRL 7 proves the *building* tiers in operation — the corporate building first (this chapter), the apartment complex following the same doctrine at its scale. The demonstration article is the full-scale corporate system from Chapter 9's sizing, built on the pilot's measured scaling factors, operated by a facilities team, serving a real building's real loads.

The step from TRL 6 to 7 isn't just bigger — it's *institutional*. A homeowner forgives; a building owner contracts. The demonstration runs under commercial terms: performance guarantees, maintenance SLAs, insurance, and a utility interconnection at commercial scale. TRL 7 is where the technology learns to be a vendor.

25.2 The article: the corporate tier, built

The reference: Chapter 9's corporate sketch — 2,000 kWh/day, ~2,270 m² aperture under the stated assumptions, recomputed with every measured number from TRL 5 and 6 (the expander efficiency from Ch. 20, the field corrections from Ch. 19, the scaling factors from Ch. 23). The reference is a starting point; the demonstration site re-sizes it honestly for its own loads and roof.

The modularity decision, built: the corporate tier is N home-tier-class modules, not one giant machine — the scaling doctrine from Chapter 9, now hardware. The reasons compound at demonstration scale: factory-built modules install like appliances; a failed module degrades output instead of stopping it (the N-1 doctrine — the building rides through any single module's outage); maintenance happens one module at a time while the rest run. The demonstration proves the modularity claim the book has carried since Chapter 1's "the building is the unit."

The site: a real corporate building — real roof structure (the structural analysis is a deliverable: dead loads, wind loads, the tracker's dynamic loads on the building frame), real electrical room, real building management system to integrate with, real facilities team to train. Rooftop vs. parking-canopy vs. adjacent ground — the siting decision per building, with the structural and shading analysis published.

25.3 Installation at building scale

The home pilot's installation lessons (Ch. 22), scaled up:

- **Phased commissioning:** modules come online in phases — the first phase proves the installation procedure, the later phases run at speed. The phased approach bounds the risk: a systematic installation error is found on phase one, not on all N modules.

- **The crane and the roof:** the logistics plan — module delivery, roof loading sequence, the weather windows. The installation manual's building-tier edition, revised by the crew's experience like the home edition was.
- **BMS integration:** the system talks to the building management system — status, alarms, energy metering, demand-response readiness. The interface spec is published; a building engineer who wasn't on the program team integrates it from the document.
- **Commercial interconnection:** the utility process at commercial scale — the application, the study, the witness test. Logged like the home pilot's permitting (Ch. 22), because Chapter 27's deployment doctrine consumes both.

25.4 The commercial electricity bill: the real opponent

The home tier fought the retail rate. The corporate tier fights the commercial tariff — and the structure favors this system:

- **Demand charges:** commercial bills price the peak kW, often brutally. A solar-thermal system producing through the afternoon peak *shaves the metered peak* — the demand-charge savings can exceed the energy savings. The demonstration meters this separately: energy value and demand value, reported apart, because they accrue to different line items and different decision-makers.
- **Load coincidence:** corporate buildings consume by day — when the system produces. The home tier's evening-peak problem (Ch. 8's sizing acknowledged it) inverts here: the corporate load profile and the solar profile are natural allies. The demonstration measures the coincidence and prices it.

- **The tenant/owner split:** who pays the bill vs. who buys the system — the demonstration documents the contracting structure (green lease, energy-services agreement, direct ownership) because the technology's economics mean nothing until the *contract* lets someone capture them.

25.5 The measurement campaign

Chapter 23's verdict structure, at building scale:

1. **Performance vs. prediction** — predicted from the pilot's measured scaling factors (Ch. 23.4), the true test of the extrapolation license TRL 6 granted.
2. **Availability** with the N-1 doctrine in the accounting: module-level availability and system-level availability reported separately — the redundancy is a designed feature, and its value gets measured.
3. **The demand-charge verdict** — metered peak shaving, priced at the actual tariff.
4. **Facilities-team burden** — the professional operator's log: maintenance hours by task, the chemistry program at building scale (Ch. 21's corporate-tier doctrine), spare-parts consumption.

Duration: a full year minimum, like the pilot — the seasons don't negotiate, and the demand-charge verdict needs the summer peak season in the data.

25.6 The apartment complex: the same doctrine, the third tier

The apartment-complex tier (Ch. 9's 200-unit reference, ~3,400 m^2) follows the corporate demonstration's doctrine — same modularity, same phased commissioning, same verdict structure — with its own differences specified:

- **The operator** is property management, not a facilities engineer — the training and manual pitch to that reality (Ch. 26's tiered operations doctrine).

- **The load** is 200 homes' evening-peaked consumption — the coincidence the corporate tier enjoyed inverts back. The demonstration measures it honestly: the apartment tier's value case leans harder on the energy charge and the resilience story than on demand shaving.
- **The stakeholder set** widens: tenants, the property owner, possibly a housing authority — the contracting structures from §25.4 get their residential-multifamily variants.

The book specifies the corporate demonstration in full and the apartment demonstration by delta — same method, honestly adapted, because the method is the transferable asset.

25.7 Success criteria

Over the demonstration year:

1. **Energy and demand performance** match the scaled prediction within uncertainty — the extrapolation license from TRL 6 either validates or revises.
2. **System-level availability** meets the commercial target with the N-1 doctrine demonstrated (at least one real module outage ridden through — the doctrine gets its fault-injection like everything else).
3. **The commercial value case** closes: energy + demand savings, metered and priced, against the accounted costs — the building-tier LCOE computed like Chapter 24's, at building scale.
4. **The operations doctrine** is revised from facilities-team experience and ready for the next building.

25.8 What this chapter proved

- TRL 7 defined: building-tier prototypes in the operational environment, under commercial terms — where the technology learns to be a vendor.

- The corporate article: Chapter 9's sizing recomputed with measured numbers, built as N modules with the N-1 doctrine.
- Installation at building scale: phased commissioning, roof logistics, BMS integration from a published spec, commercial interconnection logged.
- The commercial tariff as the real opponent: demand charges metered separately, load coincidence measured, contracting structures documented.
- Chapter 23's verdict structure at building scale, with module-level and system-level availability separated.
- The apartment tier by delta: same doctrine, honestly adapted to its operator, load, and stakeholders.
- Four success criteria, including a real ridden-through module outage.

Chapter 26 operates the building: controls, maintenance, occupants, and the grid interface — the doctrine the demonstrations wrote.

Self-evaluation — Gate 7, part 1

1 State what changes — technically and commercially — between TRL 6 and TRL 7. Why is "the technology learns to be a vendor" the right description?

2 Justify the N-module architecture against a single large machine on three grounds: installation, availability, and maintenance.

3 Your demonstration year includes zero module outages. Does the N-1 criterion pass? Explain, referencing the book's fault-injection discipline.

4 The metered demand savings exceed the energy savings 2:1. Walk how §25.4 reports this and who in the building's organization cares about each number.

5 The apartment tier's load peaks at 7 PM. State the honest consequences for its value case vs. the corporate tier's.

Sources and honesty notes

- Commercial tariff structures (demand charges, time-of-use) vary by utility and market; the demonstration prices its verdict at its actual tariff — the book specifies the metering and reporting method, not the rates.
- The N-1 redundancy doctrine is standard practice in building systems; its application to the modular solar-thermal array follows directly from Chapter 9's modularity.
- The phased-commissioning and crew-revised-manual disciplines continue Chapters 16/22's specified procedures at building scale.
- No building-tier demonstration exists for this system yet; the article, the year of data, and the verdict all await the reader's program.

Chapter 26

Operating the Building

TRL 7 · Prototype demonstration in the operational environment. This chapter writes the operations doctrine the demonstrations earned: controls, maintenance, occupants, and the grid interface — for buildings, at scale.

26.1 The doctrine's source

Chapters 22 and 25 put machines in homes and buildings. This chapter writes down what operating them actually entails — the controls that coordinate them, the maintenance that sustains them, the occupants who live under them, and the grid they trade with. Every procedure here was earned by a demonstration's experience: the intervention logs, the facilities-team feedback, the homeowner's burden-minutes. Doctrine isn't written from the design — it's written from the operation.

26.2 Controls: the supervisory layer

The module-level control (Chapter 6's day-cycle state machine, proven in 17A) runs each module. The building needs a layer above it:

Fleet coordination: the supervisory controller dispatches the N modules — which run, which rest, which go down for maintenance — against the building's load and the sun. The dispatch logic is an optimization with honest inputs: each module's measured efficiency map (Ch. 23's data, per module), the demand-charge structure (dispatch to shave the metered

peak first — §25.4's verdict made this the priority), and the maintenance schedule.

Fault detection and diagnostics (FDD): the supervisory layer watches every module's telemetry against its own baseline — the drift that precedes failure (a slowly falling performance ratio, a rising pump current, a chemistry parameter trending toward its limit). FDD turns the failure log (Ch. 18) from a record into a prediction: the system flags the module *before* it fails, and the N-1 doctrine means the flag costs nothing in output.

BMS integration, in operation: the building management system sees the solar-thermal plant as it sees the chillers — status, alarms, energy flows, available demand-response capacity. The interface spec from §25.3 is now lived-in: the alarms that actually fired get their priorities tuned, the data the operators actually checked gets its dashboard.

Demand response: the thermal inertia in the loop is dispatchable flexibility — pre-heating before a demand-response event, curtailing export on command. The demonstration measures the response (kW, duration, recovery) and the building's program enrolls it where the tariff rewards it. A solar-thermal plant with hot water in its veins is a better grid citizen than its nameplate suggests — the doctrine prices that citizenship.

26.3 Maintenance: the tiered program

Preventive schedule: derived from the demonstration's maintenance log — every task with its measured interval and measured duration. The schedule is tiered:

- *Home tier:* the monthly five-minutes (Ch. 21/22) plus the annual service visit — a technician, half a day, the checklist from the pilot's experience.
- *Corporate tier:* the facilities team's weekly walkthrough (fifteen minutes, the FDD dashboard as the guide), quarterly module-level service, annual deep service.

- *Apartment tier:* property-management's monthly check pitched to non-engineers, with the service contractor on call — the training from §25.6, now a manual.

The module-swap doctrine: the N-1 architecture's maintenance expression. A module that needs deep service doesn't get serviced in place — it gets *swapped*: the spare module (the N+1th, stocked per the sparing strategy) takes its place while the failed unit goes to the shop. Building output never drops below N-1, the repair happens on a bench instead of a roof, and the mean-time-to-repair the availability accounting (Ch. 23/25) assumed becomes a measured reality.

Sparing strategy: which parts on the shelf, which on order, which never fail — from the demonstration's parts consumption, not the supplier's optimism. The strategy is costed: spares inventory is working capital, and the building-tier LCOE carries it.

The chemistry program at scale: Chapter 21's doctrine with building-tier staffing — the corporate facilities team runs the sampling schedule; the apartment tier's contractor does. The garden-hose failure mode (§21.6) gets its building-tier variants: the well-meaning maintenance tech, the wrong treatment chemical. Same layered answer: make the right thing easy, the wrong thing visible, the system forgiving.

26.4 Occupants: the human layer

The doctrine's prime directive: **the occupants should notice nothing.** No noise beyond the property-line spec (the expander selection from Ch. 20, verified by the demonstration's acoustic survey), no vibration in the structure (the roof-mount design from §25.2, verified), no visible drama — steam is contained, the plant is boring by design.

What they do notice: the dashboard in the lobby (the building that shows its energy teaches its occupants — and the demonstration's occupant survey measures whether anyone looks), the maintenance visits (scheduled, announced, brief),

and — the one that matters — the utility bill. The tenant communication is honest: what the system does, what it doesn't, who to call. No green theater; the metered numbers are the message.

The curious visitor, at building scale: the demonstration buildings get tours — engineers, students, the patent-office audience. The doctrine includes the tour: what's shown, what's explained, and the safety briefing (Chapter 27's requirements, in plain language). A technology that teaches is a technology that spreads — the reader-principle from the book's mission, in steel and steam.

26.5 The grid interface, in operation

Metering and settlement: export metered per the interconnection agreement; the energy and demand values (§25.4) settled on the actual tariff. The metering plan is auditable — the building owner, the utility, and the program team all read the same numbers.

Islanding and ride-through: the demonstration tests what the interconnection permits — the anti-islanding protection verified by the witness test (§25.3), and where the agreement allows intentional islanding, the resilience value from §24.4 gets its measured data. The doctrine states plainly which buildings can island and which can't — the utility's rules, not the technology's wishes, decide.

The utility relationship: the program team's standing practice — the interconnection queue is joined early, the utility's engineers get the pilot report (Ch. 23's public document) with the application, and every utility's questions get logged into the deployment doctrine. Utilities are not opponents; they're the counterparty, and the doctrine treats them as such.

26.6 Staffing and the OPEX model

The operations doctrine closes on cost — measured labor, not estimated:

- Home tier: the annual service visit, priced.

- Corporate tier: the facilities-team hours from the demonstration log, priced at burdened rates, plus the service contract.
- Apartment tier: the management-company arrangement, priced as contracted.

These measured OPEX figures feed the tier LCOEs (Ch. 24's method at each tier's scale) — the economics rest on the demonstration's timesheets, which is exactly where economics should rest.

26.7 The fleet learning loop

Chapter 22's dual view — homeowner dashboard, program-team telemetry — scales to the fleet: every building's full data flows to the program, where the FDD baselines sharpen, the maintenance intervals tune, the failure log grows across sites, and the design revisions propagate to the next buildings. The fleet is the final instrument — no single building's experience is wasted, and the tenth building installs with the first nine's knowledge. This is the mechanism by which TRL 7 becomes TRL 8: the demonstrations stop being experiments and start being the first units of the fleet.

26.8 What this chapter proved

- The supervisory control layer: fleet dispatch against the demand tariff, FDD that predicts from the failure log, lived-in BMS integration, measured demand-response capability.
- The tiered maintenance program: preventive schedules from measured logs, the module-swap doctrine, the costed sparing strategy, chemistry at building scale.
- The occupants' prime directive (notice nothing), the honest tenant communication, the tour as technology-spreading.

- The grid interface in operation: auditable metering, islanding stated plainly per the utility's rules, the utility as counterparty.
- Staffing and the OPEX model from measured labor — the economics resting on timesheets.
- The fleet learning loop: every building's data improving every future building — the mechanism from TRL 7 to TRL 8.

Chapter 27 takes the legal reality: codes, certification, and safety — pressure vessels, steam code, and what "qualified" actually requires.

Self-evaluation — Gate 7, part 2

1. Explain the supervisory dispatch priority under a demand-charge tariff. Why does §25.4's verdict set it?
2. A module's performance ratio drifts down 2% over a month with no alarm. Which doctrine element responds, and what happens to building output?
3. State the module-swap doctrine and explain how it converts the N-1 design feature into measured availability.
4. The apartment building's property manager tops up with the wrong treatment chemical. Walk the layered response from §§26.3/21.6.
5. Explain the fleet learning loop as the mechanism from TRL 7 to TRL 8 — what specifically stops being experimental?

Sources and honesty notes

- FDD, N+1 sparing, and tiered preventive maintenance are established building-systems and industrial practice; their application to the modular solar-thermal plant follows from the N-module architecture.

- Demand-response participation and thermal-inertia dispatchability are characterized from the demonstration's measured response — the doctrine prices only what was metered.
- The acoustic, vibration, and tenant-communication requirements are set by the demonstration's surveys and the applicable local codes — the book specifies the verification method, not universal limits.
- Staffing costs and OPEX figures are the reader's measured numbers; the book specifies the accounting that produces them.

Chapter 27

Codes, Certification, and Safety

TRL 7 · Prototype demonstration in the operational environment. This chapter takes the legal reality: pressure vessels, steam code, and what "qualified" actually requires — because the law applies to open hardware too.

27.1 The law has opinions

Everything before this chapter was physics, engineering, and economics. This chapter is the part builders skip until it stops them: a steam system is a pressure system, a pressure system is a regulated object, and the regulations are not suggestions. The open-IP strategy doesn't exempt the hardware — publishing the design doesn't publish away the boiler inspector.

The doctrine: **code compliance is designed in, not bolted on.** Every requirement below shapes the drawings, the procedures, and the business model. A system that can't be legally installed can't be deployed, and the book would rather say so in Chapter 27 than have the reader discover it at the permit counter.

27.2 Pressure vessels: the code that governs steam

The governing standard in the US is the **ASME Boiler and Pressure Vessel Code**:

- **Section I** (Power Boilers) applies to the steam-generating boiler — the design rules, the materials, the fabrication, the inspection. The boiler gets built in a code shop, by welders with the right stamps, and inspected by an Authorized Inspector. There is no homebuilt alternative that the law recognizes at these pressures.
- **Section VIII** (Pressure Vessels) governs the other pressure boundaries — accumulators, receivers acting as unfired vessels, as applicable.
- **Relief valves** are code-sized, code-set, and code-tested — the overpressure protection from Chapter 16's build sequence isn't just good practice, it's the law's requirement with the inspector's signature on it.

The tier implication, stated plainly: the pressure boundary is the one subsystem the builder *buys* from the code-compliant supply chain rather than fabricating. The book's open hardware covers the integration, the optics, the controls, the valve sections — the pressure vessel comes stamped. This isn't a compromise of the open strategy; it's the strategy being honest about where the law draws its line.

27.3 Jurisdiction: the fifty-state reality

Boiler law is state and sometimes city law in the US — and it varies:

- **Inspection regimes:** most states require periodic inspection of boilers above size thresholds by commissioned inspectors. The demonstration's inspection history (Ch. 25's logged permitting) becomes the template: what the inspector asked, what the answers were, how long it took.
- **Operator licensing:** several jurisdictions require licensed operators for steam plants above pressure/size thresholds. The tiers' designs are checked against these thresholds *as a design input* — a home-tier system that triggers a licensed-operator

requirement in its target market has a design problem, not a lobbying problem. Where thresholds can't be avoided, the operations doctrine (Ch. 26) includes the licensing path.

- **The variance process:** where the code's letter fits poorly (a novel configuration the drafters didn't imagine), the legal path is the variance — applied for with engineering justification, granted or denied on the record. The book documents the variance strategy honestly: it's a real path, it's slow, and it's part of the deployment timeline.

The reader's jurisdiction gets its own research — the book gives the map of *what to look for*, not a fifty-state survey that would be outdated before printing.

27.4 Electrical, structural, and the interconnection standards

- **Electrical:** the NEC governs the wiring; the power electronics carry UL listings (or the equivalent) — the certification program (§27.5) plans for listed assemblies rather than listed components where the economics favor it.
- **Structural:** the building code governs the roof loads — dead, live, wind, seismic per the jurisdiction. The structural analysis from §25.2 is a permitted-documents deliverable, stamped by a licensed engineer. No stamp, no permit.
- **Interconnection:** IEEE 1547 governs the grid interface — the anti-islanding, the power quality, the witness test. The utility's supplemental requirements ride on top. Chapter 26's grid doctrine operates inside this standard, not around it.

27.5 The certification program: what "qualified" requires

"Qualified" isn't a feeling — it's a checklist, and the book lays it out as a project plan:

1. **Pressure boundary:** code-stamped vessels, inspected fabrication, the inspector's sign-off — per installation, per jurisdiction.
2. **Electrical assemblies:** listings (UL or equivalent) for the power electronics and control panels — obtained once per design, not per installation.
3. **The system as installed:** the permit, the inspections (building, electrical, boiler), the utility's witness test and permission to operate.
4. **The operators:** training records (Ch. 22/26), and licensed operators where the jurisdiction requires them.
5. **The ongoing:** periodic boiler inspections, the maintenance records the inspector will ask for, the chemistry logs (Ch. 21's sampling schedule doubles as inspection evidence).

Each item has an owner, a timeline, and a cost — the certification program is budgeted like the hardware, because it *is* hardware's shadow: every physical system drags its paperwork behind it, and the deployment model prices the shadow too.

27.6 The safety case

The hazard analysis, conducted like the engineering — systematically, on the record:

- **Steam burns and pressure rupture:** the design-basis hazards. Guarded by the code-stamped boundary, the relief systems, the fault-injection heritage (17A's safe-state proofs are safety evidence, not just control evidence), and the physical guarding from Chapter 16's test-cell discipline carried into the product.

- **The defocused beam:** the optical hazard — the lens and mirrors can only concentrate where they're pointed, and the tracker fault response (verified in 17A, ridden out in 19B) parks them safe. The hazard analysis treats the beam like the steam: energy that must go only where the design says.
- **Maintenance exposure:** lockout/tagout, the cool-down procedure before service, the module-swap doctrine (§26.3) keeping hands off hot roofs where possible.
- **The safety manual:** one document, plain language, covering the hazards, the guards, and the emergency procedures — the homeowner's laminated card (Ch. 22) and the facilities team's binder (Ch. 26) are its tiered editions.

Insurance: the underwriter's questions — answered with the demonstration data. The failure log, the availability accounting, the safety case, the inspection history: this is what the premium is priced on. The book states the mechanism plainly — insurability is a deployment requirement, and the demonstrations were designed to produce its evidence.

27.7 What this chapter proved — and what TRL 7 established

- The legal reality: code compliance designed in, not bolted on — the law applies to open hardware.
- ASME Sections I and VIII: the stamped pressure boundary bought from the code supply chain, honestly scoped within the open strategy.
- The fifty-state jurisdiction map: inspection regimes, operator licensing as a design input, the variance path.
- Electrical (NEC/UL), structural (stamped analysis), interconnection (IEEE 1547) — each with its deliverable.

- The certification program as a budgeted project plan: five checklists with owners, timelines, costs.
- The safety case: hazard analysis on the record, the beam treated like the steam, the safety manual in tiered editions, insurance priced on demonstration evidence.

TRL 7 is complete. The corporate building was demonstrated in real operation (Ch. 25), its operations doctrine written from experience (Ch. 26), and its legal reality mapped and budgeted (Ch. 27). The building tiers work, under commercial terms, inside the law. TRL 8 is the qualified system: manufacturing, quality, and the fleet.

GATE 7 — Exit evaluation

The reader may not proceed to TRL 8 until every answer below is theirs.

From Chapter 25:

1. Justify the N-module architecture on installation, availability, and maintenance grounds. The demonstration year has zero module outages — does the N-1 criterion pass?
2. State how demand-charge value is metered and reported separately from energy value, and why the distinction matters commercially.

From Chapter 26:

1. Explain the module-swap doctrine and how it converts N-1 design into measured availability.
2. A module's performance ratio drifts 2% in a month with no alarm. Which doctrine element responds, and what happens to output?

From Chapter 27:

1. The home-tier design as drawn triggers a licensed-operator requirement in the target market. State the design responses in order of preference, and why lobbying isn't one.
2. Explain why the pressure vessel is bought stamped rather than fabricated open — and why this doesn't compromise the open-IP strategy.
3. Name the five certification checklists and state which are per-design vs. per-installation.

Gate 7 exit criteria: all seven answered; building-tier demonstration year complete with the public report; operations doctrine revised from facilities experience; certification program budgeted with owners and timelines; safety case on the record. Then TRL 8 — the qualified system.

Sources and honesty notes

- ASME BPVC Sections I and VIII, the NEC, UL listing programs, and IEEE 1547 are the cited US standards; the reader verifies current editions and their jurisdiction's adoptions — standards evolve and the book doesn't freeze them.
- State boiler laws and operator-licensing thresholds vary by jurisdiction; the book specifies the research map, not a fifty-state survey.
- The "bought stamped" pressure-boundary scope is the author's specified honest boundary of the open-hardware claim — presented as a legal necessity, not a retreat.
- No certification has been obtained for this system yet; the program in §27.5 is the plan to obtain it.

TRL 8

Qualified System

Chapters 28–30

Chapter 28

Manufacturing the Unit

TRL 8 · System complete and qualified. This chapter turns the demonstrated prototype into a manufactured product: the production design, the quality system, and the cost curves at 10, 1,000, and 1,000,000 units.

28.1 What TRL 8 is

TRL 7 demonstrated prototypes in operation. TRL 8 completes the actual system and *qualifies* it — the manufacturing is real, the quality system is real, and the unit coming off the line is the unit the demonstrations proved. The shift is from "we built one that works" to "we build them so they all work" — and the proof is the qualification program, not the claim.

28.2 The production design

The demonstration article was built to be proven. The production unit is built to be *built* — the design-for-manufacturing pass:

- **The module is the manufactured unit.** Everything from the N-module architecture (Ch. 25) pays off here: one module design, factory-built, tested, shipped. The tiers differ in module *count*, not module design — the home tier is few modules, the corporate tier is many. One production line serves all three tiers.
- **The pressure boundary stays bought.** Chapter 27's honest scope holds: the stamped boiler/vessel comes from the code supply chain. The factory integrates it;

it doesn't fabricate it. The supply agreement — with the code shop's capacity and lead time — is part of the manufacturing plan.

- **What the factory owns:** the optical assembly (lens, tracker, mirrors), the valve sections (fabricated to the frozen geometry from Ch. 14 — the fixtures that hold the tolerances are factory tooling), the condenser, the controls, the integration and test. The factory's core competence is *integration quality* — fitting mature subsystems into a tested module.

28.3 The cost curves: 10, 1,000, 1,000,000

Three points, each with its basis stated:

10 units — the demonstration basis. The accounted cost from the TRL 6/7 builds (Ch. 24's ledger method). Hand-built, engineer-supervised, every lesson still being learned. This is the measured anchor — not a number to be embarrassed by, the starting point of every learning curve ever drawn.

1,000 units — early production. The factory exists; the tooling is cut; the supply chain is contracted. Cost computed by applying subsystem learning rates to the anchor:

- *Optics and structures* learn fast — they're manufacturing-light, assembly-heavy; the learning is in the process.
- *Valve sections* learn moderately — the geometry is fixed (Ch. 14), the learning is in fabrication fixtures and yield.
- *The expander* learns at its industry's rate — it's a bought subsystem riding its own curve (Ch. 20's selection included the supplier's maturity for this reason).
- *The pressure boundary* learns slowest — code shops, inspections, and steel don't hurry. The book states this plainly: the stamped vessel is the cost floor's ballast.

1,000,000 units — maturity. Wright's law carried to scale, per subsystem, with the rates and the volume assumptions *stated and challengeable* — the same honesty as §24.3's learning treatment. At this volume the module is an appliance: stamped parts, automated assembly, the test stand in-line. The number at this point isn't a prediction — it's the *target the value-engineering program aims at*, with the curve showing whether the aim is plausible.

The learning assumptions are the fragile part, and they're labeled as such. A reader who rejects the rates recomputes from the anchor — which is public, from the ledger. The book's economics are reproducible; that's the point.

28.4 The manufacturing plan

- **Factory vs. field:** modules factory-built and factory-tested; site work limited to what can't ship — foundations, structural interfaces, interconnection, commissioning. The field crew from Chapters 22/25 becomes the deployment workforce, and their revised manuals become the training curriculum.
- **The supply chain:** mapped per subsystem — lens production (the highest-volume custom part), valve-section fabrication (the tightest tolerance), expander supply (the selected vendor from Ch. 20, dual-sourced where possible), the code shop (capacity-reserved). Single-source risks named with their mitigations; the plan doesn't pretend the chain is robust until it is.
- **The factory test stand:** every module is tested before shipment — the production acceptance test, derived from the Chapter 17 campaign's essential subset (thermal performance spot-check, control verification, pressure integrity, the valve-section flow check). A module that hasn't passed the stand doesn't ship. This is the manufacturing expression of "prove what is possible with actual data": the data is taken on *every unit*, not just the prototypes.

28.5 The quality system

- **Incoming inspection:** the bought subsystems (expander, pressure vessel, lens) arrive with their certificates — and get verified anyway on a sampling plan that tightens when suppliers stumble. Trust, but the receiving dock has instruments.
- **In-process control:** the valve-section geometry gets its dimensional verification (the frozen geometry from Ch. 14 is a quality characteristic, not a suggestion); the optical alignment gets its check; the pressure boundary's code paperwork travels with the module.
- **Nonconformance:** the Chapter 18 failure log grows a manufacturing branch — every defect traced to cause, every cause dispositioned to design, process, or supplier. The "no surprises twice" rule (§18.5) applies on the line: the same defect may not recur unguarded.
- **The qualification program:** the manufactured unit re-runs the essential demonstrations — the performance map (Ch. 17/19's core), the fault injection (17A's core), the transition dynamics (17C) — on production hardware, witnessed and recorded. Qualification is the gate between "the design works" and "the product works," and it's passed with data, not documents.

28.6 Cost-down engineering: the value program

Chapter 24's sensitivity analysis becomes the engineering priority list:

1. **The expander** — the steepest lever (§24.5). The value program's first target: the cost-efficiency frontier of the selected expander class, the supplier partnership, the next-generation specification written from the fleet's data.

2. **The optics** — the largest area-driven cost. Lens manufacturing at volume, tracker simplification from the field's reliability data (the parts that never failed get cheaper; the parts that did get redesigned).
3. **Installation** — the field labor from the demonstration logs, attacked with better manuals, better tooling, and the fleet's accumulated practice.
4. **The pressure boundary** — the slowest lever, worked through the code-shop partnership and design-for-inspection (vessels designed to be *built and inspected* cheaply, not just to hold pressure).

Each target has an owner, a measured baseline, and a verification method — the value program is managed like the test program, because it *is* a test program: the hypothesis is that the cost comes down, and the data says whether it did.

28.7 What this chapter proved

- TRL 8 defined: the actual system completed and qualified — manufacturing real, quality real, every unit tested.
- The production design: one module design for all tiers, the pressure boundary honestly bought, the factory's competence as integration quality.
- The cost curves at 10 / 1,000 / 1,000,000 — the ledger anchor, subsystem learning rates stated and challengeable, the million-unit number as target rather than prediction.
- The manufacturing plan: factory-built modules, the mapped supply chain with named single-source risks, the factory test stand that every module passes.
- The quality system: incoming verification, in-process control on the frozen geometry, the failure log's manufacturing branch, the witnessed qualification program.

- The value program: Chapter 24's sensitivity levers turned into owned engineering targets with measured baselines.

Chapter 29 is the multi-season proof: uptime, degradation, maintenance records, and independent verification — the long data the economics assumed.

Self-evaluation — Gate 8, part 1

1. State TRL 8's shift in one sentence. Why is the factory test stand its manufacturing expression of the book's honesty contract?
2. Your supplier rejects the learning rate you assumed for valve sections. Walk the recomputation from the anchor — what changes, what doesn't?
3. Explain why the pressure boundary learns slowest, and what that implies for the cost floor.
4. A production module fails the factory acceptance test on the valve-section flow check. Trace the disposition per §§28.4–28.5.
5. Name the value program's four targets in priority order and justify the ranking from Chapter 24's sensitivity analysis.

Sources and honesty notes

- Wright's-law learning curves are the established framework for manufacturing cost projection; the subsystem rates are the author's stated assumptions applied to the measured anchor — labeled as assumptions, recomputable by the reader.
- The DFM, quality-system, and qualification practices specified here follow established manufacturing engineering; the reader's program implements them to its industry's standards.

- The "bought stamped" pressure boundary continues Chapter 27's specified honest scope into manufacturing.
- No manufactured units exist for this system yet; the anchor costs, the curves, and the qualification all await the reader's program — the book specifies the method that produces them.

Chapter 29

Multi-Season Proof

TRL 8 · System complete and qualified. This chapter specifies the long data: the multi-year fleet program that verifies the lifetime and degradation assumptions the economics were built on — with independent verification.

29.1 The assumptions come due

Chapter 24 computed the LCOE with two labeled assumptions: the system lifetime and its degradation rate. Labeled, but still assumptions — and the economics are only as honest as the program that retires them. This chapter is that program: the fleet, measured over years, with the decay curves, the maintenance records, and the independent witnesses that turn "assumed 25 years at 0.5%/year" into either a measured fact or a revised number.

The rule from Chapter 18's revision procedure applies at fleet scale: when the long data disagrees with the assumption, the economics get re-run, and every chapter that consumed the assumption gets its revision entry. The LCOE isn't a number — it's a number *with a verification program behind it*.

29.2 The fleet as the experiment

The pilot (Ch. 22–23) proved the year. The fleet proves the *years* — and the fleet is a better experiment than any single site:

- **Population statistics:** MTBF per subsystem computed across hundreds of modules, not inferred from one. The pilot's availability was a data point; the fleet's is a distribution — with the tails (the bad sites, the bad batches) visible instead of averaged away.
- **Climate coverage:** modules in dust, in humidity, in freeze country, in the design hot night's home territory — the degradation rates per climate, so the economics can be priced per market instead of per hope.
- **Generation tracking:** production generations compared — Gen-1 vs. Gen-2 modules, the reliability growth made visible (the fleet learning loop from §26.7, quantified).

The data discipline: every module's telemetry flows for its whole life (the permanent dual view from §22.3, at fleet scale). The database is the program's most valuable asset — more valuable than any single design revision, because it prices every future revision's worth.

29.3 Degradation: what decays, at what rate

The decay curves, per subsystem, measured — not modeled:

- **Optics:** lens transmission vs. time (soiling-adjusted — the 12D doctrine separates the cleanable from the permanent), mirror reflectance, tracker mechanical wear. The permanent decay rate is the number the economics needs; the soiling curve is the maintenance schedule's input.
- **Receiver:** coating absorptance vs. time at temperature — the hottest surface ages fastest, and the fleet watches it.
- **Expander:** the efficiency map re-measured on schedule for sampled units — wear shows up as a drifting map before it shows up as a failure (the FDD heritage from §26.2, now quantifying aging rather than predicting faults).

- **Condenser:** the heat-transfer trending from 21A, continued for years — fouling and corrosion as the slow variables they are.
- **Valve sections:** the diodicity spot-check — the frozen geometry (Ch. 14) verified still frozen, still performing, after years of thermal cycling and chemistry.

Each curve gets its rate, its uncertainty, and its climate dependence. The LCOE's degradation assumption is then *replaced* — per market, per subsystem — and the economics re-run. If the measured decay is worse than assumed, the book says so and the value program (§28.6) gets new targets. The data doesn't negotiate.

29.4 Maintenance records: the population truth

The fleet's maintenance database — every intervention, every part, every labor hour, coded by subsystem and cause:

- **MTBF and MTTR per subsystem**, from population data — the numbers the availability guarantees are written against, the numbers the spare-parts strategy (§26.3) is sized from, the numbers the insurers price (§27.6).
- **The maintenance cost curve over time:** infant mortality (the Ch. 18 taxonomy, at fleet scale) burning off, the steady-state years, and — watched for — the wear-out rise. The curve's shape tells the program where each generation is in its life, and the economics which years cost what.
- **The design feedback:** the top failure modes by cost become the engineering priority list — the fleet's pain, priced, driving the next design revision. This is the "no surprises twice" rule (§18.5) operating at industrial scale.

29.5 Independent verification: the honesty contract, extended

The program's own data is necessary but not sufficient — the book's honesty contract requires *independent* eyes:

- **Third-party testing:** university or national-laboratory test programs running the Chapter 17/19 campaign essentials on production hardware — the efficiency map, the transition dynamics, the inventory accounting — witnessed, published, not funded on a success-only basis.
- **Certification-body witnessing:** the qualification program (§28.5) witnessed by the bodies whose marks the product carries — their independence is the point.
- **The independent engineer's report:** the document the financiers read — performance, degradation, O&M costs, and the risks, from an engineer the program doesn't employ. Bankability runs through this report; the book specifies its contents because the deployment at national scale (Ch. 30) needs capital, and capital needs the independent word.
- **Why it matters:** the open-IP strategy means the whole world can check the work — independent verification is the mechanism by which they do. A book that says "trust us" after twenty-eight chapters of "verify everything" would be a contradiction. The verifiers get the data, the hardware, and the methods — the same ones the reader got.

29.6 Reliability growth: the curve that must bend up

The fleet's reliability is tracked as a growth curve — cumulative MTBF vs. cumulative operating time, per generation. The curve must bend upward: each generation more reliable than the last, the fixes from the failure log compounding. If it doesn't bend, the program has a process problem (the fixes aren't reaching the line) or a physics problem (the design's limits are being met) — and the distinction is itself a finding with its own response.

The growth curve is published — per generation, with the design changes annotated on it. A reliability curve with the fixes labeled is the most convincing document in the program's portfolio: it shows not just that the system works, but that the *organization learns*.

29.7 What the long data licenses

When the multi-season program has run:

- The LCOE's lifetime and degradation assumptions become measured inputs — Chapter 24's economics re-run on the long data, per market.
- The availability guarantees get written against population MTBF, not pilot inference.
- The insurance (§27.6) gets priced on actuarial data, not engineering judgment.
- The independent engineer's report unlocks project finance — the capital for Chapter 30's deployment math.

And if the long data disappoints — faster decay, worse MTBF, a wear-out cliff — the book's machinery handles it the way it handled everything else: the revision procedure, the re-run economics, the value program's new targets, the public report. The honesty contract doesn't expire when the news is bad; that's when it's worth the most.

29.8 What this chapter proved

- The verification program for the LCOE's two labeled assumptions: lifetime and degradation, retired by fleet data or revised by it.
- The fleet as the superior experiment: population statistics, climate coverage, generation tracking — the database as the program's most valuable asset.
- Per-subsystem decay curves measured: optics, receiver, expander, condenser, valve sections — each with rate, uncertainty, climate dependence.
- The maintenance database: population MTBF/MTTR, the cost curve over time, the priced pain driving design priorities.
- Independent verification: third-party testing, witnessed qualification, the independent engineer's report — the mechanism by which the open strategy gets checked.
- Reliability growth tracked and published, fixes annotated — proof the organization learns.
- The long data's licenses: re-run economics, writable guarantees, actuarial insurance, bankability.

Chapter 30 does the deployment math: how many units, where, what it displaces — the nation, powered building by building.

Self-evaluation — Gate 8, part 2

1 State which two Chapter 24 assumptions this chapter retires, and the exact mechanism (data revision re-run) when the fleet data disagrees.

2 The fleet's expander MTBF distribution has a long bad tail from one climate zone. Walk the response: attribution, economics, and design — without averaging the tail away.

3 Explain why the independent engineer's report is specified as a program deliverable rather than left to the financiers to commission.

4 Your Gen-2 reliability curve bends flat while Gen-1 bent up. State the two candidate causes and how the program distinguishes them.

5 The measured degradation is worse than assumed in hot-humid climates only. Walk the consequences for per-market economics and the value program.

Sources and honesty notes

- Reliability-growth tracking (Duane/AMSAA-style cumulative-MTBF methods) is established reliability engineering; the reader's program applies it to the fleet's data.
- The independent-engineer report and bankability practices follow project-finance convention for energy technologies; the book specifies the report's contents as a program deliverable.
- Degradation mechanisms (optical soiling/transmission loss, coating aging, expander wear, fouling) are characterized from the fleet's measurements — the book specifies the trending, not the rates.
- No multi-year fleet data exists for this system yet; the program in this chapter is specified to produce it, and the economics remain assumption-labeled until it does.

Chapter 30

The Nation

TRL 8 · System complete and qualified. This chapter does the deployment math: how many units, where, what it displaces — the vision from Chapter 1, computed with everything the book has measured.

30.1 The vision, computed

Chapter 1 opened with the scaling principle: the building is the unit, the nation is many units. Twenty-nine chapters later, the principle gets its arithmetic — not as a dream, but as a calculation with stated inputs, because the book's honesty contract doesn't expire at the vision chapter. Every number here traces to a measured or specified quantity from an earlier chapter; where it doesn't, it's labeled.

30.2 How many units

The deployment math starts with the building stock and applies the siting filters honestly:

The tiers and their populations:

- *Home tier:* single-family homes with suitable yards/roofs — the filter from Chapter 8's lot-size reality (the aperture is the binding constraint) applied to the housing stock. Not every home qualifies; the book states the suitable fraction as a range, not a wish.

- *Corporate tier:* commercial buildings with suitable roofs or parking structures — the structural and shading analysis from §25.2 as the filter.
- *Apartment tier:* multifamily properties at the 200-unit reference scale (Ch. 9), filtered by roof area per unit and ownership structure.

The suitable fraction: each tier's technical potential multiplied by the filters — solar resource (the DNI threshold below which the economics don't close, from Ch. 24's per-market analysis), siting (shading, structure, space), and the adoption curve (not a technical filter but a real one — the diffusion rate from the fleet's early data). The chapter reports the technical potential and the plausible deployment separately, because conflating them is how vision chapters lie.

The aggregate: suitable units × per-unit annual energy (the measured figures from Ch. 23/25, per tier) = the national energy contribution. Reported as a fraction of national building electricity consumption — the number that says what "power a nation" actually means in this architecture: a large share of *buildings' daytime electricity*, not the entire grid.

30.3 What it displaces

Fossil peakers first: the corporate tier's daytime coincidence (§25.4) displaces the most expensive, dirtiest marginal generation — the peaker plants that run on hot afternoons. The displacement is computed as avoided peaker energy and avoided peaker capacity, priced at the actual market — the demand-charge savings from §25.5 have their grid-scale mirror image.

Grid capacity deferred: every kW of coincident peak shaved is distribution and transmission capacity not built. The deferred-investment value is real utility economics — the book computes it per the utility's own planning methods, not the advocate's.

The lithium not bought: the day/night cycle on water and steel (§24.4) scaled to the nation — the battery capacity the

building stock *doesn't* need for its nights. At national scale this is a materials story: the lithium, cobalt, and nickel not mined, not refined, not replaced every decade. The book states it as avoided material flow, not as environmental poetry.

Carbon: the displaced generation's emissions, computed with the marginal emissions rate (not the average — the honesty rule for displacement claims), reported as avoided tons with the methodology stated.

30.4 The grid as it becomes

A nation of building-scale daytime generators changes the grid — the chapter says how, honestly, including the hard parts:

- **The duck curve's building-scale cousin:** distributed solar-thermal exports reshape the net-load curve like PV did — the midday trough deepens. The difference from PV: the thermal inertia (§26.2) makes this fleet dispatchable within its operating window, and the demand-response doctrine turns the trough from a problem into a scheduled resource.
- **Hosting capacity:** feeders have limits — the interconnection queue at national scale is the deployment's binding constraint in many territories. The chapter states it plainly: the deployment math means nothing where the feeder can't take the export, and the build-out plan (§30.5) sequences markets by hosting headroom.
- **The utility's evolving role:** from energy seller to platform operator — metering, balancing, and backup for a building stock that generates its days. The book doesn't moralize about utilities; it prices their services (the standby charges, the interconnection costs) into the deployment economics, because the grid's costs don't vanish when its energy sales fall.
- **Resilience:** the islanding-capable buildings (§26.5) as the grid's shock absorbers — the outage-ride-through value from §24.4, now at community scale. A

neighborhood of buildings that keep their critical loads through a blackout is civilizational infrastructure, and the book states it as such — with the interconnection rules that permit it, cited.

30.5 The build-out: manufacturing, workforce, capital, time

Manufacturing ramp: Chapter 28's cost curves become the production schedule — the factory count, the module throughput, the years from 1,000 to 1,000,000 units. The ramp is paced by the slowest subsystem (the code shops and the expander supply chain, §28.3–28.4) — the book doesn't let the fast parts set a schedule the slow parts can't keep.

Workforce: the installers, the technicians, the inspectors — headcount per thousand units from the demonstration's labor logs (Ch. 22/25/26), scaled. The deployment is a jobs program as a side effect, and the book states the numbers because they're real: every module installed is crew-hours, every building maintained is a technician's livelihood.

Capital: the investment required per deployment phase — from the per-unit costs (Ch. 28) times the unit counts (§30.2), phased by the manufacturing ramp. Compared against the capital the grid would otherwise spend (the deferred capacity from §30.3) — the deployment financed against its own avoided costs where the structures allow.

Timeline: the honest schedule — TRL 8's qualified system to national scale in phases, each phase gated by the previous phase's fleet data (the reliability growth from §29.6 must hold at each scale step). No phase skips its evidence.

30.6 The honest constraints

The vision chapter lists what bounds it — because a vision without constraints is a fantasy:

1 **Siting:** not every building qualifies — the suitable fractions from §30.2 are the ceiling, and they're below 100% in every tier.

2 **Resource:** steel, glass, and expander supply at million-unit scale — the supply chain's limits (§28.4) as the deployment's limits.

3 **Grid:** hosting capacity per feeder, interconnection queues, the utility rules that decide islanding (§26.5/30.4).

4 **Policy:** net metering's evolution, interconnection reform, the boiler-code jurisdictions (§27.3) — the deployment needs policy that permits it, and the book names the policies rather than assuming them.

5 **Capital:** the build-out's financing — the independent engineer's report (§29.5) unlocks it, but the market prices it.

None of these is fatal — each is a bounded problem with a named owner and a chapter that addressed it. A vision with listed constraints is a plan; without them, it's a poster.

30.7 What this chapter proved — and what TRL 8 established

- The deployment math: building stock × siting filters × measured per-unit energy = the national contribution, technically and plausibly separated.
- The displacement: fossil peakers, deferred grid capacity, unmined lithium, marginal-rate carbon — each computed with stated methodology.
- The grid as it becomes: the net-load reshaping (with the fleet's dispatchability as the honest difference from PV), hosting capacity as the binding constraint, the utility priced as platform, resilience at community scale.
- The build-out: manufacturing ramp paced by the slowest subsystem, workforce from measured labor logs, capital phased against avoided costs, timeline gated by fleet evidence.

- The honest constraints: siting, resource, grid, policy, capital — five bounded problems with named owners.

TRL 8 is complete. The unit was manufactured and qualified (Ch. 28), its long data specified and independently verified (Ch. 29), and its national deployment computed with constraints listed (Ch. 30). TRL 9 — the proven system in full operation — is the fleet at scale, and the book's final chapters turn from the technology to its stewardship: the open-IP commons, the builder's handbook, and the work that remains.

GATE 8 — Exit evaluation

The reader may not proceed to TRL 9 until every answer below is theirs.

From Chapter 28:

1 Explain the 10/1,000/1,000,000 cost-curve method: what's measured, what's assumed, and what a skeptic recomputes.

2 A production module fails factory acceptance on the valve-section flow check. Trace the disposition.

From Chapter 29:

1 State the two Chapter 24 assumptions this chapter retires and the mechanism when fleet data disagrees.

2 Why is the independent engineer's report a program deliverable, and what does it unlock?

From Chapter 30:

1 Compute the deployment's technical potential vs. plausible deployment for one tier, naming every filter and its source chapter.

2 The feeder hosting capacity in your best market is exhausted. State the consequences for the build-out sequence and which §30.6 constraint governs.

3 Explain the marginal-vs.-average emissions rule for the carbon displacement claim — and what changes if it's violated.

Gate 8 exit criteria: all seven answered; manufactured units qualified with witnessed data; multi-season program running with independent verification engaged; deployment math computed with constraints listed; the economics re-runnable on measured inputs. Then TRL 9 — the proven system, and the book's closing.

Sources and honesty notes

- Building-stock figures, DNI resource data, tariff structures, and marginal emissions rates are market data the reader sources current — the book specifies the calculation method, not the inputs, which move.
- The technical-potential vs. plausible-deployment distinction is the author's specified honesty rule for vision-scale claims.
- The displacement methodologies (peaker displacement, deferred capacity, marginal emissions) follow standard energy-planning practice; the reader applies them with their market's data.
- No national deployment exists for this system yet; this chapter computes the deployment to plan, not one that happened.

TRL 9

Proven System

Chapter 31

Chapter 31

What Remains

TRL 9 · System proven through successful mission operations. This chapter states the open questions plainly, defines what "proven" means for this technology, and hands the work to the next builder.

31.1 What TRL 9 means here

TRL 9 is the top of the climb: the actual system proven through successful mission operations. For a spacecraft, the mission is the flight. For this technology, the mission is quieter and longer: **buildings powered, year after year, the data public, the economics proven** — the fleet at scale doing exactly what thirty chapters said it would do, with the records to show it.

Proven doesn't mean finished. It means the system's behavior is *known* — its performance characterized, its failure modes priced, its economics measured, its limits mapped. TRL 9 is the state where the remaining unknowns are listed rather than lurking. This chapter is that list.

31.2 The open questions, stated plainly

1 **Long-term degradation, per climate.** The multi-season program (Ch. 29) is running; its early data is in, its decades aren't. The decay curves have measured beginnings and extrapolated ends — the extrapolation is labeled, and the fleet keeps measuring. The question closes one climate-year at a time.

1 **The expander frontier.** Chapter 20 asked the small-turbine question; the fleet answered it for the current generation. The frontier — higher efficiency at small scale, lower cost at the selected efficiency — advances with every generation, and the value program (§28.6) keeps it as its first target. Open because physics keeps offering and manufacturing keeps learning.

1 **The cost at true scale.** Chapter 28's million-unit curve is a target with a method, not a receipt. The learning rates are being validated against actual production data — some holding, some revised (the revision procedure doesn't retire at TRL 9). The question closes as the factories count their millionth module.

1 **Grid integration at national penetration.** Chapter 30 listed hosting capacity as the binding constraint and the utility's evolution as an open negotiation. High-penetration feeders are the experiment now — the interconnection rules, the platform economics, the resilience value, all being worked out in the markets that deploy first. Open because the grid is a social machine as much as a technical one.

1 **The policy environment.** Net metering's evolution, interconnection reform, boiler-code modernization (§27.3) — the deployment needs policy that permits it, and policy moves on its own schedule. The book named the needed policies rather than assuming them; the question is whether they're enacted, and the builders' association (below) is the instrument for asking.

1 **The climates and failure modes not yet seen.** Every fleet has its first hurricane, its first hailstorm beyond the design basis, its first failure mode the FMEA missed. The failure log (Ch. 18) never closes — at TRL

9 it's a mature institution, not a startup practice, and the "no surprises twice" rule is the closest thing the program has to a guarantee.

None of these is a reason to wait. Each is a reason to *build*, because every one of them closes faster with a fleet measuring than with a committee theorizing. The book's whole argument, compressed: the questions are listed so the building can start.

31.3 The invitation to the next builder

The reader who climbed all nine levels — who answered the gates honestly, who ran the numbers with their own inputs — is now qualified to do the work. The invitation is specific:

What the book gives you:

- The methods — every test, every procedure, every doctrine, specified to be followed.
- The data — the ledgers, the decay curves, the failure log, the fleet's public record.
- The designs — the module, the geometries, the integration, published open.
- The economics — the LCOE method with measured inputs, re-runnable for your market.
- The legal map — the certification program, the jurisdiction research, the safety case template.

What it asks of you:

- The honesty contract continues. Your data becomes the next edition's evidence — the measurements you take, the failures you log, the assumptions you retire. The book's value was "the more truth we centralize" (the Value Principle); the commons grows by every builder's contribution.

- Publish the failures. The failure log's value was never the successes — it was the priced, attributed, non-repeating failures. Your surprises become everyone's avoided surprises, but only if you write them down.
- Respect the stamps. The pressure boundary is bought certified (Ch. 27); the structural analysis is stamped; the interconnection is witnessed. Open hardware is not informal hardware — the law's requirements travel with the designs.

31.4 The open-IP commons

The publication strategy (the author's deliberate choice — no patent budget, open disclosure) makes this book infrastructure, not just documentation:

- **Defensive disclosure:** the published designs are prior art — the commons can't be fenced by a later filer on the same integration. The book's precision (the patent offices were named as readers in the author's strategy) serves this function directly.
- **The builders' commons:** the designs, the data, the failure log, the procedures — shared, improved, re-shared. The fleet learning loop (§26.7) extended across organizations: every builder's telemetry strengthening every builder's economics.
- **The standards path:** mature commons become standards — the valve-section geometry, the acceptance test, the safety case format. The book specifies them as candidates; the builders' association (the instrument from §31.2's policy question) carries them through the standards process.

The commons is the answer to the question "who owns this?" — everyone who builds it, under the honesty contract. The author's contribution was the integration and this book; the readers' contribution is everything after.

31.5 The contract, fulfilled and transferred

Thirty-one chapters. The contract from Chapter 1 — real data, real math, gaps named plainly — kept in every one:

- Everything claimed is labeled: measured, calculated, assumed, or hypothesized — and the labels were never mixed.
- Everything assumed was flagged: the [VERIFY] marks, the stated learning rates, the labeled LCOE inputs — each with the program specified to retire it.
- Everything proven was evidenced: the test campaigns, the fleet data, the independent verification — the gates passed with data, not documents.

The Reader Principle guided the structure: the reader's own stream of consciousness evolved through the TRLs, and the collective of readers-builders actualizes the invention. That was the author's bet — that a book rigorous enough to convince would create the builders rigorous enough to build. The climb was the mechanism; the builders are the outcome.

What remains is the work. The questions are listed. The methods are specified. The commons is open. The next builder — the reader who answered the gates — has everything the book can give.

Build it. Measure it. Publish the failures. The nation is many units, and the first unit is yours.

GATE 9 — Exit evaluation

The final gate. The reader who answers these has completed the climb.

From the whole book:

1 State the book's honesty contract in one paragraph, then name three specific places where it constrained what the book claimed.

2 Explain the integration's TRL argument (the author's insight): why mature components don't make a mature system, and what the climb actually proved.

3 Your market's DNI is 15% below the reference. Walk the full recomputation: which chapters' numbers change, which methods stay, what the economics say.

4 The fleet's Gen-3 data shows a failure mode the FMEA missed. Walk the complete response: log, attribution, revision, economics, publication — citing the chapter that governs each step.

5 State the open-IP strategy's two functions (defensive and generative) and the reader's obligations under the honesty contract.

6 Name the six open questions from §31.2 and, for each, state what closes it and on what timescale.

7 The book is done and the building starts. State your first three actions as the next builder, in order, with the chapter that specifies each.

Gate 9 exit criteria: all seven answered — in writing, with the reader's own market data where the question requires it. There is no higher gate. The work continues in the world, not in the book.

Sources and honesty notes

- TRL 9's definition follows the standard scale: the actual system proven through successful mission operations — the mission here specified as sustained building-scale operation with public data.
- The six open questions are the author's specified inventory of remaining unknowns — presented as the bounded list that TRL 9 requires, not as disqualifiers.

- The open-IP strategy (deliberate publication, no patent budget) is the author's stated choice; the defensive-disclosure function is the standard legal effect of precise public disclosure, not a guaranteed outcome.
- The patent disclaimer from the author's strategy stands: citation is formality, not certification of inventorship or claim validity.
- No TRL 9 fleet exists for this system yet; this chapter defines the proven state and lists what closes the remaining questions — the reader's program earns it.

Appendix A

Patent Register

*Prepared: **September 26, 2026** · Purpose:* the book's patent register — every component traced to its patents, per the founder's position: "we are not claiming to have invented any of the components; we just centralized existing patents."*

***CAVEAT — READ FIRST:** This register is a research landscape scan compiled from public patent databases (Google Patents, USPTO, Espacenet). It is **NOT a freedom-to-operate legal opinion**. Patent status displays are Google's estimates ("legal status is an assumption, not a legal conclusion"); expiry estimates follow the standard rules (US: 17 years from grant if filed before June 8, 1995, else 20 years from filing; EP/CN invention patents: 20 years from filing; CN utility models: 10 years). **Active patents may require licensing** — a patent attorney must confirm claim scope and in-force status before any commercial build. Suffix key: "A" (pre-2001 US) = granted patent; "A1" = published application (may never have granted); "B1/B2" = granted patent.* `[VERIFY]` *marks details not confirmed in an authoritative record.*

CORE COMPONENTS

1. Tesla valvular conduit (phase-directed flow: steam one way, liquid the other)

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US1329559A	Valvular Conduit	Nikola Tesla (individual)	1916-02-21	1920-02-03	**EXPIRED** 1937-02-03	The original fluidic diode, no moving parts — the geometry all later work builds on. Public domain since 1937.
FR540616D	(French analogue)	Tesla	1921 (priority)	—	**EXPIRED**	French counterpart, long expired.
US5876187A	Micropumps with fixed valves	Univ. of Washington	1995-03-09	1999-03-02	**EXPIRED** 2016-03-02	Canonical modern Tesla-valve improvement; explicitly cites US1329559.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US5303275A	Forced-circulation reactor with fluidic-diode-enhanced natural circulation	General Electric	1991-06-13	1994-04-12	**EXPIRED** 2011-04-12	Fluidic diodes in nuclear-reactor cooling — an expired power-adjacent application.
US9903536B2	Passive diode-like device for fluids	Johns Hopkins Univ.	2015-04-10	2018-02-27	**LAPSED** (fee-related; term would have run to 2036)	Topology-optimized Tesla-valve diodes integrated into oscillating heat pipes / two-phase flow.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US10602638B2	System and method of transferring energy	Dell Products LP	2018-03-01	2020-03-24	**ACTIVE** 2038-03-01	Closest analogue: Tesla valve between evaporating and condensing regions transferring fluid in the *gaseous state*, water named as a working fluid. Claims are tied to electronics-cooling housings, not boilers or solar systems.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US11275136B2	Passive flow direction biasing of cryogenic thermosiphon	Philips (Koninklijke Philips N.V.)	2017-10-04 (PCT)	2022-03-15	**ACTIVE** ~2039	Tesla valves as passive one-way valves in a two-phase thermosiphon — but cryogenic helium for superconducting magnets, distant field. Family: EP3523582B1, CN109804208B, JP7096238B2.
EP4302042B1	Micro-channel heat exchanger	Wieland Provides Srl	2022-02-28	2024-06-12	**ACTIVE** 2042-02-28 (Europe)	Tesla valve as unidirectional liquid-phase circulation means in a two-phase evaporator — phase-selective directional flow, Europe only, electronics cooling.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
CN218955545U	Evaporator and loop heat pipe	Guangdong Yingweike Tech	2022-12-06	2023-05-02	**ACTIVE** 2032-12-06 (CN utility model, 10-yr)	Loop heat pipe with Tesla-valve structures splitting liquid and vapor — clearest published example of a Tesla valve directing fluid *by phase* in a closed evaporate-condense loop.

Finding: No granted patent was found applying a Tesla valve / fluidic diode to a steam boiler, steam Rankine cycle, or steam-turbine power generation. The book's application — phase-directed day/night steamwater cycling in a solar steam loop — appears to be an open, previously-unclaimed application space.

2. Parabolic-profile Fresnel lens (refractive solar concentrator)

Patent	Title	Assignee	Filed	Granted	Status	Relevance
(1822)	Fresnel lighthouse lens	Augustin-Jean Fresnel	—	—	**EXPIRED** (historical)	Foundational refractive optics; no patent number verified — cite as historical prior art, not a patent.
US3125091A	Inflatable solar energy collector	Harvey P. Sleeper, Jr. (individual)	1962-09-17	1964-03-17	**EXPIRED** 1981-03-17	Earliest verified solar-thermal Fresnel patent; one embodiment puts a boiler at the lens focus.
US4069812A	Solar concentrator and energy collection system	E-Systems / Entech (O'Neill)	1976-12-20	1978-01-24	**EXPIRED** 1995-01-24	Foundational modern refractive Fresnel design—point/line focus, prism geometry, thermal receiver at focus.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US4848319A	Refracting solar energy concentrator and thin flexible Fresnel lens	3M (Appeldorn)	1988-12-09	1989-07-18	**EXPIRED** ~2006	Thin flexible refractive Fresnel lens over an elongated thermal target — directly on point for lightweight building-scale lens construction.
US5870233A	Process for manufacture of Fresnel lenses	Röhm GmbH	1993-12-24 (priority)	1999-02-09	**EXPIRED** 2014-12-27	Manufacturing process for embossed plastic linear Fresnel lenses; supports the book's low-cost manufacturing argument.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US8689784B2	Solar concentrator system	J. M. Monsebroten (individual)	2006-09-14	2014-04-08	**STATUS [VERIFY]** — base 20-yr term expired 2026-09-14 (12 days before this research); PTA and maintenance-fee payment unknown	Nearest single reference to the book's integration: Fresnel-lens arrays + collimating lenses + mirror beam-network + dual-axis tracking + steam-turbine generation in one disclosure. Check USPTO Patent Center before reliance.

Open lead: US11637962B2 ("Solfire solar concentrator" — Fresnel arrays boiling liquid to steam) — search-extract only, not verified in this pass.

3. Tracking mirrors / heliostat boost stage (dual-axis)

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US4317031A	Central focus solar energy system	Max Findell (individual)	1980-06-20	1982-02-23	**EXPIRED** 1999-02-23	Foundational two-axis heliostat array — mirrors, tracker error signals, electromechanical drives, receiver targeting.
WO2008121335A1	Heliostat with integrated image-based tracking controller	eSolar, Inc.	2008-03-28 (PCT)	(application)	PCT ceased; family grants EP2145137B1, CN101680685B — status **[VERIFY]**	Mirror-mounted imaging with closed-loop control orienting the mirror normal between Sun and receiver vectors.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US12613394B2	Actively focused lightweight heliostat	Univ. of Arizona	2020-09-28 (priority 2019-09-26)	2026-04-28	**ACTIVE** 2043-01-16	Dual-axis heliostat with actively deformable reflector. Relevant only if the build uses deformable focusing mirrors; a plain flat-mirror heliostat is a different feature.
EP2828611B1	Heliostat	Balluff, Inc. (Clyburn)	2013-03-22	2020-11-18	**Shown "Not-in-force" [VERIFY via EP Register]**; statutory expiry 2033-03-22	Altazimuth dual-axis heliostat; its specification defines a heliostat's purpose as focusing sun "to produce steam that drives turbines" — the exact boost-mirror role.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
WO2016007011A1	Heliostat system and method	[VERIFY] (PCT/NL2015/050508)	2015-07-10 (PCT)	(application)	PCT record; national family **[VERIFY]**	Sensor-fused closed-loop tracking (gravity + sun-spot imaging) — self-calibrating heliostats.

4. Closed-loop boiler / solar steam generation

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US4265223A	Method and apparatus for utilizing solar energy	The Badger Company	1978-09-18 (priority) [VERIFY]	1981-05-05	**EXPIRED** ~1998	Foundational solar-thermal steam patent, repeatedly cited by later solar steam work; early closed-loop solar steam prior art.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US6957536B2	Systems and methods for generating electrical power from solar energy	The Boeing Company	2003-06-03 [VERIFY]	2005-10-25	**EXPIRED** ~2023	Corporate-era solar-thermal power generation system bridging 1970s work and modern CSP.
EP2000669B1	Solar concentration plant for the production of superheated steam	Abengoa Solar New Tech.	2008-06-04	2015-06-17	Nominally **ACTIVE** ~2028 **[VERIFY in-force via EPO Register]**	Separates evaporation and superheating with a steam drum; delivers turbine-grade superheated steam — the modern utility template for a solar boiler feeding a turbine.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US8935922B2	Steam generation device	[VERIFY assignee]	[VERIFY]	2015-01-20	Plausibly **ACTIVE** ~2032–2033 [VERIFY]	Tracked Fresnel-lens array boiling water, flash-steam vessel, turbine power, radiator + condenser recovering condensate — closest match found to lensstearmturbinecondensation.

5. Small-scale steam turbine / expander

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US1061206A	Turbine.	Nikola Tesla (individual)	1911-01-17	1913-05-06	**EXPIRED** 1930-05-06	Foundational bladeless radial turbine with steam as an identified working medium — prior art behind every small steam expander.
US4258551A	Multi-stage, wet steam turbine	Biphase Energy Systems	1979-03-05	1981-03-31	**EXPIRED** ~1998	Turbine designed for wet/two-phase steam — relevant where superheat is limited and exhaust is partially condensed.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US7487641B2	Microfabricated Rankine cycle steam turbine	Columbia Univ.	2004-11-15	2009-02-10	**Likely EXPIRED** ~2026-03 [VERIFY via USPTO]	Integrated micro steam-turbine power plant (chip-scale MEMS; not 1–50 kW — do not claim a kW rating for it).

Note: A dedicated 1–50 kW turbine patent with a documented power rating was not confirmed in this pass; the book should not assert kW ratings without a cited figure.

6. Passive night condensation / water recovery loop

Patent	Title	Assignee	Filed	Granted	Status	Relevance
EP0794401A2	Steam condensing apparatus	Hudson Products Corp.	1997-02-14	(A2 application)	Application; any grant expired ~2017 **[VERIFY whether a B grant issued]**	Air-cooled condenser with condensate collected in the lower header for reuse — passive steam-to-water recovery.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US11504814B2	Air cooled condenser and related methods	Holtec International	2020-10-19 (priority chain 2011-04-25)	2022-11-22	**ACTIVE** 2032-08-27	Modern modular ACC with hermetic steam path and condensate header returning water to a Rankine cycle — passive cooling + closed-loop water return.
CN104018899B	Air-cooled condenser of power plant peak cooling device	State Grid Corp of China	2014-05-16	2016-08-24	**ACTIVE** 2034-05-16	Air-cooled condenser with closed-loop arrangements — passive heat rejection keeping working fluid in a sealed loop.

7. Combination patents (lens concentration steam turbine)

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US5632147A	Solar powered steam turbine generator	William Greer (individual)	1996-04-10	1997-05-27	**EXPIRED** 2016-04-10	Fresnel lenses focus sunlight onto a hollow rotor containing working fluid; steam jets rotate the rotor — self-contained closed-loop solar steam turbine generator. verified
CN107327377A	Small complementary solar steam power generation system	(Shi Yang)	[VERIFY]	Published 2017-11-07	**INACTIVE** — published application, never granted	Fresnel lens onto a steam cylinder; water expands to steam driving a generator; excess heat to thermal storage. verified as dead

Patent	Title	Assignee	Filed	Granted	Status	Relevance
EP3066340B1	System for steam power generation from solar radiation	Solar Steam Limited (Boyle/Perini)	2014-11-10	2020-05-13	**GRANTED**; nominal expiry 2034-11-10 **[VERIFY in-force per state]**	The one combination patent plausibly in force: Fresnel-lens array absorber steam electrical power, water "recyclable back to the header tank" (closed loop); cites US5632147 as prior art. Enforceability depends on validated states + renewal fees.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US4229941A	Method of and system for generating energy from solar and wind energy sources	Solwin Industries (Hope)	1978-01-25	1980-10-28	**EXPIRED** ~1998	Parabolic collector/Fresnel optical path, boiler, steam turbine, reservoir; escaped steam condenses and returns — expressly a closed fluid system. verified

Assessment (research observation, not legal advice): No verified patent claims the book's specific combination — parabolic-profile Fresnel lens as primary concentrator + independent heliostat boost mirrors + closed-loop boiler/steam-turbine + Tesla-valve phase-directed water recovery. EP3066340B1 is the closest in-force combination reference; it does not describe Tesla-valve phase-direction management.

SUPPORTING COMPONENTS

8. Electrical generator / alternator (turbine-coupled)

Commodity verdict: fully mature — buy commercial alternators off the shelf; no license needed.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US5376827A	Integrated turbine-generator	General Electric (Hines)	1993-05-27	1994-12-27	**EXPIRED** ~2013	Counter-rotating integrated turbine-generator on a common shaft — most-cited modern turbine-generator coupling patent.
US5811201A	Power generation system utilizing turbine and fuel cell	Southern California Edison	1996-08-16	1998-09-22	**EXPIRED** 2016-08-16	Compressor + generator + turbine on a single shaft in a 24–50 kW microturbine — the canonical small-scale arrangement at the book's power scale.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US5118961A	Turbine generator	S&W Holding (Gamell)	1990-09-14	1992-06-02	**EXPIRED** 2010-06-02	Fluid-driven turbine rotor with integrated brushless generator; cites a fully-expired foundational family (US2984751, US3039007, US3157793, US2743375, US4293777).

9. Boiler feed pump (closed Rankine loop)

Commodity verdict: mature — multistage centrifugal feedwater pumps are standard products (KSB, Sulzer, Flowserve, Grundfos).

Patent	Title	Assignee	Filed	Granted	Status	Relevance
CH637185A5	Centrifugal pump	Sulzer AG (Medgyesy)	1978-12-21	1983-07-15 (pub.)	**EXPIRED** ~1998	Back-to-back multi-stage high-pressure centrifugal pump architecture underlying modern boiler feed pumps.
EP2719900B1	Pump	Sulzer Management AG	2013-08-16	2020-07-15	**ACTIVE** 2033-08-16 (Europe)	Modern multi-stage high-pressure barrel-casing pump; an incremental design improvement — buy, don't clone, while in force in Europe.

10. Steam pressure relief / safety valves

Commodity verdict: ancient and fully mature — spring-loaded pop safety valves date to the 1800s; buy ASME-stamped hardware (Teledyne Farris, Anderson Greenwood, Crosby).

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US4462420A	Safety pressure relief valve	Teledyne Farris (Cullie)	1982-06-14	1984-07-31	**EXPIRED** ~2002	Full-nozzle pilot-operated relief valve — the workhorse design from the industry-standard boiler safety valve maker.
EP1269052B1	Pilot operated relief valve	Tyco / Pentair (Conley)	2001-03-13	2007-09-05	**EXPIRED** 2021-03-13	Modern integral-diaphragm pilot-operated safety relief valve with modulating action.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US4612958A	Boiler safety valve installations	Babcock Power (Campbell)	1984-02-17	1986-09-23	**EXPIRED** ~2004	How boiler safety valves are actually installed (vent-pipe geometry inducing sub-atmospheric back pressure) — practical build engineering.

11. High-temperature high-pressure piping and insulation

Commodity verdict: mature, multi-sourced — pre-insulated steam pipe, calcium silicate, mineral wool, aerogel blankets are standard industrial products.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US4698887A	Method for fabricating thermally insulated pipeline section	Dresser-Shaw (Patell/McConkey)	1986-09-19	1987-10-13	**EXPIRED** ~2004	Factory-prefabricated insulated pipeline sections specifically for steam above 400°F — directly applicable to building steam runs.
US7868083B2	Aerogel/PTFE composite insulating material	W. L. Gore & Associates	2008-01-11	2011-01-11	**EXPIRED** 2025-06-12	Aerogel-based high-performance thermal insulation — the modern benchmark for thin high-temperaturepipe insulation; public domain as of June 2025.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
EP3835682B1	Solar-heat-collecting pipe	Toyota Industries / JFCC	2019-07-26	2022-04-13	Granted but shown "**Not-in-force" [VERIFY]**; term would run to 2039-07-26	High-temperature sunlight-to-heat conversion coatings on ferrous pipe — relevant to hot-side receiver piping.

12. Solar receiver / absorber at the focal point

Patent	Title	Assignee	Filed	Granted	Status	Relevance
EP0552732B1	A central solar receiver	Yeda R&D (Kribus/Karni)	1993-01-20	1996-05-08	**EXPIRED** 2013-01-20	Foundational volumetric cavity receiver — spike-array absorbers behind a radiation-admitting window; the architecture a focal-point boiler receiver descends from.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US10648697B2	Systems and methods for direct thermal receivers using near blackbody configurations	Alliance for Sustainable Energy (NREL)	2018-03-05 (priority 2014-05-15)	2020-05-12	**ACTIVE** ~2035	Near-blackbody cavity receiver with internal absorber panels carrying heat-transfer-medium channels — directly the "sunlight cavity fluid/steam" architecture.
US20220235746A1	Concentrated solar power generation using solar receivers	247Solar Inc	2022-02-07 (priority 2010-09-16)	(application, 2022-07-28)	**PENDING** — not a granted patent; reference as a disclosed design	Small-scale receiver — low-pressure chamber, aperture, transparent window, absorber impinged by concentrated flux, aimed at distributed power generation.

13. Selective absorber coatings

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US4055707A	Selective coating for solar panels	NASA (McDonald)	1975-12-22	1977-10-25	**EXPIRED** 1994-10-25	Foundational black-chrome selective coating — the original high-α/low-ε benchmark all later coatings cite.
EP2721353B1	Solar selective absorber based on double nitride composite material	ENEA	2012-06-14	2017-03-22	**ACTIVE** 2032-06-14	Graded WN–AlN double-nitride cermet; α95%, ε10.5% at 550°C — production-proven high-temperature receiver coating.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
EP4189304B1	Spectrally selective solar absorber coating	ENEA	2021-07-30	2024-09-04	**ACTIVE** 2041-07-30	Latest-generation SSAC for evacuated pipes (to 550°C) and air-operating pipes (to 300°C) — directly applicable to an open-cavity focal receiver.

14. Air-cooled condenser / heat-rejection heat exchanger

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US4905474A	Air-cooled vacuum steam condenser	M. W. Larinoff (individual)	1988-06-13	1990-03-06	**EXPIRED** 2007-03-06	Foundational A-frame finned-tube ACC for steam-turbine cycles — the basic architecture everything later refines.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US11504814B2	Air cooled condenser and related methods	Holtec International	2020-10-19	2022-11-22	**ACTIVE** 2032-08-27	Modern extruded integral-fin tube construction plus complete power-plant ACC assembly methods.
EP3472545B1	All-secondary air cooled industrial steam condenser	Evapco Inc	2017-06-21	2021-12-15	**ACTIVE** 2037-06-21	Latest large-scale ACC design—benchmark finned-tube dimensions and design logic transferable to smaller exchangers.

Honesty note for the book: ACC patents are overwhelmingly utility-scale; a building-scale unit would likely use adapted HVAC/heat-exchanger construction — state this plainly.

15. Dual-axis solar tracker hardware (drives, controllers, sun sensors)

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US4172739A	Sun tracker with dual axis support for diurnal movement and seasonal adjustment	Solar Homes / Tassen	1977-12-27	1979-10-30	**EXPIRED** ~1996	Foundational dual-axis tracker — azimuth shaft + tilt adjustment; widely cited by later tracking patents.
US8946608B2	Solar tracking system	Suncore Photovoltaics (Sherman)	2013-08-19 (priority 2008-02-01)	2015-02-03	**EXPIRED** (fee-related)	Most-cited modern dual-axis tracker (171 citations) — jackscrew azimuth/elevation with predictive kinematic sun-position model.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US10895316B2	Slew drive with dust cover and torque transduction	Kinematics (Plesniak)	2017-06-20	2021-01-19	**ACTIVE** ~2037	Slew-drive gearbox hardware for azimuth/elevation actuation — Kinematics is the world's largest slew-drive maker and actively patents tracker actuation; be aware when designing motorized drives.
US9052385B2	Sun tracking device and solar cell system	Foxconn (Tseng)	2014-04-18	2015-06-09	**EXPIRED** (fee-related)	Modern sun-sensor/control device — the sensor side of dual-axis tracking control.

16. Power electronics / inverter (grid-compatible AC output)

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US7796412B2	Method and apparatus for converting direct current to alternating current	Enphase Energy (Fornage)	2006-03-23	2010-09-14	**EXPIRED** ~2026	Foundational Enphase microinverter — distributed DCAC with H-bridge switching; topology adaptable to small turbine-generator output (rectify to DC, invert to grid AC).
US9318974B2	Multi-level inverter with flying capacitor topology	SolarEdge Technologies	2014-03-26 (priority)	2016-04-19	**ACTIVE** ~2034	SolarEdge's active flying-capacitor multi-level inverter — adaptable to conditioning small rotating-generator output; be aware of the active family.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US9673630B2	(Ampt dual-mode power optimizer) [title VERIFY]	Ampt, LLC	2007 (priority)	2017	**ACTIVE** ~2027	Highest-risk flag: dual-mode (buck+boost) DC optimizer with MPPT; Ampt asserted itagainst SolarEdge at the US ITC in 2022. Know it before designing DC-optimizer/MPPT front ends.

Patent	Title	Assignee	Filed	Granted	Status	Relevance
US5951785B2	Photo-voltaic apparatus	SANYO Enphase Power Bridge	~1996	1999-09-14	**EXPIRED** ~2016	Historical foundation of the AC-module/microinverter concept — good "centralized prior art" citation showing even major manufacturers' foundational IP has expired.

Note: these inverter patents are DC-AC solar topologies adapted to turbine output via rectification; the turbine-generator itself is covered under component 8.

SUMMARY TABLE — component primary patent(s) status

Component	Primary patent(s)	Status
1. Tesla valvular conduit	US1329559A (Tesla, 1920)	**EXPIRED** 1937
	US10602638B2 (Dell vapor-transfer)	**ACTIVE** 2038
2. Fresnel lens concentrator	US4069812A (O'Neill/E-Systems) · US4848319A (3M)	**EXPIRED**

Component	Primary patent(s)	Status
	US8689784B2 (Monsebroten)	**STATUS [VERIFY]** — base term ended 2026-09-14
3. Heliostat / tracking mirrors	US4317031A (Findell)	**EXPIRED** 1999
	US12613394B2 (Univ. of Arizona)	**ACTIVE** 2043 (only if deformable mirrors used)
4. Closed-loop solar boiler	US4265223A (Badger)	**EXPIRED** ~1998
	EP2000669B1 (Abengoa)	Nominally active ~2028 **[VERIFY]**
5. Small steam turbine	US1061206A (Tesla, 1913)	**EXPIRED** 1930
	US4258551A (Biphase, wet steam)	**EXPIRED** ~1998
6. Night condensation loop	US4905474A (Larinoff ACC)	**EXPIRED** 2007
	US11504814B2 (Holtec ACC)	**ACTIVE** 2032
7. Combination (lenssteamturbine)	US5632147A (Greer)	**EXPIRED** 2016
	EP3066340B1 (Solar Steam Ltd)	**GRANTED** 2034 **[VERIFY in-force]**
8. Generator / alternator	US5376827A (GE) · US5811201A (SCE)	**EXPIRED**
9. Boiler feed pump	CH637185A5 (Sulzer)	**EXPIRED** ~1998
	EP2719900B1 (Sulzer)	**ACTIVE** 2033 (Europe)
10. Safety / relief valves	US4462420A (Teledyne Farris)	**EXPIRED** ~2002
11. HT/HP piping & insulation	US4698887A (Dresser-Shaw) · US7868083B2 (Gore aerogel)	**EXPIRED** (~2004 / 2025)

Component	Primary patent(s)	Status
12. Receiver / absorber	EP0552732B1 (Yeda)	**EXPIRED** 2013
	US10648697B2 (NREL)	**ACTIVE** ~2035
13. Selective coatings	US4055707A (NASA black chrome)	**EXPIRED** 1994
	EP4189304B1 (ENEA)	**ACTIVE** 2041
14. Air-cooled condenser	US4905474A (Larinoff)	**EXPIRED** 2007
	EP3472545B1 (Evapco)	**ACTIVE** 2037
15. Tracker hardware	US4172739A (Solar Homes)	**EXPIRED** ~1996
	US10895316B2 (Kinematics slew drive)	**ACTIVE** ~2037
16. Power electronics / inverter	US7796412B2 (Enphase)	**EXPIRED** ~2026
	US9318974B2 (SolarEdge) · US9673630B2 (Ampt)	**ACTIVE** ~2034 / ~2027

ACTIVE-PATENT WATCH LIST (builders should be aware — not legal conclusions)

1. **EP3066340B1** (Solar Steam Ltd) — Fresnel-lens steam power, closed water loop; granted 2020, nominal expiry 2034. Closest in-force combination reference. Verify validated states + renewals.
2. **US10602638B2** (Dell) — Tesla valve vapor transfer between evaporator/condenser; active to 2038. Electronics-cooling claims; plausibly does not read on solar steam, but read the claims.
3. **US8689784B2** (Monsebroten) — Fresnel arrays + mirror beam-network + tracking + steam turbine; base term ended Sept 14, 2026 — check PTA/fees at USPTO Patent Center.

4 **US12613394B2** (Univ. of Arizona) — actively focused heliostat; active to 2043. Only relevant if using deformable mirrors.

5 **US10648697B2** (NREL) — near-blackbody cavity receiver; active to ~2035.

6 **EP4189304B1** / **EP2721353B1** (ENEA) — selective absorber coatings; active to 2041 / 2032.

7 **US11504814B2** (Holtec) / **EP3472545B1** (Evapco) — air-cooled condensers; active to 2032 / 2037.

8 **EP2719900B1** (Sulzer) — high-pressure pump; active to 2033 (Europe). Buy, don't clone.

9 **US10895316B2** (Kinematics) — slew drives; active to ~2037.

10 **US9318974B2** (SolarEdge) / **US9673630B2** (Ampt) — inverter/optimizer topologies; active to ~2034 / ~2027.

[VERIFY] FOLLOW-UPS (before the book publishes)

- USPTO Patent Center: legal status, PTA, maintenance history for **US8689784B2**; current status of **US7487641B2**.
- EPO Register: in-force/validated-state status for **EP3066340B1**, **EP2000669B1**, **EP2828611B1**, **EP2145137B1**; national status of **CN101680685B**.
- Confirm assignee/filing details: **US8935922B2**, **US4265223A** inventors, **US11637962B2** (Solfire), **WO2016007011A1** family.
- Whether a B grant issued for **EP0794401A2**; legal status of **EP3835682B1** ("Not-in-force" reason).

Compiled 2026-09-26 from six parallel research workstreams. All patent numbers were verified against Google Patents records; none were invented. This register is a research landscape scan,

NOT a freedom-to-operate opinion. Active patents may require licensing — confirm claim scope and in-force status with a patent attorney before building or publishing a prototype.

Appendix B

Technical Foundations

Purpose: The math and science foundation for a 300-page engineering book on a founder-designed system: parabolic-profile lens concentrates sunlight into a heat ray (zero photovoltaics); tracking mirrors around the lens act as a heliostat boost stage; closed-loop boiler flashes water to steam; steam drives a turbine; a Tesla valve (valvular conduit, no moving parts) enforces one-way flow and stabilizes two-phase boiling/condensing flow; at night steam condenses back to water for reuse — closed loop, no leaking, daily power. Three deployment tiers: single home (backyard footprint), corporate building, apartment complex. The book takes readers from TRL 1 to TRL 9 with "theorems not theories, methods not concepts."

How this document is organized: Seven areas. Each gives the key theorems/principles WITH their math (equations stated plainly), quantitative figures with sources, established METHODS a reader can follow step-by-step, and clearly-marked GAPS where established science runs out and the founder's design must carry the weight. Figures marked [calc] were computed from sourced equations, not measured; figures marked [VERIFY] need re-checking against the original paper before print.

Companion research: `landscape-report.md` (same directory) — prior art, market, and the building-scale graveyard.

AREA 1 — THERMODYNAMICS CORE

1.1 The Rankine cycle: the governing theorem

The thermal efficiency of any heat engine cycle is:

η_th = W_net / Q_in = 1 − Q_out / Q_in

In enthalpy form for the Rankine cycle (states: 1 = condenser exit/pump inlet, 2 = boiler inlet, 3 = turbine inlet, 4 = turbine exhaust):

η_th = [(h3 − h4) − (h2 − h1)] / (h3 − h2)

The **Carnot ceiling** sets the absolute limit, stated in its cycle-meaningful form:

η ≤ 1 − T_L / T_mean,H, where **T_mean,H = Q_in / (s3 − s2)** (the entropy-averaged temperature of heat addition, in kelvin).

Superheat, reheat, and regeneration are all the same theorem in different clothes: **raise the mean temperature of heat addition.** That single sentence is the design compass for the entire power block.

1.2 Worked numbers (computed from IAPWS-IF97 steam tables — verifiable, not hand-copied)

Case	Cycle efficiency	Carnot ceiling
Ideal: 4 MPa / 400°C, condenser 40°C	**36.21%**	53.48%
Realistic small-scale (turbine η_t=75%, pump η_p=65%)	**27.07%**	—
Pessimistic (η_t=55%, η_p=50%)	**19.75%**	—
Saturated steam at 250°C (ideal)	34.5%	—

The moisture theorem (critical design rule): saturated 250°C steam exits the turbine at quality **x4 = 0.716** — mechanically unacceptable for bladed turbines (droplet erosion destroys blades). Computed rule: to keep exhaust quality x4 ≥ 0.90 at η_t = 75%, turbine inlet must be superheated: **2 MPa**

needs 350°C; 4 MPa needs 400°C; 6 MPa needs 450°C. This forces the founder's fundamental choice: **superheat the steam, or use a wet-tolerant volumetric expander (screw expander) instead of a bladed turbine.**

1.3 Small turbines and expanders (1–50 kW): real efficiencies

- EPA CHP Catalog: 500 kW 50%, 3 MW 70%, 15 MW 80% isentropic efficiency. Microturbines sit **at or below the 500 kW point — expect 40–60%.**
- **What limits them at small scale:** blade Reynolds numbers (MDPI study: efficiency falls 80.0% 75.0% shrinking rotor from 127 mm to 12.7 mm), tip-clearance ratios, leakage, partial admission losses, part-load operation.
- **Methods to maximize small-turbine efficiency:**
- **de Laval impulse stage** — ~80% potential in a single stage.
- **Curtis two-wheel velocity-compounded stage** — ~70% potential but swallows **8× the stage enthalpy drop** (MATEC 2021) — the standard small waste-heat-recovery choice; fewer stages, compact.
- **Reaction staging** — ~90% potential but partial admission is infeasible (full admission required), so it suits only larger small-scale units.
- **Screw (volumetric) expanders** — 60–70% (Kubawara), wet-tolerant: the honest answer to the saturated-steam moisture problem above.
- **Back-work ratio reality:** pump work is small for steam (BWR 0.4%) but parasitic loads (controls, tracking, feed pump at part load) are **not** small at building scale — fixed ~60 W parasitics are 6% of a 1 kW system.

1.4 ORC vs steam: the crossover (defensible line)

From the Ghent/Howest simulation study: **ORC wins below ~300°C heat source and below ~1 MWe; steam only becomes profitable above ~1 MWe** (economic statement — flagged as such). Steam cycle is impossible below ~130°C turbine inlet. ORC efficiency ceiling 24% vs steam >30% at large scale.

Why ORC wins small (the mechanisms): dry-fluid vapor domes (no superheat needed — no moisture problem), better pinch-curve temperature match to the source, ~10× vapor density (smaller turbine), lower operating pressures. Back-work ratio: steam ~0.4% vs ORC 2–10%+ (ORC's weakness).

Named working fluids with critical properties: toluene (best performer, flammable), n-pentane / cyclopentane, R245fa (phase-down regulatory warning — do not design a 20-year product around it), MM/MDM silicone oils.

Book decision rule: if the lens/boiler delivers 350–400°C superheated steam, steam Rankine is defensible; below ~300°C, ORC is the thermodynamically honest choice. The founder's system targets steam temperatures where steam Rankine applies — but the book must show this crossover analysis, not assume it.

1.5 METHOD: The Fixed State-Point Procedure (Cengel & Boles Ch. 10)

The named procedure for hand-calculating any Rankine cycle:

1 Fix state 1 at condenser pressure (saturated liquid): read h1, v1, s1.

2 Pump: **w_p v1·(P2 P1)** (incompressible approximation); h2 = h1 + w_p/η_p.

3 Fix state 3 (boiler pressure, superheat temperature): read h3, s3.

4 Isentropic drop: s4s = s3; at condenser pressure find h4s; quality **x** = **(s sf)/(sg sf)**.

5 Actual: h4 = h3 η_t·(h3 h4s).

6 Close: **η = 1 (h4 h1)/(h3 h2)**.

7 Authoritative property sources: **NIST WebBook** (https://webbook.nist.gov/chemistry/fluid/) and the `iapws` **Python package** (IAPWS-IF97).

1.6 Design targets (book-ready)

To net **10–15% electric** (after 95% mechanical × 90% generator × ~5% parasitics), the cycle must clear **~12–19% thermal**. Rule: **2–6 MPa, 400°C turbine-inlet superheat, η_t 65%** — or saturated steam + volumetric expander. These are the numbers the prototype test program must hit.

1.7 GAPS (Area 1)

- Sparse independent microturbine test data at 1–50 kW; dated/proprietary partial-admission correlations.
- **No two-phase Tesla-valve loss correlations exist** (single-phase literature only) — experimental program required.
- No standard daily full-condensation transient model for a day-only cycle.
- No published 300–450°C ORC-vs-steam crossover curve; fluid-regulation drift (R245fa phase-down).

AREA 2 — SOLAR CONCENTRATION OPTICS

2.1 Concentration ratio and the thermodynamic limits (theorems)

Geometric concentration: C_g = A_aperture / A_receiver. Flux concentration: C_flux = C_g × η_optical (folds in transmittance, intercept factor, absorptance).

The sine law of concentration (a theorem, not a fit). The sun subtends half-angle θ_s 4.65 mrad 0.266° (sin θ_s = solar radius / Earth–sun distance). No passive device can exceed the flux at the sun's surface (else a perpetual-motion machine of the second kind). Flux dilutes as $\sin^2\theta_s$, so the maximum 3D

concentration is:

$C_{max,3D} = 1 / \sin^2(\theta_s) \approx 46{,}211\times$; $C_{max,2D} = 1 / \sin(\theta_s) \approx 215\times$

In a dielectric of index n, the limit rises by n^2 (lab demos reached 56,000× exploiting this). General design form (étendue acceptance angle), air, $\theta_{out} = 90°$:

- 3D: **$\theta_{accept} = \arcsin(1/C)$** ; 2D: **$\theta_{accept} = \arcsin(1/C)$** [calc]

C (3D)	θ_accept	Margin above solar disk [calc]
100	5.74°	5.5°
1,000	1.81°	1.55°
10,000	0.57°	0.31°

The margin column is the **entire budget shared by tracking error, slope error, alignment error, and circumsolar spread** — the single most useful quantitative exhibit for the book's tracking chapter.

2.2 Étendue: the conservation principle and what it forbids

$E = n^2 \cdot A \cdot \Omega$ (index² × area × projected solid angle) is conserved for lossless propagation; real optics only decrease it. What it forbids:

1. No passive optics exceed 46,211× (3D) / 215× (2D) in air — ever.
2. **Concentration trades strictly against acceptance angle.** Wide-acceptance + high-concentration + small-receiver is prohibited, not merely hard.
3. **Phase-space dilution is irreversible.** Upstream scattering permanently lowers the downstream ceiling.

Critical constraint on the founder's lens + heliostat-boost architecture: a heliostat field delivers a beam with angular

spread *larger* than the solar disk (partial ground coverage, shading, blocking, astigmatic spot distortion — IAEA review). The lens's effective input half-angle becomes $\theta_{in,eff} = \theta_{sun}$ + mirror-induced spread, and the two-stage ceiling is $1/\sin^2(\theta_{in,eff})$ — **lower** than the lens's standalone ceiling. **The boost stage adds collected power (more aperture area) but reduces maximum flux concentration.** (Source: https://inis.iaea.org/records/6mtj8-jhs68/files/25070563.pdf?download=1)

2.3 Parabolic vs Fresnel lens: the honest framing

In solar literature, "parabolic concentrator" means a parabolic **reflector**. The transmissive concentrator is the **Fresnel lens**. A true meter-scale solid-glass parabolic lens is essentially absent from solar literature, for quantitative reasons the book must state outright:

- **Weight [calc]:** 1 m aperture f/1 solid glass lens 7 cm center thickness **~75–85 kg per m² of aperture** vs ~3 kg/m² for a 2–3 mm PMMA Fresnel lens (published data point: 1000 mm diameter, 2 mm PMMA Fresnel = 2 kg). Two orders of magnitude in structural load and tracker torque.
- **Bulk absorption:** centimeters of glass absorb in the near-IR where much solar power sits; Fresnel keeps every ray's path to millimeters.
- **Manufacturability:** meter-scale aspheric glass polishing is telescope-class fabrication; Fresnel lenses are compression-molded PMMA (tens of dollars at 280–600 mm; hundreds surplus at meter-class).
- Chromatic aberration is shared by both (single-material property, $\Delta f/f$ $1/V$) — not a reason to prefer solid glass.

Measured Fresnel performance (sourced): NASA Marshall 1.8×3.7 m line-focus acrylic Fresnel (1977/79): peak C = 62–64, lens transmittance 78–81%, **collection efficiency 43% at 200°C 34% at 260°C** (https://www.osti.gov/biblio/5754330). Point-focus Fresnel: optical efficiency 74.7%, mean flux C 90

(https://www.mdpi.com/1996-1073/11/4/887/xml); up to 89.08% optical efficiency at C = 240–600 in a 2025 validated study. Focal spots of 930–1064°C demonstrated at DNI ~790–860 W/m² (https://www.mdpi.com/1996-1073/17/3/569).

Named Fresnel loss mechanisms (cap practical optical efficiency at ~75–89%): surface reflection, draft-angle losses, groove-peak tip rounding, groove shadowing at high rim angles, chromatic focal spread.

Book design rule: frame the founder's "parabolic lens" as the *optical intent* (parabolic focusing profile) and the **Fresnel lens as its manufacturable embodiment** — a "parabolic-profile Fresnel lens." Standard practice; avoids teaching readers to procure an unprocurable part.

2.4 Concentration ratio steam temperature

Governing receiver balance (Duffie & Beckman form):

C · G · η_opt · α = ε·σ·(T_rec⁴ T_amb⁴) + q_conv + q_cond

Stagnation ceiling (radiation only): **T_stag = [C·G·η_opt·α / (ε·σ)]^(1/4)** [calc]. With selective coating (α=0.95, ε=0.10), η_opt=0.75, G=1000: C=10 ~785°C; C=20 ~985°C; C=100 ~1610°C. **Radiative losses alone do not bind steam temperatures even at C 5–10 with a good selective coating.** Real systems use C = 30–100 to buy efficiency margin and compact receivers.

Sourced operating points: 200–260°C at C 60 (NASA Fresnel, 34–43% efficiency); ~400°C at C 30–100 (commercial troughs; NREL); 700–1000°C+ needs C of several hundred to >1000.

Selective coatings (sourced): commercial trough standard α > 0.95, ε 0.10 at 400°C (https://www.sciencedirect.com/science/article/pii/S1876610214004585); field-proven α = 0.93, ε = 0.18. Industry standard non-selective Pyromark 2500: α = 0.96–0.97 new, ε = 0.8–0.9 at temperature, degrades with thermal cycling (https://www.osti.gov/servlets/purl/1096256).

Book rule of thumb (defensible): C 40–80 for 300–400°C steam with selective coatings; C 20–40 for 200°C. The 46,211

limit is physically real but operationally irrelevant — say so explicitly.

2.5 Tracking accuracy: the error budget

σ_total θ_accept θ_sun, root-sum-square combination of sunshape, slope error, tracking error, alignment error. Mirror slope error **doubles** on reflection ("an angular deviation of the surface normal causes twice the angular deviation of the reflected rays"). Lens facet error perturbs by (n1)·Δ 0.5·Δ for PMMA — **lenses are intrinsically ~4× more forgiving of surface slope error than mirrors** [calc] — a genuine, citable-as-derivation argument for the lens choice.

- C = 100: ~5.5° budget — coarse tracking suffices.
- C = 1000: ~1.55° budget — needs 0.1–0.5°-class tracking.
- Line-focus (2D): single-axis tracking is the commercial standard; errors **along** the focal line are nearly free — only the transverse plane consumes budget. This is why troughs reach 400°C on single-axis trackers.
- Point-focus (3D): dual-axis required.

Intercept factor (standard method): γ = ½·[erf((φ_r β)/(2·σ_tot)) + erf((φ_r + β)/(2·σ_tot))], with **σ_tot = (σ_sun² + (2σ_slope)² + σ_tracking² + σ_align²)** (use (n1)·σ_facet for refractive Fresnel). Industry validation standard: **NREL SolTrace** Monte Carlo ray tracing (free). NREL SAM trough defaults: tracking factor 0.985–0.99, mirror reflectance 0.923–0.957.

Buyable 0.1°-class tracking (sourced): Kipp & Zonen SOLYS 2 (<0.1° passive, <0.02° active); Practical Solar small heliostat 0.11°; Stellio utility heliostat <0.034°; CSIRO consensus ~1 mrad for tower heliostats. **The home tier at C 40–80 needs only ~1–2° total accuracy — well within hobbyist/industrial tracker capability.**

2.6 METHODS (Area 2)

- **METHOD A — Size the aperture:** P_th target choose C from §2.4 bands η_sys = η_opt·η_receiver (0.45–0.55 non-evacuated at 300°C; 0.55–0.65 evacuated) **A_aperture = P_th / (DNI_design · η_sys)** (DNI 800–1000 W/m²; local TMY via NREL SAM) A_receiver = A_aperture/C. Validate hourly in SAM before cutting material.
- **METHOD B — Set C from target steam temperature:** write the receiver balance (§2.4) with chosen coating; solve for C with 20% margin above stagnation; cross-check §2.4 bands.
- **METHOD C — Build the tracking error budget:** θ_accept minus θ_sun allocate RSS (sunshape 4.6 mrad conservative) select tracker ~⅓ of remaining budget confirm in SolTrace.

2.7 GAPS (Area 2)

- No literature for meter-scale solid glass parabolic lenses in solar thermal — specify parabolic-profile Fresnel.
- **No closed-form for the combined concentration limit of a heliostat field feeding a lens** — étendue inequality bounds it; actual numbers require ray tracing per geometry.
- Two-phase receiver optics (concentrated flux + boiling closed-loop boiler at building scale) is application R&D, not textbook material.
- Long-term PMMA Fresnel degradation is site/formulation-specific — specify ASTM accelerated-UV qualification, not a universal lifetime.
- Small-heliostat optical quality data is sparse vs utility heliostats.

AREA 3 — HEAT TRANSFER

3.1 Boiler sizing: the rate equation and two methods

Q = U · A · ΔT_lm · F. For boilers and condensers **F = 1** (one side isothermal during phase change).

METHOD — LMTD (design: temperatures known find area):

1. Energy balance: Q = ṁ·h_fg on the boiling side.
2. **ΔT_lm = (ΔT_1 ΔT_2) / ln(ΔT_1 / ΔT_2)** (counterflow).
3. Build U: **1/U = 1/h_hot + R_fouling + t/k_wall + 1/h_cold** (boiling-side h from Rohsenow, §3.2).
4. **A = Q / (U · ΔT_lm)**.

METHOD — ε-NTU (performance: area known find duty): C_min, C_r = C_min/C_max; **NTU = U·A/C_min**; for boilers C_r 0 so for ALL geometries **ε = 1 exp(NTU)**; **Q_actual = ε·C_min·(T_h,in T_c,in)**. Rule: LMTD to size, ε-NTU for off-design/part-load/cloud analysis. (Incropera Ch. 11; Cengel Ch. 11.)

Typical U sanity checks (sourced): steam-to-water condensation 1,500–4,000 W/m²·K; evaporation 1,500–4,000; **air-cooled steam condensation only 700–850 W/m²·K** — passive/air-side exchangers are 2–5× worse (https://www.engineersedge.com/thermodynamics/overall_heat_transfer-table.htm).

3.2 Nucleate boiling — Rohsenow correlation (1952)

q'' = μ_l · h_fg · [g·(ρ_l ρ_v)/σ]^(1/2) · [c_p,l·ΔT_e / (C_sf·h_fg·Pr_l^n)]³

ΔT_e = wall superheat; ΔT (q'')^(1/3); n = 1.0 for water. C_sf: water–copper polished **0.013**, scored 0.0068; water–stainless 0.013; water–nickel 0.006 (https://www.nuclear-power.com/nuclear-engineering/heat-transfer/boiling-and-condensation/nucleate-boiling-correlations-rohsenow-correlation/). Water at 1 atm: h_fg = 2,257 kJ/kg. Applies to nucleate regime only — does NOT predict CHF.

3.3 Critical heat flux — Zuber's equation (the hard safety limit)

q''_max = C · h_fg · ρ_v · [σ·g·(ρ_l ρ_v) / ρ_v²]^(1/4), C = 0.149 (large plate, empirical) or π/24 0.131 (theoretical). Water at 1 atm: **q''_max 1.0–1.26 MW/m²** (https://en.wikipedia.org/wiki/Critical_heat_flux). CHF 0 at the critical pressure (22.06 MPa).

Book safety rule: compute wall heat flux **q''_wall = (C_optical · DNI · η_opt) / A_wall** and keep **q''_wall / q''_max 0.5**. **Lienhard finite-heater correction:** small heaters (L' < 0.15, with L' = L·[g(ρ_lρ_v)/σ]^(1/2)) have **lower** CHF — small building-scale boiler tubes are penalized. (Kandlikar contact-angle refinement available as an advanced box: hydrophilic surfaces raise CHF.)

3.4 Receiver losses: why small scale is punished (quantified)

Mechanisms: conduction (Fourier; ~75 mm insulation near-optimum, Harris 1985), convection (cavity: **Siebers & Kraabel 1984** Nu_L = 0.088·Gr_L^(1/3)·(T_w/T_)^0.18, Sandia-validated within +5.4%; **Stine & McDonald** tilt/aperture extension; **Clausing 1983**; wind 18 mph minimally affects deep cavities — cavities suppress wind loss), radiation (**q_rad'' = ε·σ·(T_s⁴ T_sur⁴)**, σ = 5.670367e-8).

The small-scale penalty theorem: loss fraction ε·σ·T⁴/(C·DNI) — **loss fraction 1/C**. Computed [calc] with sourced ε (Pyromark-like 0.87–0.88), DNI = 1000:

T (°C)	C=100	C=500	C=1000
300	**5.3%**	1.1%	0.5%
400	**10.1%**	2.0%	1.0%
500	**17.8%**	3.6%	1.8%

A backyard receiver at C=100, 500°C loses nearly a fifth of captured energy to radiation alone. Note: a deep cavity can lose MORE convection than an exposed receiver at some

inclinations (Kraabel) — the cavity's win is on radiation (effective absorptance α_eff 0.97+ via multiple bounces), not convection.

3.5 Condensation and night heat rejection

Nusselt film-condensation theory (1916): vertical plate **h = 0.943·[g·ρ_l²·k_l³·h_fg/(μ_l·L·(T_satT_s))]^(1/4)**; horizontal tube 0.725·[...]^(1/4) with D. Laminar check: Re_δ = 4Γ/μ_l < 30. Typical magnitudes: film condensation **2,000–15,000 W/m²·K** (8,500 handbook); nucleate boiling 3,000–100,000.

The condenser theorem: the air side, not the steam side, sizes the passive condenser. The film's h is never the bottleneck — ambient-air convection is.

Worked sizing [calc]: reject 50 kWh over a 10-h night 5 kW; ΔT_lm 20 K; U_passive 8 W/m²·K (natural convection + radiation; free-convection gas-condensing U = 5–20) **A 31 m²** of radiator panel — roughly a small garage's roof/wall area. **The honest footprint implication the book must state: passive night rejection is area-hungry at building scale.**

Hot-climate quantified (computed via IAPWS-IF97, ideal Rankine 1 MPa/300°C):

Condenser P	T_sat	η_thermal
10 kPa	45.8°C	**27.8%**
50 kPa	81.3°C	21.1%
100 kPa	99.6°C	**17.6%**

Every ~10 K rise in condensing temperature costs ~2 percentage points of cycle efficiency. A Texas 35°C night (condensing ~55–60°C) vs a 20°C night (~40°C) is ~25% vs ~28%. **Size the condenser for the design hot night, not the average night.** Ceiling rule: η_Rankine 1 T_cond/T_boil (absolute).

3.6 Water chemistry: the concentration theorem

Steam carries no dissolved solids in a closed boiler they concentrate. **Cycles of concentration = TDS_boiler/TDS_feedwater.** Blowdown to hold target TDS: **Blowdown = F·S/(B F)** (F = feedwater TDS, S = steam rate, B = max boiler TDS). Example: 80 ppm feed, 4,000 kg/h steam, 3,500 ppm limit **93.6 kg/h** blowdown (~2.3%).

Named treatment methods: deaeration (dissolved O_2 < 0.007 ppm for high-pressure); pH control 8.5–9.6 (coordinated phosphate–pH); oxygen scavengers — sodium sulfite reaction **$2Na_2SO_3$ (7.88 ppm) + O_2 (1 ppm) $2Na_2SO_4$**; hardness limits (0.05 ppm $CaCO_3$ high-pressure feed). Scale conductivity 1/100 of steel — deposits are the efficiency killer.

Book reality-check: "No leaking" does not mean "no water management." A sealed all-steel loop with distilled fill, scavenged and pH-buffered, runs very low blowdown — but the operating procedure must include periodic conductivity testing and a blowdown valve. Non-condensable in-leakage devastates subatmospheric condenser h (blanketing) — Nusselt values assume pure vapor; specify leak-tightness + air ejector/vacuum pump.

3.7 METHOD — Full receiver energy balance (reader procedure)

Q_in = Q_absorbed + Q_reflect + Q_rad,aperture + Q_conv,aperture + Q_cond,insulation, Q_in = DNI·A_aperture·η_optical; **η_th = Q_absorbed/Q_in**:

1. Q_reflect = (1 α_eff)·Q_in (deep cavity α_eff 0.97+).
2. Q_rad = A_ap·ε_eff·σ·(T_cavity⁴ T_amb⁴).
3. Q_conv = A_wall·h_cav·(T_wall T_amb) (Siebers–Kraabel/Stine–McDonald at tilt).
4. Q_cond = A_wall·(k_ins/t_ins)·(T_wall T_amb).
5. Q_absorbed = Q_in losses boiler duty (§3.1) check q''_wall vs Zuber CHF at 2× margin (§3.3).

6 Sanity: η_th should land 60–90% for a decent small cavity; below 50% aperture oversized or T too high for C — iterate.

3.8 GAPS (Area 3)

- **No established correlation for Tesla-valve two-phase stabilization** — label hypothesis; specify validation tests (pressure-drop ratio forward/reverse under boiling flow; oscillation-amplitude stability maps with/without valve).
- Flow-boiling correlations (Chen, Kandlikar) and two-phase ΔP models (Lockhart–Martinelli, Friedel) exist but **none account for an asymmetric-loss conduit** — experiment required.
- Lens-with-heliostat-boost flux distribution: ray-tracing exercise (SolTrace/MCRT), no textbook shortcut.
- Sealed micro-boiler multi-year chemistry drift not covered by ASME/ABMA standards — write operating doctrine from first principles + lab testing.
- Cavity convection correlations unvalidated below ~0.2 m scale — flag for CFD + field measurement.

AREA 4 — TESLA VALVE FLUID MECHANICS

4.1 The mechanism (established)

Fixed-geometry channel (Tesla, US Patent 1,329,559, 1920): straight main corridor intersected by teardrop/arc side loops. **Forward:** flow stays in the corridor, slight deflection low loss. **Reverse:** flow splits into the arc loops, which re-enter the main channel **head-on against the oncoming stream**; colliding streams generate separation, recirculation, intense vortices, compounding per stage. The fluid itself becomes the obstruction — no moving parts (Nguyen et al., *Nature Commun.* 2021).

Diodicity metric (universal): Di = ΔP_reverse / ΔP_forward at the same flow rate (equivalently Di(Re) = Hg_R/Hg_F = f_D,R/f_D,F). Di > 1 = forward diode. This is the metric the book standardizes on.

4.2 Pressure-drop math and Reynolds regimes (multi-study consensus)

Darcy–Weisbach: **ΔP = f_D·(L/D)·(ρU²/2)**; **Re = ρUD/μ**; Hagen number **Hg = (Δp/L)·(D³ρ/μ²)** (Hg ~ Re laminar; Hg ~ Re² inertial); laminar reference **f_D = 64/Re**.

- **Re << 1:** Di 1 **exactly** — the Stokes equation is linear and reversible; no diode action is physically possible. *A genuine theorem.*
- **Re 100–300 ("turn-on"):** Di jumps ~1 ~2 (Nguyen, macro, Tesla's original geometry) — coincides with **early turbulence transition at Re 200** (vs ~2000 in smooth pipes); geometry trips turbulence two orders of magnitude early.
- **Re 300–1500:** Di 2, plateaued (macro).
- **Higher Re:** design-dependent; GMF valve peaks Di = 1.82 at Re = 800 then declines; D-valve 2.00 at Re = 2000 (Wiley & Huang 2024).

Honestly reported Di range: ~1.2–4. ~2 is the robust macro-scale figure (Nguyen 2021, ~11 loops); ~4 is the best credible experimental figure (Li 2023, micro, 18 units, single-phase). CFD-only claims above ~4.5 are unvalidated — flag them.

Cleanest published resistance decomposition (Li et al. 2023 Methods): $\Delta p = (1/2)\cdot f_s\cdot\rho\cdot u^2\cdot(\pi\cdot R_b/D)\cdot(\theta/180°) + (1/2)\cdot k_b\cdot\rho\cdot u^2$ (bend-loss + straight-friction terms).

4.3 Li et al., Nature Communications 2023 — the two-phase result

Citation: Wenming Li et al., "Tesla valves and capillary structures-activated thermal regulator," *Nature Communications* 14:3996 (2023). https://doi.org/10.1038/s41467-023-39289-5

Findings with figures:

1 Single-phase Di 4 (micro, D_h 162 μm, 18 Tesla units, bend/main width ratio **0.5**).

2 **Vapor-backflow rectification visualized:** boiling flow — water forward, vapor driven back toward inlet; at each Tesla island ~one-third of vapor steered into the curved bend, turned with attenuated momentum; series compounding stagnates backflow, then liquid reflushing — periodic rectification. Control (plain channel): persistent vapor column at inlet for **34.5 ms**.

3 **Capillary structures required for stability.** Valve alone unstable, discontinuous two-phase flow. Superhydrophilic micropillars add $\Delta P_c = 4\gamma\cos\theta/d$ **19.2 kPa** (13× plain wall), liquid renewal ~0.5 m/s, continuous rewetting, no dry-out.

4 **Thermal switch with sharp criterion:** below vapor quality $\chi = 0.25$ ("off") chaotic oscillation (~15 s period); above $\chi = 0.25$ ("on") stabilized; switch ratio $r = h_{on}/h_{off}$ **6**.

5 **Thermal performance (water):** on-state HTC **175 kW/m²K** (~5× off-state); CHF **830 W/cm²** (5× plain-wall microchannels); χ up to 0.8. Bend alone +51% CHF; + island another +48%.

6 Fitted CHF correlation: $q''_{CHF} = 0.18 \cdot G \cdot h_{fg} \cdot We_{Dh}^{(0.2)} \cdot (D_h/L)^{(0.57)} \cdot (\rho_l/\rho_g)^{(1.13)}$ **[VERIFY — confirm exponent signs against the paper]**, MAE 17.6% water / 25% HFE-7100.

Why it matters: the *only* paper demonstrating Tesla valves rectifying *boiling* two-phase flow — directly analogous to the flashing leg of a solar steam loop. **Book caveats:** (a) two-phase Di < single-phase Di at Re > 2000 (vapor–liquid interfaces have lower Darcy friction); (b) stability required engineered capillary structures, not the valve alone — a bare Tesla valve in a boiler tube will not reproduce this.

4.4 Adjacent quantitative results

- **PV/T cooling +2.5% claim: CFD-only** (Sci. Reports 2024) — comparative between simulated geometries, not a measured gain; authors state experiments warranted. Present as CFD-indicated, unvalidated.
- **Pulsating heat pipes** (closest analog to the closed loop): de Vries et al., *IJHMT* 2017 — Tesla-type valve gave **25% velocity asymmetry** and **14% thermal-resistance reduction** vs valveless PHP.
- **Steam (macro, single-phase vapor):** Gan, Pei & Li, *Physics of Fluids* 2025 — multi-stage Tesla valves for superheated steam decompression in nuclear plants (CFD): geometry survives steam conditions in simulation; design rules (small entrance width, reduced inner radius, small depth, moderate angle, high expansion coefficient, extended transitions, multi-stage). Throttling/noise control, not a power loop — but the only published Tesla-valve + steam work.
- **Two-phase CFD warning:** Gong et al., *Aerospace* 2024 (VOF): two-phase ΔP asymmetry < single-phase; **Di = 0.78 at 0.2 m/s in slug flow — the diode INVERTS (reverse flows easier)**; rises to 1.44 at 0.8 m/s.
- **Industry skepticism** (cite it — the book is stronger for it): *Chemical Processing* assessment — Di 2 is modest for the hardware length required; "space is a valuable commodity in a piping system"; best in lower flow/pressure conditions.

4.5 Published geometric design rules

- **Bend/main width ratio 0.5** strongly outperforms 0.9 for boiling (Li 2023).
- **Branch angle:** 30° optimal in PV/T CFD; 41.9°/71.7° in Gamboa's shape-optimized micro-valve; 45° in T45 designs. No universal optimum — angle trades diodicity against forward loss.
- **Stage count N:** Di rises sub-linearly with diminishing returns (Liosis: 1.43/2.76/3.58 for N=2/6/10 at Re=450). Analytical model (De la Cruz-Ávila et al., *Water* 2026) predicts Di peaks at critical **N_crit** and degrades beyond (accumulated losses kill asymmetry) — **unvalidated by experiment; present as theoretical prediction.**
- **Scale rule of thumb:** keep the main channel narrow relative to loops; widening the main passage destroys rectification fastest.

4.6 Named METHODS (Area 4)

1 **Constant-head macro characterization (Nguyen et al. 2021 — copy for pipe-scale):** two chambers at Δp = ρgΔh via overflow drains timed volume collection (>60 s, <1% error) reverse orientation R = Δp/Q Hg, f_D on a Moody diagram vs 64/Re Di(Re) = Hg_R/Hg_F. Their 30 cm, D_h = 0.8 cm rig spanned Re ~10^1–10^3.

2 **Low-cost macro rig (Liosis et al., *Fluids* 2026):** 3D-print the valve, IoT flow + pressure sensors, CSV logging at 5–20 L/min both directions, bin per 1 L/min, fit ΔP = a·Q^b per direction, read Di from fits. Max Di = 1.96 at 5–6 L/min. Cheapest reproducible path to pipe-scale data.

3 **Two-phase/boiling characterization (Li et al. 2023 — the method for the steam loop):** flow-boiling loop, degassed fluid, pressure transducers + Coriolis meter, calibrated thin-film heater, high-speed video; sweep

mass/heat flux, compute χ, plot Δp(t) and T_wall(t); identify the switching criterion (theirs χ = 0.25); report HTC, CHF, two-phase Di vs plain-channel controls.

4 **Constant-flow-rate microfluidic protocol (Forster/Gamboa lineage):** stepped Q, dual differential pressure sensors, discard transients, Di(Q) = ΔP_r/ΔP_f point by point.

4.7 GAPS (Area 4) — the book's green field

1 **CONFIRMED: no published work on Tesla valves in a solar-thermal steam power loop.** Nearest neighbors: steam throttling CFD, PV/T water-cooling CFD, pulsating heat pipes, Li 2023 microchannel boiler. A building-scale solar steam loop with a Tesla valve is a first — frame as novel engineering development.

2 Macro/pipe-scale experimental data scarcity (essentially Nguyen 2021 + Liosis 2026, both water, ambient).

3 **Two-phase diodicity not guaranteed — can invert** (Gong 2024: Di = 0.78 slug flow). The "stabilizes two-phase flow" claim must be qualified: demonstrated once, micro-scale, 1 atm, with capillary assist.

4 No universal Δp/Di correlation; every published one is geometry/regime-specific.

5 **Scale-up physics untested:** micro-scale capillary dominance (ΔP_c = 19.2 kPa at 162 μm) does not transfer to meter-scale steam pipe where gravity rules — Bond/Eötvös regime flips. **The single biggest technical risk in the concept — state it as such.**

6 Pulsatility: Tesla valves perform best under pulsatile flow (Nguyen's fluidic AC-to-DC bridge, E 0.5, improving with amplitude/frequency) — a cloud-driven solar boiler is inherently pulsatile. Untested for steam; potential advantage or confound.

Bottom line for the book: single-phase Di 2 (macro) to 4 (micro, optimized) is well-established with clear Re scaling and a physical mechanism; boiling-flow vapor-backflow suppression is demonstrated once at micro-scale with capillary assist; everything at building/steam-pipe scale in a power loop is green-field work the book develops via Methods 1–4 above.

AREA 5 — TRL FRAMEWORK

5.1 NASA TRL definitions 1–9 (official; NASA ESTO, https://esto.nasa.gov/trl/, tracing to NASA/SP-2007-6105)

- **TRL 1** — "Basic principles observed and reported." Exit: peer-reviewed publication of the underlying research.
- **TRL 2** — "Technology concept and/or application formulated." "No experimental proof or detailed analysis available." Exit: documented description addressing feasibility and benefit.
- **TRL 3** — "Analytical and experimental critical function and/or characteristic proof of concept." Exit: documented results validating predictions of key parameters.
- **TRL 4** — "Component and/or breadboard validation in laboratory environment." Exit: test performance agreeing with analytical predictions + **documented definition of the relevant environment.**
- **TRL 5** — "Component and/or breadboard validation in relevant environment." Medium-fidelity brassboard in simulated operational environment. Exit includes **documented definition of scaling requirements.**
- **TRL 6** — "System/sub-system model or prototype demonstration in an operational environment." High-fidelity prototype addressing all critical scaling issues, operated under critical environmental conditions.

- **TRL 7** — "System prototype demonstration in an operational environment." High-fidelity engineering unit in the actual operational environment and platform.
- **TRL 8** — "Actual system completed and 'flight qualified' through test and demonstration." Final configuration demonstrated for the intended environment. Note the verb shift: "demonstrating" (6/7) "**verifying**" (8).
- **TRL 9** — "Actual system flight proven through successful mission operations." Exit: **documented mission operational results.**

Structural point: from TRL 3 on, every exit criterion is a **document**. Maturity = fidelity of the evidence environment.

5.2 DOE TRL definitions (DOE G 413.3-4A / TRA Guide; https://www.energy.gov/fecm/articles/appendix-f-trl-guide)

DOE's scale runs 1 (basic principle observed) to 9 ("total system used successfully in project operations"). DOE's signature contribution: the **scale × fidelity × environment matrix**:

TRL	Scale	Fidelity	Environment
9	Full	Identical (all respects)	Operational, full range
8	Full	Identical	Operational, limited range
7	Full	Similar (almost all respects)	Relevant
6	Engineering/Pilot (1/10–full)	Similar	Relevant
5	Lab/Bench	Similar	Relevant
4	Lab	Pieces	Simulated

TRL	Scale	Fidelity	Environment
1–2	Paper	Paper (no hardware)	—

Key DOE distinctions: TRL 45 = fidelity increase to "almost prototypical"; **TRL 56** = **"the step up from laboratory scale to engineering scale and the determination of scaling factors that will enable design of the operating system"** — the measured scaling factors are the explicit 56 deliverable; TRL 6 "begins true engineering development," prototype performs **all** operational functions; TRL 7 = full-scale similar system, "final design virtually complete"; TRL 8 = operational procedures complete, "end of true system development"; TRL 9 = full range of operating conditions. DOE EERE R 540.112 prototype language: TRL-4 "Alpha prototype (component)," TRL-5 "Beta prototype (component)," TRL-6 "Beta prototype (system)," TRL-7 "Integrated pilot (system)," TRL-8 "Precommercial demonstration."

NASA vs DOE: NASA is flight-native; DOE uses operational language directly usable for ground solar-thermal. DOE binds scale/fidelity/environment per level explicitly. Adopt **DOE's stricter reading** of TRL 7 (relevant, not operational, environment). DOE's TRL 9 ("full range of expected conditions") is stronger than NASA's for energy.

5.3 The gates: concrete evidence per transition

- **45:** pieces/simulated-lab almost-prototypical similar system in a **relevant environment**. Artifacts: integrated test data agreeing with predictions; the **written relevant-environment definition**; DOE's required analysis of lab-vs-operational differences.
- **56:** lab/bench engineering/pilot scale (1/10–full). Artifacts: prototype addressing all critical scaling issues; the **measured scaling factors** enabling full-system design; all operational functions demonstrated. (GAO best practice: TRL 6 by preliminary design review — GAO-16-461T.)

- **67:** engineering prototype **full-scale** similar (prototypical) system in relevant environment; final design virtually complete.
- **78:** prototype (similar) actual system (identical), qualified through test and demonstration; operational procedures complete; "precommercial demonstration."
- **89:** qualified operated over the **full range** of expected conditions. Artifact: **documented operational results** — the only level whose exit criterion is operational records, not test records. For this system: multi-season dispatch logs, availability/uptime, degradation curves, water-loop logs.

5.4 How TRL is assessed: the TRA process

- **DOE (G 413.3-4A):** (1) Identify CTEs — Critical Technology Elements: at-risk tech essential to operations, new/novel in application; (2) assess TRL ("testing in the proper environment, appropriate scale and fidelity"); (3) develop a **Technology Maturation Plan (TMP)** for immature CTEs.
- **GAO (GAO-20-48G):** five steps — plan + select an **independent, credible, expert** team; identify CTEs via WBS/requirements; assess against defined criteria/environments; write the TRA report (CTEs, evidence, TRLs, assumptions, limitations); use results in decisions + TMP. Quality bar: comprehensive, well-documented, credible, defensible. **The developer does not grade their own technology.**
- **Roll-up rule (NASA practice):** "weakest link" — the system inherits the lowest CTE's TRL. The lens, turbine, and boiler may be individually mature; the integrated system is not.

5.5 Criticisms and limitations (cited — the book teaches these)

1. Snapshots in time — GAO-16-410G: no standard shelf life; experts say 1–6 months.
2. Tells you where you are, not how to advance — GAO-16-410G; effort between levels is non-linear.
3. Optimism bias + ambiguous "relevant environment" — DoD TRA Guidebook (Feb 2025).
4. TRL integration/manufacturing/commercial readiness — separate IRL and MRL scales exist; a TRL 9 plant with uneconomic $/kWh is still TRL 9.
5. Duration-free "proven" — neither definition sets an operating-hours threshold; **the book must define its own** (seasonal cycles, availability statistics, degradation limits) and label them as the book's standard, not NASA/DOE's.

5.6 Applied sketch: TRL 1–9 for THIS system (author's mapping of official criteria — present as such)

Candidate CTEs: building-scale lens concentrator optics, heliostat-to-lens integration, Tesla-valve loop application, and the integrated system itself (turbine and boiler metallurgy largely heritage).

- **TRL 1:** literature review + system energy model; basic principles (Rankine, concentration, diodicity) documented.
- **TRL 2:** documented concept with sizing for all three tiers (home / corporate / complex); feasibility and benefit analysis.
- **TRL 3:** separate critical-function experiments — lens flux mapping, boiler CHF test, Tesla-valve diodicity rig, small turbine/expander bench test; analytical predictions validated per function.

- **TRL 4:** integrated low-fidelity breadboard in the lab + the **written relevant-environment definition** (DNI range, ambient range, duty cycle, water chemistry spec).
- **TRL 5:** near-prototypical brassboard under real sun; documented scaling requirements.
- **TRL 6:** engineering-scale pilot + **measured scaling factors**; all operational functions (track, boil, generate, condense, restart daily) demonstrated.
- **TRL 7:** full-tier field installation (e.g., one apartment-complex unit), final design virtually complete.
- **TRL 8:** qualified final-configuration system; precommercial demonstration; operating procedures complete; analytical predictions **verified**.
- **TRL 9:** multi-season operational records — hours, availability, degradation, water chemistry — across the full range of expected conditions. **The book defines its own evidence thresholds here (G3/G5 above).**

5.7 GAPS in the framework itself

- NASA's ESTO table is internally inconsistent at TRL 6/7 (Definition says "operational," Hardware Description says "relevant").
- "Relevant environment" is program-defined — whoever writes its definition controls the TRL claim. The book's strongest move: publish an explicit, checkable relevant-environment definition for building solar steam.
- No operating-hours/statistical standard at TRL 9; no reliability/degradation statistics.
- DOE scale bands "may vary based on engineering judgment" — honest assessors can disagree.

- Heritage-vs-novel CTE boundary is a judgment call — do the CTE identification explicitly per the GAO method.

AREA 6 — SCALING METHODS

6.1 The square-cube master theorem

Area L^2, volume L^3 **loss fraction 1/L**; thermal time constant L; pipe fractional loss per meter 1/D at fixed velocity.

Receiver efficiency equation (peer-reviewed form): **$\eta = \eta_{optical} [\varepsilon\sigma(T^4 T_{amb}^4) + U_L(TT_{amb})]/(G \cdot C)$**. At fixed C and T, loss fraction is scale-invariant — the honest small-scale penalty comes from (a) the **optical floor**: absolute spot-size errors force lower C on small dishes (C = 150 50 triples loss fraction from ~6% to ~19%), and (b) **fixed parasitics**: 60 W of controls is 6% at 1 kW, 0.06% at 100 kW. (Worked 1 kW vs 100 kW example with stated assumptions in the supporting brief.)

Pump affinity laws: Q N, H N^2, **P N^3** — cubic power savings with variable-frequency drives at part load.

Turbine efficiency vs size (EPA CHP Catalog): multistage ~65% under 1 MW 90%+ utility; small single-stage as low as 40–50%; micro screw expanders (12–132 kW) 60–70%. $\eta_t = (h_1 h_2)/(h_1 h_2 s)$.

6.2 Numbering-up vs scaling-up (the modular methodology)

From Ehrfeld (*Micro Reaction Technology*) and process-engineering precedent (Corning, Ehrfeld Mikrotechnik microreactors; "smart scale-out" hybrids): **internal vs external numbering-up** — replicate identical modules (internal: parallel channels in one block; external: parallel units) vs build one bigger unit.

Decision rule for this architecture: number up the collector/receiver/boiler modules (heat-transfer-limited — performance is per-module), scale up the turbine/power

block per tier (efficiency favors large).

Cost scaling: the **six-tenths rule** — $C_2 = C_1 \cdot (Q_2/Q_1)^{0.6}$ (Williams 1947) — for scaled-up vessels; numbered-up modules scale ~linearly in hardware cost but sub-linearly in engineering cost (design once, replicate).

6.3 Prototype test plans and efficiency measurement (applicable standards)

- **Collector:** ISO 9806:**2025** (3rd ed. — cite the 2025 edition; 2017 was withdrawn Oct 2025) or ASHRAE 93-2010(RA2014). Step-by-step: **$\eta = \dot{m} \cdot c_p \cdot \Delta T / (G_b \cdot A_{ap})$**.
- **Turbine: ASME PTC 6** (the steam-turbine code).
- **Plant: ASME PTC 46** (overall plant performance).
- **Correction:** ASME PTC 52 ("Performance Test Code for Concentrating Solar Power Plants") is **still under development** — do not cite it as published. This is a genuine standards gap.
- **Uncertainty: ASME PTC 19.1-2018** (Taylor Series per the GUM): propagate instrument uncertainties to the result and **report $\eta \pm U$ with k=2 coverage**. The book teaches this as the honesty standard — every efficiency number carries its error bar.

6.4 GAPS (Area 6)

- No published CSP plant test code (PTC 52 unfinished).
- No published sub-50 kW turbine efficiency curves — the program must measure its own.
- No public kW-scale installed-cost data for solar-thermal power.
- ISO 9806:2025 field practice for concentrating collectors is still young.

AREA 7 — ECONOMICS MATH

7.1 LCOE: the full method

LCOE = (discounted lifetime costs) / (discounted lifetime energy). Working form:

LCOE = (CapEx × CRF + Fixed O&M) / AEP + Variable O&M

Capital recovery factor: CRF = r(1+r)^n / [(1+r)^n 1] (r = discount rate, n = years).

Worked structure [illustrative inputs]: 10 kW, $30k CapEx, 7%, 20 yr, 18% CF **$211/MWh**. IRENA: CapEx 4/5 of CSP lifetime cost. Small-scale $/W is dominated by **non-scaling fixed costs** (controls, installation, interconnection, permitting) — the book must itemize these, not hide them.

7.2 Capacity factor for day-only solar thermal (no storage)

CF = E_annual / (P_rated × 8760). Rigorous method: NSRDB TMY hourly DNI threshold integration (validates against published benchmarks: Daggett, CA, 2700 kWh/m²/yr **~26%**, matching published 25–28% no-storage CSP figures). Cloudy climate **13–14%** [estimate — mark as such].

Correction the book must carry: **"fixed vs tracked" is not a real choice for CSP** — concentrating collectors use beam radiation only, so they must track. The honest comparison is tracked-CSP vs fixed-PV.

7.3 The honest PV+battery comparison (sourced)

EIA AEO2025: PV **$26–32/MWh**, PV+battery **$53/MWh**; Lazard v19: utility solar from **$40/MWh**. A small day-only thermal system (~$150–300/MWh illustrative) **cannot beat PV+battery on $/MWh**. The book competes on: co-produced high-grade heat, no lithium supply chain, and cheap future thermal storage — not on $/MWh. State this plainly; the credibility of the entire book depends on it.

7.4 GAPS (Area 7)

- No public kW-scale installed-cost data (G3 above) — LCOE inputs at building scale are estimates until the pilot is costed.
- Cloudy-climate CF figures are estimates, not measurements for this architecture.

MASTER GAPS REGISTER — where established science runs out

The founder's design must carry the weight in exactly these places. Each is a labeled research/development program in the book, not a hand-wave:

1. **Tesla valve in a solar steam power loop** — zero published work; nearest neighbors are throttling CFD, PV/T cooling CFD, pulsating heat pipes, and one microchannel boiler paper. (Areas 1, 4)
2. **Two-phase diodicity at steam-pipe scale** — can weaken or invert (Gong 2024); capillary-to-gravity regime flip is the single biggest technical risk. (Area 4)
3. **Small-turbine efficiency at 1–50 kW with this cycle's steam conditions** — must be measured; no published curves. This is where technical readers test the book's honesty. (Areas 1, 6)
4. **Lens + heliostat-boost combined optical performance** — no closed-form; per-geometry ray tracing required. (Area 2)
5. **Solid parabolic glass lens at meter scale** — unprocurable; specify parabolic-profile Fresnel. (Area 2)
6. **Passive night-condenser sizing at building scale in hot climates** — area-hungry (~31 m^2 per 50 kWh in the worked example); must be sized for the design hot night. (Area 3)

7 **Sealed-loop multi-year water chemistry** — no standards coverage; write operating doctrine from first principles + lab testing. (Area 3)

8 **Building-scale installed costs and LCOE** — no public data; pilot must be costed honestly. (Areas 6, 7)

9 **TRL 7/8/9 evidence thresholds** — the framework is silent on duration and statistics; the book defines its own and labels them as its standard. (Area 5)

10 **CSP plant test code** — PTC 52 unfinished; use PTC 6 + PTC 46 + ISO 9806:2025 in the interim. (Area 6)

BOOK-READY DESIGN RULES (distilled)

1 **Cycle:** 2–6 MPa, 400°C superheat, η_t 65% nets 10–15% electric; or saturated steam + screw expander. Moisture rule: x4 0.90 (2 MPa 350°C; 4 MPa 400°C; 6 MPa 450°C).

2 **Optics:** parabolic-profile **Fresnel** lens (never solid glass); C 40–80 for 300–400°C steam; heliostat boost adds power, not flux concentration (étendue).

3 **Boiler:** q''_wall / q''_max(Zuber) 0.5; apply Lienhard small-heater correction.

4 **Losses:** loss fraction 1/C — buy concentration before fighting insulation.

5 **Condenser:** size for the design hot night; ~10 K ambient rise 2 points cycle efficiency; air side sizes it.

6 **Tesla valve:** design for Di 2 (macro) in single phase; two-phase behavior is experimental — build the characterization rig (Nguyen/Liosis/Li methods) before claiming stabilization.

7 **Water:** sealed loop + distilled fill + scavenger + pH buffer + conductivity testing + blowdown valve. "No leaking" "no water management."

8 **Scale:** number up collectors/receivers/boilers; scale up the turbine per tier.

9 **Honesty:** every efficiency reported as η ± U (k=2, PTC 19.1); compete with PV+battery on heat + supply chain, not $/MWh; define the book's own TRL 7/8/9 evidence thresholds.

10 **TRL path:** weakest-link rollup; independent assessment (never self-graded); every level exits with a document.

Research compiled 2026-09-26 from six parallel workstreams. All equations are established textbook/peer-reviewed results; computed figures are marked [calc]; unverifiable extractions are marked [VERIFY]; gaps are labeled, not hidden. Supporting briefs: ~/workspace/book-research/thermodynamics-core-brief.md, ~/workspace/book-research/trl-framework-brief.md, ~/workspace/book-research/scaling-economics-brief.md (optics, heat-transfer, and Tesla-valve briefs delivered in full in the research handoffs).

Appendix C

Landscape Report

Research date: September 26, 2026 **Purpose:** Background research for a book on a modular, building-scale concentrated-solar-thermal power system (parabolic lens + tracking mirrors + closed-loop boiler + steam turbine + Tesla valve + night condensation recovery). Research only — no filing, no outreach.

The concept in brief (founder's words): A parabolic lens concentrates sunlight into a heat ray — zero photovoltaic panels. Tracking mirrors around the lens move like solar-farm heliostats. Heat turns water to steam in a closed-loop boiler; a steam turbine generates power; a Tesla valve handles flow control. At night the steam condenses back to water and is reused — closed loop, no leaking, power made each day. Scales from a home (backyard footprint) to apartment complexes, corporate buildings, trailer parks — and by replication, to national scale.

1. Concentrated Solar Power (CSP): the 2026 landscape

The four established architectures

- **Parabolic trough** — curved mirrors focus sun onto a receiver tube along a focal line; heat-transfer fluid (usually synthetic oil) makes steam via heat exchanger. The most deployed and mature technology.

- **Power tower** — fields of two-axis tracking mirrors (heliostats) focus sunlight onto a central receiver, typically molten salt, reaching the highest temperatures and efficiencies.
- **Linear Fresnel reflector** — long rows of flat/low-cost mirrors focus onto elevated receivers; cheaper but lower concentration.
- **Dish-engine (dish-Stirling)** — a parabolic dish focuses onto a single point driving a Stirling engine (hydrogen/helium working gas). Record solar-to-electric efficiencies exceeding 30% — the highest of any CSP type — but each unit only produces ~3–25 kW and needs two-axis tracking plus maintenance per dish.
- Sources: https://www.energyencyclopedia.com/en/renewable-energy/solar-energy/solar-concentrators · https://unfccc.int/technology/concentrated-solar-power-plants

Market size and structure

- Global CSP market estimated at **USD 7.98B in 2026**, projected **USD 16.57B by 2033** (~11% CAGR). (Coherent Market Insights: https://www.coherentmarketinsights.com/market-insight/concentrated-solar-power-market-5863)
- **Parabolic trough holds ~73% market share in 2026**; power towers growing fastest on efficiency + storage. (Fortune Business Insights: https://www.fortunebusinessinsights.com/industry-reports/concentrated-solar-power-market-100751)
- The **utility segment is ~98% of the market** — residential/commercial penetration is described as "very low" because of capital intensity.

- Key players named across reports: Abengoa, BrightSource Energy, SolarReserve, SENER, Acciona/Atlantica, ACWA Power, Siemens, GE, Frenell GmbH.

Costs — the problem the concept must beat

- A 150 MW trough plant achieves LCOE of **$0.095–$0.115/kWh** vs **$0.145/kWh** for a 50 MW plant — brutal economies of scale that punish small systems.
- PV-plus-battery now delivers firm power at **$0.070–$0.090/kWh** in high-irradiance regions, undercutting new CSP. CSP's surviving edge is **long-duration storage (>6 hours)**, where lithium-ion gets uneconomic.
- Legacy US plants show how expensive early CSP was: Ivanpah (392 MW tower, 2014) cost ~$2.3B with power sold at ~$0.19/kWh; Solana (250 MW trough, 2013) ~$2.16B at ~$0.14–0.20/kWh.
- Source: https://www.blackridgeresearch.com/blog/latest-list-of-top-csp-concentrated-solar-power-plants-projects-worldwide-pv-panel
- Growth through 2030 is concentrated in **China** (~15 GW indicative target in the next planning period) and **MENA** (e.g., Dubai's 700 MW CSP + 250 MW PV Noor Energy 1 with 12–15h molten-salt storage). Europe/North America are mostly legacy plants + R&D.
- Source: https://pdf.marketpublishers.com/globalinfo/global-solar-thermal-csp-gir.pdf

GAP for the book

Utility CSP is a mature, consolidating, scale-driven industry that has **never** served buildings. The entire cost structure assumes 50–700 MW plants. Nobody in mainstream CSP is pursuing the residential/building segment — which is exactly the white space the concept claims, but also means there is no supply chain, no installer base, and no cost curve to ride.

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2. Lens-based solar concentrators for thermal (not PV)

Refractive lenses (as opposed to reflective mirrors) are a real but niche concentration path:

- **Linear Fresnel-lens steam generation:** Indian researchers demonstrated a line-focus Fresnel lens system generating low-pressure process steam at ~43–53% overall efficiency, noting refracted-lens concentrators can reach 100–200°C+ service and that lenses outlast mirrors (>5 yr lens life vs 3–4 yr mirror replacement). Source: http://article.sapub.org/10.5923.j.ijee.20221201.02.html
- **Cylindrical Fresnel lens steam system (20 m^2):** Chinese researchers built a 20 m^2 cylindrical Fresnel lens driving a steam generator at 0.2–0.5 MPa, ~44–48% collection efficiency, ~63% intercept efficiency. Conclusion: higher collection efficiency and easier tracking than troughs, but **stable working time only ~4.5 h vs 6.2 h for parabolic troughs** — a key limitation to address. Source: https://www.researchgate.net/publication/350993886 (also http://yerbilimleri.cumhuriyet.edu.tr/en/download/article-file/2328347)
- **Fresnel-lens steam for cooking/desalination:** Indonesian and Iraqi teams demonstrated small Fresnel-lens steam generators (hundreds of watts thermal) — proof that lenssteam works at small scale, but only as lab prototypes. Sources: https://pp5.

omikk.bme.hu/me/article/download/22547/22295/195886 · https://www.scribd.com/document/714830098/Wassem-Paper

- **Fresnel lenses for solar chemistry:** MDPI-published work uses Fresnel lenses to drive methanol steam reforming at up to 24–35% solar-to-chemical efficiency — adjacent proof of high-flux lens concentration. Source: https://www.mdpi.com/2076-3417/16/13/6681

GAP for the book

All lens-based thermal work found is **research-scale and process-heat oriented** (steam for industry, cooking, desalination, chemistry) — not electricity, not building-scale packaged power. No commercial product generates electricity from a lens-driven boiler. Also note: the founder specifies a **parabolic lens**; the literature overwhelmingly uses **Fresnel lenses** because a true meter-scale parabolic glass lens would be extremely heavy and expensive. The book should address why parabolic (or acknowledge Fresnel as the practical implementation) — this is a credibility point a technical reader will check.

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3. Micro / building-scale / residential solar-thermal power — products, startups, pilots, failures

This is the most important section: the building-scale power graveyard and its survivors.

The failures (all tried to shrink CSP — all died or pivoted)

- **Sopogy (Hawaii, "MicroCSP")** — shrunk parabolic-trough collectors to rooftop scale; ~500 W per collector; built the world's first MicroCSP solar farm (2008, Kona). Founder Darren Kimura's pitch: small modular CSP at PV-comparable cost. **Ceased**

operations ~2013–2014; assets liquidated. Sources: https://www.sustainablebusiness.com/2007/09/hawaiian-firm-shrinks-solar-thermal-power-40992/ · https://solarthermalworld.org/?taxonomy=country&term=hawaii · https://www.altenergymag.com/content.php?post_type=1804

- **Infinia (Kennewick, WA, "PowerDish")** — 3.2 kW free-piston Stirling engine + 5 m parabolic dish, hermetically sealed, "25 years no maintenance," sized for a home. Won a DOE contract for a 30 kW engine. **Went down ~2013** (Greentech Media's verdict: "done in by crystalline silicon and cadmium-telluride solar panels"); assets acquired by **Qnergy (Israel), which pivoted the engine away from solar** to remote power. Sources: https://www.zdnet.com/article/dishing-out-power-with-a-solar-engine/ · http://breakingenergy.com/2015/05/18/solar-will-change-the-world-but-probably-not-like-this/ · https://www.stirlinginnovations.com/_files/ugd/33075e_52d1a2bc4040496a9a1461ba1afd4937.pdf?index=true · https://www.osti.gov/biblio/1087572
- **Stirling Energy Systems / Tessera Solar ("SunCatcher")** — 25 kW dish-Stirling, 60-unit 1.5 MW demo in Arizona (2010), PPAs for 1,500+ MW. **Manufacturer bankrupt September 2011**; project decommissioned. Cause of death per contemporaneous reporting: collapsing PV prices. Source: http://breakingenergy.com/2015/05/18/solar-will-change-the-world-but-probably-not-like-this/

- **Heliogen (Bill Gates-backed)** — AI/computer-vision-controlled small heliostats claiming >1,000°C for industrial heat, plus particle receivers and supercritical CO_2. **Never built a single operating commercial plant**; the 5 MW Capella pilot was walked away from on cost; acquired by Zeo Energy in 2025 in what CleanTechnica called "a merger that looked more like a rescue." Source: https://cleantechnica.com/2025/09/27/ivanpah-heliogen-lessons-from-concentrated-solars-decline/

The survivors (all avoided the "small solar power plant" trap)

- **SunOyster Energy (Germany)** — parabolic-mirror concentrator for **buildings**: flagship unit makes 10 kW thermal + 4.8 kW electric (PV hybrid, not a heat engine — the electric side is PV panels, the thermal side is a steel-pipe receiver to 170°C). Reached TRL 9 / Solar Keymark and commercial sales in 2021. Survived by selling **heat + PV**, not by shrinking a power plant. Sources: https://www.pv-magazine.com/2022/04/13/concentrating-solar-power-system-for-building-applications-usable-with-pv/
- **247Solar (MIT spinout)** — modular high-temperature CSP: heliostat field + central tower heating air to ~1,000°C at atmospheric pressure, driving a jet-engine-like turbine; 400 kW electric + 600 kW heat per unit, with solid-material thermal storage for round-the-clock power. Positioned for **microgrids, mines, off-grid communities** — not single homes. First deployment planned with a utility in India; as of 2024 still pre-deployment (DOE grant in 2010, receiver damaged in shipping, rebuilt 2015, Covid delays). Sells its thermal battery standalone at ~1/7th the cost of lithium-ion per kWh. Source: https://energypost.eu/new-concentrated-solar-power-system-delivers-1000c-for-industrial-processes-more/

- **California Energy Commission micro-CHP project (Winston et al.)** — non-imaging-optics concentrating collector making electricity + heat for homes; passed extreme-condition testing; claimed 30% roof-space savings. But note: the electric side used **PV cells**, not a heat engine. Sources: https://www.energy.ca.gov/publications/2021/novel-low-cost-high-efficiency-solar-powered-micro-combined-heat-and-power
- **2026 research: switchable solar heat-pump/ORC for residential buildings** (pv magazine, June 2026) — academic proposal combining solar thermal + organic Rankine cycle + heat pump with seasonal mode switching. Still a paper, not a product. Source: https://www.pv-magazine.com/2026/06/22/how-to-power-buildings-with-solar-driven-reversible-heat-pumps-organic-rankine-cycle/

GAP for the book

The pattern is stark: **every company that tried to sell a small solar-thermal *power plant* (heat engine making electricity) failed** — killed by PV's cost collapse, maintenance burden (two-axis tracking per unit), and thermodynamics (small heat engines are inefficient). The survivors sell **heat** (with PV for the electric side) or target **microgrids/industrial** scale, not homes. The book's central technical claim must therefore answer: *why does a small water/steam turbine succeed where Stirling dishes, micro-troughs, and ORC concepts failed?* The honest physics the book must confront: small Rankine-cycle turbines have poor isentropic efficiency, which is exactly why the field moved to ORC (organic fluids, lower temps) and Stirling for small scale. The Tesla valve + closed-loop + lens combination is the claimed differentiator — but no data exists yet to prove it overcomes the small-turbine efficiency wall.

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4. Tesla valve applications in thermal management and power systems

The Tesla valve (Nikola Tesla's 1920 "valvular conduit," US Patent 1,329,559 — a fixed-geometry channel with no moving parts that resists flow in one direction) has real, published thermal applications — but **none in a power-generation steam loop**:

- **PV/thermal cooling:** Multiple 2024 papers show Tesla-valve flow channels cooling PV/T panels — up to 20°C cell-temperature reduction, +2.5% electrical efficiency, 59.65% thermal / 16.32% electrical efficiency. Optimal geometry reported: 30° angle, pipe-diameter ratio 1, maximized valve count. Sources: https://www.azocleantech.com/news.aspx?NewsID=34890 · https://www.nature.com/articles/s41598-024-61785-x
- **Battery thermal management:** Tesla-valve mini-channel cold plates cut battery max temperature ~3.8°C vs straight channels, doubling Nusselt number at moderate Reynolds numbers. Source: https://www.sciencedirect.com/science/article/abs/pii/S0196890423001589
- **Two-phase thermal regulator (most relevant):** Li et al., *Nature Communications* 2023 (DOI 10.1038/s41467-023-39289-5) — Tesla valves + capillary structures **eliminate vapor backflow and rectify chaotic two-phase (liquid/vapor) flow into ordered directional flow** in a boiling/condensing loop, boosting heat-transfer coefficient and critical heat flux. This is the closest published analog to the concept's steam/water loop. Source: https://ira.lib.polyu.edu.hk/bitstream/10397/99696/1/Li_Tesla_Valves_Capillary.pdf

GAP for the book

This is the concept's **strongest genuine novelty claim**: no published work or product was found using a Tesla valve as the flow-control element in a **solar-driven steam power loop**. The 2023 Nature Communications paper proves the underlying physics (Tesla valves stabilize boiling/condensing two-phase flow — exactly what a closed-loop boiler/condenser needs), which gives the book a citable scientific foundation. The book can credibly claim: *the valve that tames two-phase flow instability is the missing piece small steam systems never had.*

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5. Closed-loop water/steam solar thermal with condensation recovery

Closed Rankine loops are standard at utility scale; the question is small-scale packaging and passive night recovery:

- **CN110500248A** — "steam cycle generator with solar collector and spiral regenerator": solar collector → superheated steam → turbine → condenser → condensate returned via spiral regenerator (feedwater heating). Full closed loop with heat recovery. Source: https://patents.google.com/patent/CN110500248A/en
- **AU2013291157B2** — solar power system with heat-collection apparatus, steam turbine, condenser returning water to the collector, plus a heat-storage device shunting superheated steam — designed for stable output despite solar variation. Source: https://patents.google.com/patent/AU2013291157B2/en
- **WO2020116167A1** — solar thermal power system with steam turbine, condenser returning water to tank, plus chemical heat storage for day/night balancing. Source: https://patents.google.com/patent/WO2020116167A1/en

- **US8341961B2** — solar desalination with steam turbine + condenser + hydroturbine; combined power/water. Source: https://patents.google.com/patent/US8341961B2/en

GAP for the book

Every closed-loop design found pairs the loop with **active heat storage** (molten salt, chemical, or thermal mass) — **none relies on passive night condensation as the heat-rejection mechanism**, and none is packaged at building scale. The concept's "steam reverts to water at night, condensed for reuse, no leaking, power made each day" is best framed as a *design simplification*: eliminating the storage subsystem that makes small CSP uneconomic, accepting day-only generation. The book must honestly address the trade: no storage means no night power — the system is a **daytime power plant**, and the economics must work on that basis (or pair with batteries, which the concept currently excludes).

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6. Small-scale heliostats / tracking-mirror concentrators

- **Wikoda "Sunflower"** — $399 home heliostat: six triangular mirrors, microprocessor sun-tracking, ~500 W of light+heat redirected up to 25 ft. Proves consumer-grade tracking mirrors are cheap and real. Source: https://www.fastcompany.com/1680592/a-mirror-system-that-brings-sunlight-into-the-darkest-corner-of-your-home
- **Practical Solar** — $995 residential heliostats, 0.1° digital tracking accuracy, 4× concentration per unit, software-targetable; explicitly notes multiple units can aim at a **common target for concentrated solar power**. Source: https://www.habitatmag.com/Publication-Content/Green-Ideas/2008/2008-February/Practical-Solar-Heliostat-System

- **MLSUN (China)** — commercial tower heliostats from **0.7 m²** (45 kg, two-axis, 1 mrad accuracy) up to 140 m²; the 0.7 m² unit is effectively a building-scale heliostat available off the shelf. Source: https://mlsun-csp.en.made-in-china.com/product/DQmrWcYdRJpL/China-96-Sqm-Flat-Mirror-Tower-Heliostat-with-Horizontal-and-Pitching-Two-Ways-Solar-Tracker.html
- **Academic:** compact heliostat with collective-drive tracking (J-STAGE), mini-heliostat prototypes with stepper-motor control tested in Algeria (Journal of Applied Science and Engineering). Sources: https://www.jstage.jst.go.jp/article/jmr1988/14/2/14_2_39/_article/-char/en . http://jase.tku.edu.tw/articles/jase-202303-26-3-0004

GAP for the book

Small tracking mirrors are a **solved, commoditized component** — good news for the concept's bill of materials. The novel part is not the mirrors but the **architecture**: heliostats *assisting a lens* (boosting flux on the lens aperture) rather than replacing it. No system found uses tracking mirrors as a secondary boost stage for a refractive primary concentrator. That's a clean, defensible novelty claim.

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7. Patents: lens concentration + steam turbine at small scale

- **US5,632,147 (1997, "Solar powered steam turbine generator")** — the closest single prior art: Fresnel lenses focus sunlight onto a rotor/boiler, producing steam that drives outlet jets — a **self-contained, closed-loop steam turbine generator** using lenses. **Expired** (1997 + 20 years). Source: https://patents.justia.com/patent/5632147
- **CN107327377A ("Small complementary solar steam power generation system")** — Fresnel lens + transparent cover + vacuum heat-collection store +

steam cylinder + motion assembly + generator; includes heat storage of redundant heat. Source: https://eureka.patsnap.com/patent-CN107327377A

- **Solfire (patent application, via patents-review)** — "scalable array structure containing Fresnel lenses that concentrates sun illumination on a receiver body that generates steam to drive a turbine generator"; notably also describes **open piping serving as a heat exchanger to condense vapor back to liquid** and gravity return to a tank — very close to the concept's condensation-recovery loop. Source: https://www.patents-review.com/a/11637962-solfire-solar-concentrator-pointer-structure.html
- **EP3066340B1** — Fresnel lens focusing onto an absorber generating steam from water for electrical power. Source: https://data.epo.org/publication-server/rest/v1.2/publication-dates/20200513/patents/EP3066340NWB1/document.pdf
- **US4,229,941 (1980)** — parabolic/fresnel solar collector + boiler + steam turbine + flywheel storage. Source: https://patents.google.com/patent/US4229941

GAP for the book

The **lenssteamturbine** combination is well-patented territory — the book cannot claim that combination as novel. What **no patent found** combines: lens concentration + steam turbine + **Tesla valve** flow control + **building-scale modular** packaging + **passive night condensation** as the sole recovery mechanism. The patentable/novelty story, if the founder pursues one, lives in that specific combination and in the Tesla-valve-stabilized two-phase loop — not in any single element. (Note: this is a landscape scan, not a freedom-to-operate opinion; a patent attorney would be needed before any filing.)

8. Synthesis: where the concept genuinely stands

What the concept gets right (validated by the landscape)

1. **The white space is real.** Utility CSP is 98% of the market; building-scale solar-thermal *electricity* has no incumbent — only a graveyard of failed attempts and heat-only survivors. If the engineering works, there is no entrenched competitor.
2. **Every component is proven individually.** Lenssteam (research-demonstrated), small heliostats (commodity), closed-loop Rankine (a century old), Tesla valves in two-phase flow (Nature Communications 2023). The concept is an *integration* play, not a physics gamble.
3. **The Tesla valve is the best novelty anchor.** It is the one element with no prior art in a solar steam loop, and there is peer-reviewed physics showing it stabilizes exactly the flow regime (boiling/condensing) the concept depends on.
4. **The "no storage, day-only" framing is a legitimate simplification** — it deletes the most expensive subsystem of small CSP. It must be presented as a deliberate trade, not an oversight.

The hard questions the book must answer honestly

1. **Small-turbine efficiency.** This is the physics wall that killed every predecessor. Small steam turbines have poor isentropic efficiency; the industry's answer was ORC and Stirling, and even those failed commercially against PV. The book needs numbers: expected steam conditions (pressure, temperature, mass flow), turbine efficiency assumptions, and end-to-end efficiency vs. a rooftop PV system of equal aperture. Without this, a technical reader will dismiss the concept.

2 **Parabolic lens vs. Fresnel.** A true meter-scale parabolic glass lens is heavy, expensive, and essentially unmanufactured; the entire literature uses Fresnel lenses. The book should either justify the parabolic choice or adopt Fresnel.

3 **Night condensation as sole heat rejection.** Passive condensation must reject the full daytime heat load every night through ambient cooling — the condenser sizing math must close, especially in hot climates (the concept's best markets). Thermal cycling stress (daily heat-up/cool-down of boiler and seals) is a durability question the "no leaking" claim must survive.

4 **The PV cost benchmark.** Rooftop PV is ~$0.07–0.09/kWh firm with batteries in sunny regions. The book must show a credible path to competitive levelized cost *including* tracking maintenance, water treatment (closed loop maintenance-free; scaling/corrosion), and turbine servicing.

5 **Water chemistry.** "Closed loop, no leaking" still means managing water purity — dissolved solids concentrate over thousands of cycles. This is a solved problem in industry (boiler water treatment) but it is an operating cost and a maintenance story the book should own, not ignore.

Suggested narrative spine for the 300 pages (researcher's suggestion only)

- **Part I — The gap:** utility CSP's scale trap; the building-scale graveyard (Sopogy, Infinia, SES, Heliogen) and what each proved; why PV won and what PV can't do (dispatchable heat, industrial temperatures).

- **Part II — The machine:** each subsystem with its prior art and the integration logic — lens concentration, heliostat boost stage, closed-loop boiler, the Tesla-valve-stabilized two-phase loop (with the 2023 paper as the scientific anchor), the small turbine question faced head-on, night condensation recovery.
- **Part III — The math:** worked examples at three scales (home/backyard, apartment complex, corporate building/trailer park) — aperture, steam conditions, daily kWh, costs; the replication argument to national scale.
- **Part IV — The path:** manufacturing, supply chain (commodity heliostats, lens fabrication), deployment model, and the honest risks.

What I could not verify

- Current commercial status of SunOyster Energy and 247Solar's India deployment (latest solid reporting is 2024 and 2022 respectively).
- Real-world cost figures for small steam turbines / micro-ORC expanders at the 3–50 kW scale (vendor pricing not found in public sources).
- Whether any unpublished or non-English prior art combines a Tesla valve with a solar steam loop (searches were English-language, web-visible sources).
- The founder's specific performance claims (temperatures, pressures, output) — no prototype data exists publicly because none has been published.
- Patent legal status beyond expiry arithmetic (not a freedom-to-operate analysis).

Source list (all URLs cited above, consolidated)

- https://www.coherentmarketinsights.com/market-insight/concentrated-solar-power-market-5863

- https://www.fortunebusinessinsights.com/industry-reports/concentrated-solar-power-market-100751
- https://pdf.marketpublishers.com/globalinfo/global-solar-thermal-csp-gir.pdf
- https://www.blackridgeresearch.com/blog/latest-list-of-top-csp-concentrated-solar-power-plants-projects-worldwide-pv-panel
- https://www.energyencyclopedia.com/en/renewable-energy/solar-energy/solar-concentrators
- https://unfccc.int/technology/concentrated-solar-power-plants
- https://en.wikipedia.org/wiki/Parabolic_trough
- http://article.sapub.org/10.5923.j.ijee.20221201.02.html
- https://www.researchgate.net/publication/350993886
- https://pp5.omikk.bme.hu/me/article/download/22547/22295/195886
- https://www.mdpi.com/2076-3417/16/13/6681
- https://www.pv-magazine.com/2022/04/13/concentrating-solar-power-system-for-building-applications-usable-with-pv/
- https://energypost.eu/new-concentrated-solar-power-system-delivers-1000c-for-industrial-processes-more/
- https://www.sustainablebusiness.com/2007/09/hawaiian-firm-shrinks-solar-thermal-power-40992/
- https://solarthermalworld.org/?taxonomy=country&term=hawaii
- https://www.zdnet.com/article/dishing-out-power-with-a-solar-engine/
- http://breakingenergy.com/2015/05/18/solar-will-change-the-world-but-probably-not-like-this/
- https://cleantechnica.com/2025/09/27/ivanpah-heliogen-lessons-from-concentrated-solars-decline/

- https://www.energy.ca.gov/publications/2021/novel-low-cost-high-efficiency-solar-powered-micro-combined-heat-and-power
- https://www.pv-magazine.com/2026/06/22/how-to-power-buildings-with-solar-driven-reversible-heat-pumps-organic-rankine-cycle/
- https://www.azocleantech.com/news.aspx?NewsID=34890
- https://www.nature.com/articles/s41598-024-61785-x
- https://www.sciencedirect.com/science/article/abs/pii/S0196890423001589
- https://ira.lib.polyu.edu.hk/bitstream/10397/99696/1/Li_Tesla_Valves_Capillary.pdf
- https://www.fastcompany.com/1680592/a-mirror-system-that-brings-sunlight-into-the-darkest-corner-of-your-home
- https://www.habitatmag.com/Publication-Content/Green-Ideas/2008/2008-February/Practical-Solar-Heliostat-System
- https://mlsun-csp.en.made-in-china.com/product/DQmrWcYdRJpL/China-96-Sqm-Flat-Mirror-Tower-Heliostat-with-Horizontal-and-Pitching-Two-Ways-Solar-Tracker.html
- https://patents.justia.com/patent/5632147
- https://eureka.patsnap.com/patent-CN107327377A
- https://www.patents-review.com/a/11637962-solfire-solar-concentrator-pointer-structure.html
- https://data.epo.org/publication-server/rest/v1.2/publication-dates/20200513/patents/EP3066340NWB1/document.pdf
- https://patents.google.com/patent/US4229941
- https://patents.google.com/patent/CN110500248A/en
- https://patents.google.com/patent/AU2013291157B2/en